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Published in:
Psychological Review

DOI:
[10.1037/0033-295x.105.4.634-677](https://doi.org/10.1037/0033-295x.105.4.634-677)

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Document Version
Publisher's PDF, also known as Version of record

Publication date:
1998

[Link to publication in University of Groningen/UMCG research database](#)

Citation for published version (APA):
Van Geert, P. L. C. (1998). A dynamic systems model of basic developmental mechanisms: Piaget, Vygotsky, and beyond. *Psychological Review*, 105(4), 634-677. <https://doi.org/10.1037/0033-295x.105.4.634-677>

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A Dynamic Systems Model of Basic Developmental Mechanisms: Piaget, Vygotsky, and Beyond

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A dynamic systems model is proposed on the basis of a general developmental mechanism adopted from the theories of J. Piaget and L. S. Vygotsky, more particularly a mechanism based on the concepts *assimilation versus accommodation* and *actual development versus zone of proximal development*. In the model, action and experience have a distributed effect on contents (skills, knowledge, rules, action patterns, etc.) ordered along an abstract developmental distance dimension. After a mathematical treatment of the model, an overview is given of empirical evidence on continuous and discontinuous change. The dynamic model is then applied to the classic Piagetian and the neo-Piagetian models, models of continuous and discontinuous domain-specific change, and to models of cognitive strategies, transitions, microdevelopment, and inter- and intraindividual variability.

Nineteen hundred ninety-six marked the centenary of the birth of two eminent developmental psychologists: the Swiss Jean Piaget and the Russian Lev Semyonovitch Vygotsky. The first emphasized the child as the maker of his or her own development, and the second emphasized the role of teaching and guidance and the formative effects of culture and society. Recently, a number of attempts have been made to show that their theories are less incompatible and mutually exclusive than is often believed (Glassman, 1994; Tudge & Winterhoff, 1993). In the end, the most important question concerns which substantial and lasting contributions, if any, both Piaget's and Vygotsky's works have made toward a better understanding of the nature and mechanisms of mental and behavioral development. Lourenço and Machado (1996) have recently emphasized that a considerable part of the criticism of Piaget does not address or hardly addresses the fundamental issues that his theory contributed and the issues that are still worth considering.

The general aim of this article is to demonstrate that the wide variety of characteristic patterns of cognitive development is in fact the expression of the intrinsic dynamics of dynamic systems that operate on the basis of major developmental mechanisms that can already be found in Piaget's and Vygotsky's works. If this can be demonstrated, the dynamic systems model based on these mechanisms can be seen as a way of integrating the fundamental aspects of a wide variety of theories, that is, as a way to arrive at a new synthesis. This synthetic theory does not deny the multitude of differences that exist between their theories but tries to find a convergence point in an abstract dynamic of basic principles of change that underlie development.

The structure of this article is as follows: First, I select and discuss a number of concepts from Piaget's and Vygotsky's theories that refer to basic developmental mechanisms; second,

I explain how these concepts can be transformed into a general dynamic model of development; third, I try to show that this computational model allows us to infer or deduce major quantitative empirical predictions or findings from a variety of models that have taken Piaget's and Vygotsky's works as starting points, or for that matter as starting points for criticism.

Dynamics, Theories, and Data

Dynamic systems theory is a recent attempt to apply general abstract principles and methods from nonlinear dynamics to the study of development (Thelen 1989, 1992; Thelen & Smith, 1994; van Geert, 1994a). The approach has been very successful in the field of motor performance and motor development in which mathematically rigorous methods have been used (Thelen, 1995a, 1995b; Thelen & Smith, 1994; Turvey & Fitzpatrick, 1993). However, other domains of development have also been the subject of dynamic systems theory formation and model building: language and communicative development (Elman, 1995; Fogel, 1993; Fogel & Thelen, 1987; McCune, 1995; Ruhland & van Geert, 1998; Sera & Smith, 1987; Smith & Sera, 1992; van Geert, 1991, 1993, 1995b), cognitive development in general (Bogartz, 1994; Case & Okamoto, 1996; Cooney & Troyer, 1994; K. W. Fischer & Bidell, 1997; Howe & Rabinowitz, 1994; Kreindler & Lumsden, 1994; Lewis, 1994, 1996; Rabinowitz, Grant, Howe, & Walsh, 1994; Smith, 1997; Thelen, 1995b; Thelen & Smith, 1994; van der Maas, 1993; van der Maas & Molenaar, 1992; van Geert, 1991, 1993, 1994a, 1994b, 1995b), and social and emotional development (Camras, 1992; Eckerman & Stein, 1990; Fogel, 1993; Fogel, Nwokah, Dedo, & Messinger, 1992; Fogel & Reimers, 1989; Lewis & Douglas, 1998).

I define a *dynamic* system as any formal specification of a relationship between a set of properties at one point in time and a set of properties at a previous point in time (Sandefur, 1990, p. 5). The systemic aspect is covered by the constraint that the temporal relationship is explained by the functional relationships that those properties entertain (Jackson, 1991).

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A central tenet of dynamic systems models, as Thelen and Smith (1994) wrote, is that "order, discontinuities, and new forms emerge precisely from the complex interactions of many heterogeneous forces" (p. 37). Developmental order comes about as a result of self-organization. In addition to mutual causality and self-organization, a third important concept is the *attractor* concept. Given the dynamic principles, the specific parameters and an initial state, a system is drawn toward a stable state called an attractor (which can also be a dynamic state such as a cycle or a particular way of visiting the points in a state space, such as a chaotic attractor).

Following in the wake of nonlinear dynamics, synergetics, chaos, and complexity theory, a number of scholars have examined the similarity between Piaget's basic theory of change on the one hand and modern conceptualizations of self-organization and discontinuities on the other (Beilin, 1994; Case, 1992a; Chapman, 1992; Garcia, 1992; van der Maas & Molenaar, 1992; van Geert, 1991). This interest from systems theory is not difficult to explain, because Piaget himself was strongly involved in systems-theoretical thinking in the late 1960s, especially notions such as *equilibrium* and *perturbation* (Vonèche & Aeschlimann, 1994). More recent theories, such as the information-processing approach to rule-governed thinking and neo-Piagetian theory, share a number of the basic developmental mechanisms and derive additional mechanisms from various sources. My view is that a dynamic systems approach, aiming at constructing a general computational procedure, is in fact very well suited for bridging the gaps between related theories and for bringing the central, common elements to the fore.

A striking feature of the empirical evidence on forms and patterns of change and development is that it is almost as if anything goes. That is, there exists evidence for a wide variety of developmental patterns (the overview of empirical studies given in the remainder of this article illustrates this point). Major, but probably unnecessary, theoretical controversies are created if one first assumes that of all this variety of empirical findings only a small subset is indicative of the "true" underlying developmental pattern and, second, that empirical findings that do not match this pattern are measurement error or are based on wrongly conceived experiments. For instance, the fact that both slow linear increase and sudden discontinuities exist, sometimes even within a single domain of development (Ruhland, 1998; Ruhland & van Geert, 1998), is a problem only if it is claimed or implicitly understood that development as a whole or in essence is a matter of discrete stages, continuous growth, or anything else that is incompatible with all the other notions in the list. If, however, a model of development is basically a model of mechanisms of change and not of alleged characteristic or prototypical patterns of development, variability, that is, the coexistence of many different patterns of development, becomes the target of the explanation. This is the stance taken in the present article.

The approach followed in this article is highly similar to that followed by theorists such as Kauffman (1992), Holland (1975), Langton (1992), Bak and Chen (1991), Bak and Sneppen (1993), Vandewalle and Ausloos (1996), Boerlijst and Hogeweg (1992), Prusinkiewicz and Lindenmeyer (1990), Rucker (1995), Vallacher and Nowak (1994), Nowak, Szamrej, and Latane (1990), and many others. What these theorists have

pursued is the minimal set of general principles necessary to computationally explain major qualitative properties of a particular problem domain, for instance, evolution, learning, or morphogenesis. Most of the authors cited above used what has turned out to be a universal apparatus for dynamically computing interesting qualitative patterns, namely the cellular automaton (CA). A CA is nothing but a set of units that update some property in answer to its current state and to some or other kind of input. It has been shown that CAs have interesting mathematical properties that make them particularly fit for modeling nonlinear dynamic phenomena (Langton, 1992; see Nowak et al., 1990, and Messick & Liebrand, 1996, for an application to social psychology). In the present article, I have used a simplified approach, limiting the CA to a one-dimensional array of contents ordered along a developmental distance dimension, which exchanges information with another array, the external world.

At present, the CA model currently used by psychologists is the neural network, such as in parallel distributed processing networks (Rumelhart & McClelland, 1986) and in adaptive resonance theory (ART; Carpenter, 1991; Grossberg, 1991; see Rajmakers, 1996, for an interesting application of ART to the issue of whether learning nets can self-organize into higher levels). Recently, an important contribution to developmental psychology was made by Elman et al. (1996), following work done by, among others, Elman (1993), Hare and Elman (1995), and Mareschal, Plunkett, and Harris (1995). Because a presentation of their work would go beyond the scope of this article, I confine myself to discussing the major differences between those models and the model I am presenting. Basically, a neural network model describes how a specified input, for instance, of natural language in a communicative context, gives rise to a targeted output, for instance, a child's ability to talk in sentences that comply with the grammar of his or her mother tongue. The mechanism that intermediates between the input and output is a self-organizing structure of interconnected nodes, the basic structure of which resembles that of the brain. However, the model put forward in the present article tries to capture developmental processes at a higher level of aggregation, so to speak. Basically, it implements general principles of developmental change—namely those described by, among others, Piaget and Vygotsky—and then uses the computational mechanism based on those principles in order to show that they lead to characteristic quantitative properties of development. The current model refrains from specifying which model at the level of actual processing of sensory input does the job, which is exactly what the neural network models try to accomplish. However, in spite of the considerable contributions that such models have made to further the understanding of developmental processes, one of their disadvantages is that the properties of the network structures that do the work are not structurally transparent (Plunkett & Elman, 1997). Thus, in order to understand how they do what they do, the networks' properties must be experimentally analyzed, which is far from a trivial task. For that reason, I believe that it is important to work also with dynamic models at higher levels of aggregation, such as the one discussed here. More precisely, they are models in which specific principles—for instance, the tension between an assimilatory and an accommodatory force—have been directly implemented in order to

see whether they do indeed lead to the expected outcomes. In that sense, neural network models on the one hand and higher order dynamic models like the one presently discussed on the other must be treated as complementary approaches that must try to establish a cooperative instead of a competitive relationship.

A direct connection between connectionist modeling and Piagetian theory has been made by, among others, Karmiloff-Smith (1996) and Parisi (1996). Karmiloff-Smith, for instance, has stated that Piaget's fundamental insight that the infant actively participates in the construction of its own cognition is seriously hampered by the fact that Piaget proposed a general learning mechanism without domain-specific preadaptations. Karmiloff-Smith has argued that a connectionist model of development may solve this problem without having to refer to innate representations per se (see also Elman et al., 1996).

Basic Principles Adopted From the "Founding Fathers"

Basic Principles Adopted From Piagetian Theory

First, Piaget viewed development as a process of *equilibration*, that is, the dynamic seeking or constructing of an equilibrium state. Piaget understood this equilibrium as the potentiality to withstand perturbations or disturbances by an act of compensation, which is characterized by reversibility (Piaget, 1970b, p. 725; 1975).

Second, according to Piaget, development takes the form of a journey through successive levels of equilibrium, that is, stages. Although we are used to ascribing four stages to Piaget's model, Flavell's thorough overview of Piaget's work up to 1962 calls preoperational thought a subperiod and thus distinguishes only three stages (Flavell, 1963). In *The Psychology of Intelligence* (Piaget, 1947), Piaget distinguished five stages (Piaget later summarized the second and third under the heading of *preoperational thinking*). In Piaget's short overview of his own theory (Piaget, 1970b), he described three major stages, each of which was further distinguished into two substages. In his *Epistémologie Génétique* (Piaget, 1970a) Piaget described the four stages known from most developmental textbooks, but made a further distinction between two substages in the second and third stage. In his discussion of the concept of *stage*, Piaget (1955) claimed that every stage includes a level or substage of preparation and one of completion, or formation versus final equilibrium. Finally, when dealing with the development of early concepts, such as *space* and *object*, Piaget (1937/1954) distinguished six stages, which are in fact substages of the sensorimotor stage. In summary, the essence of Piaget's stage concept is not the four stages described in the textbooks. His major point is that development occurs in the form of small series of discontinuities that involve a qualitative transformation and unity at the level of equilibrium. When we agree on calling a particular equilibrium level a *modus* or *mode*, Piaget's stage theory reflects a concept of *diachronous multimodality*. That is, there are successive (i.e., diachronous) equilibrium modes (multimodality).

Third, although Piaget conceived of the stages as overall encompassing structures, he discovered, in collaboration with Inhelder (Piaget & Inhelder, 1941), that the actual application to types, domains, or problems did not occur in a strictly synchronous fashion. Piaget coined the term *horizontal décalage* to

refer to the phenomenon that a certain equilibrium is reached earlier for some contents than for others (Piaget, 1955). This is an example of *synchronous multimodality*: At the age of 8 the concrete operational mode is applied to problems of number, mass, or quantity of matter and the still preoperational mode is applied to problems involving weight and volume. As the child's cognitive development proceeds, this synchronous multimodality is replaced by a unimodal form of thought (which, in this particular case, is formal operational).

Fourth, because of the qualitative and structural properties of the equilibrium levels, a transition from one level to the other must take the form of a discontinuity, or sudden jump. Concrete and formal operational thinking are two different kinds, and there exists no gradual and continuous transformation from one to the other. This qualitative distinction, however, does not logically or necessarily lead to the conviction that all development must therefore be in the form of major discontinuous shifts. If the concept of horizontal décalage is taken to its extreme, a qualitative shift will take the form of a continuous assimilation, one after the other, of all the separate contents or concepts to which the new cognitive structure eventually applies. This is of course not what Piaget believed, but the standpoint of qualitative distinctions is not incompatible with the possibility of continuous and gradual developmental change (or of change in the form of successive strategies, as with Siegler's model, 1981). Piaget himself did not particularly emphasize the transitory or discontinuous character of stages but preferred to speak about a stage as a *structure of the whole* (Piaget, 1955), that is a structure with a unitary character. The structure-of-the-whole concept is most often interpreted as claiming that all cognitive contents that an empirical subject can entertain must obey the same principles and properties. In fact, the concept holds that there are sets of cognitive contents that are connected with one another in a way that guarantees the emergence of an equilibrium. Chapman (1988) has convincingly shown that Piaget's concept of the structure of the whole never implied a factual synchrony of occurrence of all cognitive contents.

With the fifth principle we come to the most important explanatory mechanism, that of *adaptation*. Adaptation is a general, biological process that applies to the world of biological interactions as well as to the realm of cognitive operations and structures in the mind. Adaptation results from two opposing tendencies, *assimilation* and *accommodation*. Assimilation "is the integration of external elements into evolving or completed structures of an organism" (Piaget, 1970b, pp. 706–707). It amounts to the fact that the world, or whatever part of it the child is actually facing, is understood and shaped in accordance with the cognitive understanding of that moment. Piaget specified a symbolic form to the principle of assimilation:

$T + I \rightarrow AT + E$, where T is a structure, I the integrated substances or energies, E the eliminated substances or energies, and A a coefficient > 1 expressing the strengthening of this structure in the form of an increase of material or of efficiency in operation. (p. 707; italics added)

In the complementary operation of accommodation, the child incorporates or adapts himself or herself to the particularities of reality: Accommodation is "any modification of an assimilatory scheme or structure by the elements it assimilates" (Piaget,

1970b, p. 708). This modification is possible because assimilation is never absolute: However much the view of reality is shaped into the form of the child's current form of understanding, there is always an aspect that resists the transformation, in that the assimilating scheme must adapt to the exigencies and particularities of reality. Piaget associated the concept of accommodation to the biologist's *reaction norm*: "Accommodations are within a certain statistically defined norm" (p. 708). Moreover, "accommodation does not exist without simultaneous assimilation either" (Piaget, 1970b, p. 708). It follows then that, insofar as accommodation is a source of novelty and new information, new must always be close to old, that is within the postulated reaction norm.

According to Piaget (1970b), repeated exercise of a particular type of activity leads, via assimilation, to "*an increase of material or of efficiency in operation*" (italics added; Piaget, p. 707). To put it more generally and in non-Piagetian terms, experience with a particular type of activity leads to various forms of improvement of the activity in question (more precisely its underlying scheme). For instance, it may lead, first, to automatization of that activity (or parts thereof) and thus to higher processing and performance speed, second, to an increase in mastery and thus to a decrease in errors, or third, to an increase in generalization and thus to a broadening of the range of problems and occasions to which that particular type of activity is applicable.

Piaget's (1937/1954) developmental mechanism of adaptation is a specific example of a mechanism that combines a conservative with a progressive tendency: "Assimilation is conservative and tends to subordinate the environment to the organism as it is, whereas accommodation is the source of changes and bends the organism to the successive constraints of the environment" (p. 397). Assimilation is conservative in that it consolidates the current state of intellectual functioning of the child: Every encounter with reality confirms that the child's current understanding is useful (see also Piaget's assimilation equation quoted earlier).

It is important to note that the progressive principle—accommodation—is a central developmental principle, not just a matter of temporary adaptation to occasional unpredictable aspects of the environment. In his explanation of this concept, Flavell (1963) gave an example of an infant touching a ring suspended on a string. The object is assimilated to the existing schemata, such as the grasping schema, which in turn accommodate to the physical particularities of the ring (and not, for instance, to some or other abstract geometric property of rings, which is a form of accommodation that is still far beyond the infant's cognitive reach). It is important to note that this accommodation is not just a local and temporal adaptation (like a mattress adapting to the form of the sleeping body). Accommodation prefigures a later developmental state of the grasping schema, namely a form of grasping that is preadapted to a wide range of objects, that is generalized and differentiated in a way that allows the infant to adapt the schema to the form of the ring on the basis of visual information alone. To put it differently, the region of accommodation at time t specifies the developmental state of the schema at time $t + n$. Thus, with each experience involving an accommodation, the child is confronted with a

future developmental state, in which the current accommodation will be part of the assimilation aspect.

Note that I have not mentioned the notion of equilibration as a separate developmental mechanism. The reason is that Piaget (1964, 1970b, 1975) saw equilibration as a capacity, namely, the capacity to counteract perturbations by an act of compensation. Equilibration is the result of a developmental process based on the fundamental mechanism of adaptation. It can also be seen as a developmental process, in that it refers to the process of building that capacity.

In summary, the major developmental mechanism of adaptation that occurs in the form of actions and experiences involves a synchronic dialectic between a conservative mechanism consolidating the present state of development (assimilation) and a progressive mechanism preparing and establishing a future state of development (accommodation).

Basic Principles Adopted From Vygotsky's Theory

Vygotsky's major contribution to developmental psychology is his theory of the relationship between development on the one hand and instruction, play, educational help, imitation, and learning, that is, education in the broad sense of the word, on the other hand. Vygotsky's contribution to developmental psychology is now widely acknowledged (Bidell, 1988; Schneuwly, 1994; Van der Veer & Valsiner, 1991; Wertsch, 1985; Wertsch & Tulviste, 1992). A wide variety of studies have been published showing the fruitfulness of Vygotsky's major concepts in early development (Bernicot, 1994; Diaz, Neal, & Vachio, 1991; Goldin-Meadow, Alibali, & Breckenridge Church, 1993; Lillard, 1993; Meins, 1997; Pacifici & Bearison, 1991; Pellegrini, Perlmutter, Galda, & Brody, 1990; Rogoff, Malkin, & Gilbride, 1984; Saxe, Gearhart, & Guberman, 1984; Valsiner, 1984; Whitehurst et al., 1988) and in schooling (Alibali & Goldin-Meadow, 1993; Aljaafreh & Lantolf, 1994; Brissaud, 1994; Campione, Brown, Ferrara, & Bryant, 1984; De Guerrero & Villamil, 1994; Griffin & Cole, 1984; Holzman, 1995; Lidz, 1995; Lumpe & Staver, 1995; Peters, 1996; Weil-Barais, 1994).

Three basic concepts are used to explain how development comes about as a consequence of education: *zone of proximal development* (ZPD), *internalization* (or *interiorization*), and *mediation*. Of these three, *mediation* primarily refers to the way the human intellect operates, that is, by using instruments such as signs and symbols, that play an instrumental role in mediating between human participants and the objects on which they act (Moll, 1994; Van der Veer & Valsiner, 1991; Wertsch, 1985). Interiorization is the mechanism that explains why the ZPD leads to developmental progress. In this article, I concentrate on the ZPD mechanism because it focuses on the long-term dynamics of development (see van Geert, 1994b, for a dynamic growth model of the ZPD mechanism).

Vygotsky described and defined his ZPD concept in a series of lectures held in the early 1930s (Van der Veer & Valsiner, 1991). In a lecture from 1933 he defined the ZPD as a distance between the child's actual development and the child's potential development (Van der Veer, 1996). The first is expressed in the form of tasks the child can solve on his or her own, the other by tasks solved under the guidance of adults or more capable partners (Vygotsky, 1933, as cited in Van der Veer & Valsiner,

1991, p. 337; Vygotsky, 1978, p. 57). In the context of school instruction, Vygotsky also spoke of the distance between a child's real mental age and his or her ideal mental age. The latter is in fact the future real mental age that will be achieved by a process of teaching and learning that acts within an optimal distance from the current real mental age. If the distance is too small or too big, the child will profit little from the instruction given. This optimal distance, according to Vygotsky (1933, as cited in Van der Veer & Valsiner, 1991), coincides with the ZPD. In the school context, this notion of ZPD implies that if a 5-year-old child has an ideal mental age of 7, optimal instruction should address contents and issues that are 2 mental age years beyond the child's current independent developmental level (Vygotsky, 1933, as cited in Van der Veer & Valsiner, 1991, p. 340). This conception of the function of instruction does not imply that these more advanced contents are simply transmitted or "implanted" in the child's mind by the educator. The educational activity is the starting point of an internal process that transforms the instructed contents into contents (skills, knowledge, etc.) that are the child's own (Vygotsky, 1933, as cited in Van der Veer & Valsiner, 1991, p. 331). This internal transformational process is illustrated by the fact that "(the imaginary curve of) school instruction . . . [does] . . . not proceed in parallel with (the imaginary curve of) cognitive development, and thus, has its own dynamics" (Van der Veer & Valsiner, 1991, p. 335), a fact that is nicely illustrated by the results of dynamic growth modeling of Vygotsky's ZPD mechanism (van Geert, 1994b). Another illustration is the fact that children who have the same ZPD or ideal mental age can have different real mental ages (actual level of development; Van der Veer & Valsiner, 1991; Vygotsky, 1978), which explains why children with similar mental ages often widely differ in the extent to which they can profit from instruction (Vygotsky, 1933, as cited in Van der Veer & Valsiner, 1991). In addition to instruction, play is another important source of ZPD experiences (Lillard, 1993; Valsiner, 1984, 1988; Vygotsky, 1966, 1978).

Van der Veer and Valsiner (1991) pointed at a possible inconsistency in Vygotsky's ZPD concept or the concept of an ideal mental age that is relevant to the current discussion about mechanisms. Vygotsky explained how schooling bridged the gap between the actual and the potential level, but he did not directly address the issue of how the ideal or potential level itself increased. Van der Veer and Valsiner (1991) argued that the shift of the ZPD toward increasingly higher levels must be a logical consequence of Vygotsky's basic developmental principles, although Vygotsky, in their view, never made particularly clear how this shift comes about. It is likely that the principle of optimal distance, which figures so prominently in the explanation of how the gap between the actual and the potential level is bridged, may also explain the progress of the ZPD and the ideal mental age itself. That is, the dynamic relationship between the actual and the potential developmental level explains not only the change in the actual level but also in the potential level or ZPD (Valsiner, 1994; van Geert, 1994b). Given the potential dynamics of a ZPD-driven developmental process (van Geert, 1994b), it is likely that the major function of, for example, instruction or help is not to offer a learning environment merely at the ZPD but rather a learning environment at or around the ZPD that can also expand the ZPD.

In summary, the central contribution of Vygotsky is his concept of development as the dynamic interplay between two levels of development, one the actual, the other the potential, that is, the future level of development, that are both an intrinsic part of the child's developmental state. The dynamic relationship between these two aspects, expressed by the notion of the ZPD, is what makes development move about. This dynamic principle is highly similar to Piaget's notion of the interplay between accommodation and assimilation. Note, however, that this similarity does not imply that Vygotsky, like Piaget, was basically a stage theorist. Vygotsky hardly addressed the notion of stages, except for the developmental model sketched in *Thought and Language* (1962; see also Verhofstadt, Vyt, & van Geert, 1995, for an overview; Van der Veer, 1986; Zender & Zender, 1974). My main interest, both with Piaget and Vygotsky, lies with their basic developmental mechanisms—which are similar in their common emphasis on the dialectic aspect—and not with stages or continuity per se. The question of whether those mechanisms lead to continuity, discontinuity, or both, depends on the conditions in which they operate.

From Basic Principles to a Dynamic Systems Model

General Properties of Dynamic Models of Cognitive Development

The principles that I adopted from Piaget's and Vygotsky's theories are still heavily loaded with conceptual connotations that need to be clarified before the principles can be transformed into a formal dynamic systems model. In this section I discuss a number of generalizations needed for building such a model.

First, the theories implicitly or explicitly imply that the explanation of development requires a distinction between some internal and some external domains, whatever their respective natures. The internal domain is where we project the person's knowledge, skills, motor system, or whatever specifies the contents and developmental level of the person's current cognitive system. The external domain contains the sources of information, help, resources, models, and so forth. Development comes about as a result of transformations applied to contents from the internal and from the external domain. Theories differ as to the relative importance they ascribe to either of the domains.

Second, developmental theorists accept that the internal and external domains communicate with one another, that there exist links of various kinds between them. For instance, the notion of *experience* implies that something in the external domain has a relation of condition, cause, or content to something in the internal domain. The notion of *action* implies that a subject can alter properties of the outside world. The discussions among theorists are about the nature and not about the principle of that link or connection.

Third, the theories agree on the principle of developmental order. That is, behaviors, actions, problems, tasks, and experiences (or whatever else constitutes the units of analysis) can be ordered along a developmental scale in terms of less or more developmentally advanced levels.

Fourth, the theories that I discuss agree on the fact that development is possible. In view of the discussion on the learning paradox, this is not a trivial statement (Chomsky, 1980; Fodor,

1975; see also the literature on the so-called "learning paradox," Boom, 1991; Juckes, 1991). On the basis of the work of, among others, Kruschke (1996), Raijmakers and Molenaar (1994), Raijmakers, van Koten, and Molenaar (1996), Raijmakers (1996) argued that neural nets of the adaptive resonance type can construct higher levels of complexity without being issued with new resources or contents by an external source. This demonstration of the possibility of increasing complexity through self-organization is of central importance to any general theory of development.

The fifth and probably most important point has been directly adopted from Piaget's and Vygotsky's theories. It states that there exists a fundamental dialectic between a primarily subject- and a primarily object-driven force, which constitutes the motor behind the developmental process. This dialectic involves a tension between a consolidated, current level (which is the primarily subject-driven aspect of an activity or experience) and a potential, future level of development (which is the primarily object- or environment-driven aspect of an action or experience, although it is still confined by the current state of the subject). For Piaget, this dialectic takes the form of assimilation and accommodation. Vygotsky conceptualized it as the distinction between actual development and proximal development. A similar dialectic principle is also present in a variety of modern theories. Information-processing theory, especially the rule-assessment or production systems approach (Klahr, 1995; Klahr & Wallace, 1976; Siegler, 1983; Simon & Klahr, 1995), makes a distinction between rules or strategies on the one hand and the information resulting from applying those strategies on the other. The resulting information is then integrated in a way that either consolidates or modifies the available rules and strategy sets. A comparable dialectic plays a central role in Thelen and Smith's (1994) dynamic systems theory. The fundamental dialectic here is the one between the ongoing action and the possibilities that action retrieves from the environment as it acts on the latter. Because the subject incorporates these possibilities into the ongoing action, the way of acting changes and thus leads to new skills. The basic dynamic model resembles Gibson's (1979) notion of action and perception potentialities and the affordances of the environment that eventually change these potentialities. Finally, in neural network models of development (e.g., Elman et al., 1996), the potential state is represented by a training set consisting of inputs and required outputs. The learning system obtains information from the training set in terms of its own current learning state by making errors. The errors lead to updating the connection weights, which eventually reduce the distance between the current state of the system (its actual state) and the properties exemplified by the training set (the potential state).

With these five principles in mind we can now try to build a dynamic systems model that implements the basic developmental mechanisms discussed so far.

Components of the Model

Developmental space and the developmental distance dimension. If we compare a person at a specific age with that same person at an older age in terms of developmentally relevant differences, we can do so on the basis of an arbitrarily large (but in general very large) set of descriptive dimensions. Each

dimension specifies one particular aspect at which the younger person from our example differs from the older. Each dimension also allows for a developmental ordering, all other conditions being equal. For instance, given that all other conditions are similar, a more generalized application of a rule or principle is more developmentally advanced than a less generalized one. This principle will appear in the form of sequences that can in principle be detected by using Rasch models, for instance (G. H. Fischer & Molenaar, 1995).

It is important to note that a multidimensional ordering can always be transformed into a one-dimensional ordering (van Geert, 1994a, 1997). Because each of the dimensions—which we usually call variables—specifies a potential ordering, we can specify an arbitrarily chosen minimum or starting point and an arbitrarily chosen end point. Although the selection of those minima and maxima is a pragmatic matter, it must of course make theoretical and empirical sense where exactly we put them. Thus for any set of descriptive dimensions that make sense with regard to some particular developmental domain or phenomenon, we can construct a hyperspace consisting of those dimensions. For our purpose, the dimensions need not be empirically independent of one another. The hyperspace has an origin or lower bound, $\emptyset$, which has as its coordinates the set of specified minima on our dimensions, and an upper bound M , with the set of specified maxima for its coordinates. By definition, any task or action relevant to the developmental question at issue, that a particular person performs in a particular way and at a particular time—let us call this T_i , can be described in terms of the descriptive dimensions chosen. This implies that T_i can be specified as a particular point in the hyperspace, with the set of values on each of the descriptive dimensions or variables acting as coordinates. Alternatively, T_i can be specified as a developmental distance, which is nothing but the distance between T_i and $\emptyset$ in the hyperspace. If the dimensions are conditional or oblique—which we would expect in the case of psychological and developmental variables—that distance is not equal to the Euclidean distance between T_i and $\emptyset$. Irrespective of what exactly the distance is, it can always be abstractly conceived of as a dimensionless number ranging between 0 and 1. By doing so we are now able to compare developmental levels or steps belonging to a specified domain of phenomena or contents as positions on a one-dimensional, abstract distance variable that enables us to reduce developmental processes to one-dimensional quantitative patterns. It goes without saying that such reduction is a matter of theory-specific simplification without ever entailing anything like an ontological claim about the alleged simplicity of development as such. Finally, note that by developmental level, I mean a point on the $\emptyset$ - M distance dimension (as in "water level") and not a phase or stage (although levels can of course remain stable long enough to form a stage, for instance).

The internal and external system array. Because we have confined ourselves to the specification of actions, experiences, and so forth in terms of their developmental position with regard to an (eventually imaginary) origin of a descriptive hyperspace, we can represent a developing subject at time t as an ordered one-dimensional array of cells, $I(c_1, c_2, \dots, c_n, \dots)$. A cell such as c_n is nothing more than a reference to subject-specific conditions (e.g., mechanisms, skills, representations, or whatever one may wish to work with) necessary to perform an action

or solve a problem at the developmental level, n . Thus the array I represents the subjects' possible actions or experiences, et cetera in terms of their distances from the descriptive origin. Note that because the contents are only references, all contents of the developing system (e.g., a cognitive system of a particular subject) are reduced to an index specifying that content's position on a developmental ordering scale in the descriptive space.

In addition to this internal array there exists an external array that contains all the possible experiences, actions, problems, events, and so forth that a developing system may be confronted with during its lifetime. Because the developing system cannot experience the external world except in terms of something it can understand (that the developing system can activate, process, act upon, etc.), alone or with the help of others, we can in fact refrain from specifying those external events in terms of their intrinsic properties and specify them instead in terms of the internal contents they will potentially activate. Because the internal contents are represented by a developmental index, the experiences (accommodations, etc.) that activate them can also be represented by such an index, namely the index of the content they activate. In this way, experiences are subjectively and contextually defined. For instance, a particular state of events in the world, such as a plastic ring, may evoke a sensorimotor grasping scheme—say a c_n content—in a baby, and a formal geometric representation of a circle in an adult—say a c_m content (Flavell, 1963). In an alternative theoretical framework, that ring may start a soft assembly process of reaching and grasping in one case and a soft assembly process of imagining formal geometric properties in another case. It is even possible that, with the same subject, the ring invokes a c_i content at one time and a c_j content at another. We may thus specify the external array as follows, $E(c_1, c_2, \dots, c_n, \dots)$, and claim that I is a proper subset of E , namely $I \subseteq E$, which implies that the environment is viewed as a potential source of learning and development.

The system's activity function. The possibility that a subject performs a particular kind of action at level n or is capable of processing or understanding an experience at that level says nothing about the actuality or probability of that action or experience. In order to specify the probability of the activation of a content at a particular level (i.e., the probability that a person performs an activity on that developmental level), I define a weight function across the array of possible contents, that is, levels. Thus, the likelihood that a content c_i will be activated at time t is a (filtered) function of its associated weight w_i (see Figure 1).

I have described the developing system in terms of two coordinates. One is the set of positions on the developmental distance dimension; the other is the weight function that ascribes a (varying) weight for each level, $W[(c_1, w_1), (c_2, w_2), \dots, (c_n, w_n)]$.

The idea that the probability of activation of a content depends on some weight function associated with that content is a basic notion behind a host of simulation models, for instance, parallel distributed network modeling (Bechtel & Abrahamsen, 1991), neural network modeling (Carpenter, 1991; Elman et al., 1996; Grossberg, 1991), and models of competitive strategy choice (Siegler & Shrager, 1984).

In the limit, where i is a real distance measure on a continuous dimension, we may assume that the field of weights is a sum of functions of all is :

$$\sum_{p=1}^{p=n} \left[\sum_{i=1}^{i=n} (a_p \cdot e^{-(i-b_p)^{q_p}}) \right], \quad (1)$$

where the power q_p is an even number and constants $a_p, b_p, \dots$ are positive constants; the function specifies a symmetrical bell curve with a top falling on point b_p . If $p = 1$, the field of weights takes the form of a single bell curve with the array I as abscissa and a maximum a_p on b_p . If $p > 1$, the field of weights has two or more "bumps" corresponding with the maxima of the associated bell-shaped distributions.

Except for maybe the most extreme nativist or maturational theories (if any such extreme theories exist), all theories accept that development is the effect of interactions between the developing system's internal properties and the environment in which the system operates. Given the premises defined earlier, how can such an interaction or link between the internal and the external system be represented in the most general way? Recall that the developing system is represented by a one-dimensional ordered array, which can most naturally be specified as an array of numbers (each number specifying a distance, that is a developmental level, and thus also a set of possible actions, experiences, etc., corresponding with this level). Because the model does not reckon with the exact nature of the link but only with its existence, it thus suffices that it can specify the existence of a particular link. This can be done by connecting an item in the internal domain and an item in the external domain by marking them with a similar index figure. A similar index means that a link exists between a particular indexed element from the internal domain and a particular indexed element from the external domain, whatever the nature of that link and whatever the properties the linked elements share or those by which they are distinguished. In short, in the dynamic model, any event that takes place in the form of an action and an experience is in fact a confluence between an internal content and an external content that share a developmental level index.

For instance, the event at time t_1 consisting of a child who is solving a particular arithmetic problem by using a certain numerical skill, is represented by specifying at time t a similar index to the numerical skill on the one hand (a content in the internal domain) and to the actual problem on the other hand (a content in the external domain). These links are not fixed. Which internal content will be linked to which external one depends on both fundamental and contingent restrictions. The fundamental restriction is that only contents actually present in the internal domain can be addressed. The contingent restriction is that many kinds of contextual properties exist that determine the content to which an actual event will be assimilated (see the examples on the object concept and conservation discussed later).

We can now incorporate the Piaget-Vygotsky dialectical mechanism of development into the dynamic model in the following way. Recall that when the system performs an action, it means that a particular content is activated, namely the content c_n (e.g., a particular skill, rule, principle), which is a content at developmental level n that is responsible for that action. The effect of an action on the further development of the system, however, depends on two contents. One is the activated content c_n based on the system's internal array properties and that represents its actual, context-dependent developmental level, which

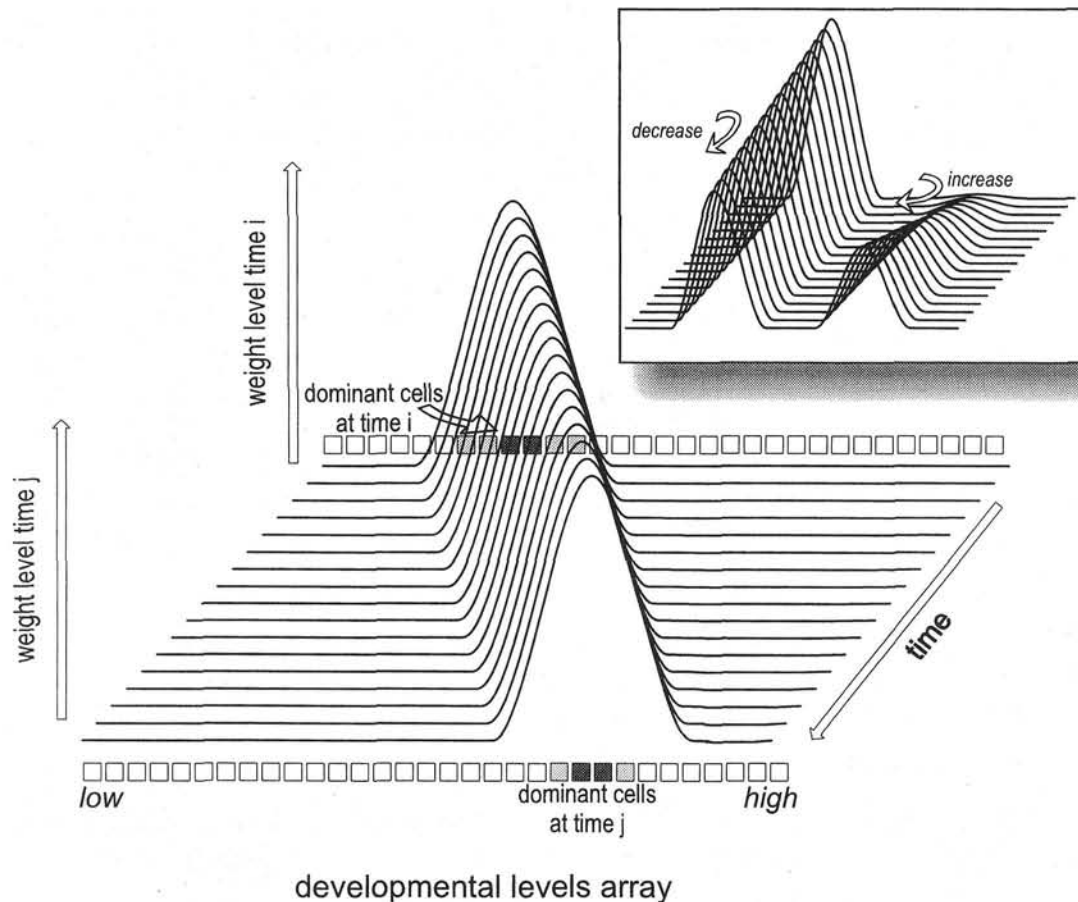


Figure 1. A simplified representation of continuous shift in a unimodal weight distribution across the internal content array. The activation probability of the cells is proportional to the weight. The inset panel shows a bimodal change in the weight vectors: The first mode declines and a second mode emerges, giving rise to bimodal responses.

Piaget would identify as assimilation and Vygotsky would identify as the actual level of development. The other content is defined by accommodation or by the help and information given or by feedback resulting from performing the action. I show later that this second content defines or specifies the system's potential level.

Let me first concentrate on the content primarily activated on the basis of the internal array properties. The developmental level attributed to a subject at time t , given the context occurring at time t , is determined by the developmental level of the action (or experience) performed at t in that context. Which action this depends first on the array of available actions and thus on the subject's developmental state and second on a stochastic aspect, namely the task, problem, or context given. We have seen that for any content there is a weight and that this weight is related to that content's activation probability. It is natural to assume, therefore, that the actual activation of a content is a stochastic function of the weight. More precisely, the activated content at time t , A_t , is the content that corresponds with the maximum value of the following stochastic function:

$$A_t = c_i \text{ if and only if } (w_i \cdot \xi_i)_t \\ = \text{MAX} [(w_1 \cdot \xi_1)_t, (w_2 \cdot \xi_2)_t, \dots, (w_n \cdot \xi_n)_t]. \quad (2)$$

(A_t corresponds with c_i if and only if Equation 2 holds, for w_i the current weight of content i and $\xi_{1,2,\dots}$ random numbers drawn from a uniform distribution with fixed minimum and maximum.) This function simply multiplies each of the weights associated with a level or content from the array of levels or contents with a different, randomly drawn number. Next, it decides which product has the largest value. It then selects the content corresponding with the winning number as the activated content. Each time the system is assumed to perform an action and have an associated experience, the random numbers are of course reset and chosen anew in order to warrant a natural distribution of outputs.

Note that the developmental level attributed to the system at time t is only the level incorporated into the system's current activity and associated experiences at time t , irrespective of the amount of accidental factors that codetermine those actions and experiences. Thus the system has no "real" developmental level

concealed behind the "accidental" levels that result as a consequence of the context. In this sense, there is no distinction between the scale of action time and the scale of developmental time in that developmental time is just the sequence of actual actions. This principle of the unity of action time and developmental time plays an important role in Thelen and Smith's (1994) dynamic model of development.

An interesting property of the content selection or activation function described by Equation 2 is that the frequency distribution of the activated contents differs from that of the weight function. The content activation function operates as a simple filter that lets the higher values of the weight function go through and holds back the lower values. Assuming Equation 1 specifies a unimodal, normally distributed array of weights with mean weight value m_w and standard deviation s_w , Equation 2 specifies a band pass filter that will withhold all weights (or, more precisely, the contents associated with those weights) that are smaller than $m_w - (s_w/2)$ or bigger than $m_w + (s_w/2)$. This means that the activated contents are always within the band(s) defined by the upper part of the weight function (which, depending on the modality of the function, could be one or more bands and, depending on the width of the function, could involve a smaller or a bigger bandwidth or range). In order to obtain a filter that lets a larger range of developmentally different contents pass, we can apply a logarithmic transformation of the weights.¹ Thus, the filter function from Equations 2 and 2a (see Footnote 1) specifies the actual expressed developmental level of the system, that is, the level that comes about by the interaction between the internal possibilities of the system and its external conditions or contexts of action.

However, there is more than just an expression of the system's actual developmental level assimilating the external world to its current developmental state. There is also, first, the input of the environment, E_i , in the form of, for example, an experience, accommodation, information, or help, and, second, the potential level P_i (which takes the form of the accommodation or affordance range, or of the activity that a child can perform with the help of more competent others).

Let me first concentrate on the environmental input experienced by the system. The environmental input processed as an experience E_i can take various forms: It presents the developing system with help and instruction, with information the system will have to accommodate, or with a structured environment that will consolidate and possibly alter the potential level. The environmental input—which takes the form of an experience—must, by definition, be accessible to the system (or else it is not considered an input) and thus is represented in the form of an index number that corresponds with an index number in the internal array. That is to say, environmental information, help, instruction, or whatever else the external world may present to the developmental system activates an internal content, in the form of a representation, understanding, an affordance for further action, et cetera. That internal content is specified by a developmental index number, that is, by a position on the developmental distance dimension. We can also say that the environmental input is an experience, that is, an environmental event that activates a specific internal content. Consequently, if there exists no content that can be activated by an event, there is no experience of that event. However, events will always activate

an internal content (and if the participant does not understand an event, that content is eventually a feeling expressing wonder, embarrassment, etc.). Formally stated, experiences must be within a range of levels that does not exceed the internal array's maximum plus a small expansion zone,

$$E_i = c_k \text{ if and only if } k \subseteq n + \phi \quad (3)$$

(an experience of externally given information at time t is at developmental level c_k if and only if Equation 3 holds, that is, if k is within the range bound by the maximal level n plus some small expansion zone ϕ).

This expansion zone is of central importance. If $\phi = 0$, the system will forever be confined to the size of its initially given content array. That is, it will never be able to go beyond what is innately given. This is in fact what strictly nativist theories postulate, namely that the size of I does not increase. We see later that development is possible even if ϕ is very small and for instance is determined by what the child can understand or grasp if helped by a more competent other or by a prestructured situation.

Let us now focus on the second aspect, the potential level. The potential level has no mysterious status. It is simply a developmental level corresponding with a set of contents (skills, rules, etc.) in the array that is most sensitive to the effect of experience brought about by the activated content c_n (this array of sensitivity to instruction is basically what Vygotsky intended with his ZPD concept; see also Meins, 1997). Why would there be one content that is more sensitive to experience than another? The existence of such an optimum is easy to explain if we reckon with two opposing tendencies that are likely to occur in learning and developing systems. The first tendency is that sensitivity, attention, effort, and similar resource factors increase with increasing novelty (all other things being equal). New things are more interesting than old ones, offer possibilities for exploration, are more explicitly emphasized by educators, and so forth. The second tendency is just the opposite, namely that resource factors increase with increasing familiarity: Familiar things are easier to understand than unfamiliar ones, their processing requires less effort, it is easier to do something one is familiar with than something new, and so on (there exists a considerable literature on the novelty–familiarity issue; examples of recent studies in the field of infant perception and cognition are Bogartz, 1997, and Spence, 1996; examples of studies that address the existence of different mechanisms for processing novel vs. familiar information are Metcalfe, 1993, and Tulving et al., 1994, 1996). Assume that both tendencies—preference for novelty and preference for familiarity—can be expressed in the form of an exponential function. The familiarity

¹ The logarithmic transformation of w_i takes the form $\log[\omega_i \cdot (1 + w_i)]$ and thus leads to the following transformation of Equation 2:

$$A_i = c_i \text{ if and only if}$$

$$\{\log[\omega_2 \cdot (1 + w_i)] \cdot \xi_i\}_i = \text{MAX} \{[\log[\omega_2 \cdot (1 + w_1)] \cdot \xi_1], \\ (\log[\omega_2 \cdot (1 + w_2)] \cdot \xi_2), \dots, (\log[\omega_2 \cdot (1 + w_n)] \cdot \xi_n)\}_i \quad (2a)$$

for $\xi_{1,2,\dots}$ random numbers, $w_{1,2,3,\dots}$ the weights, and ω_2 a constant.

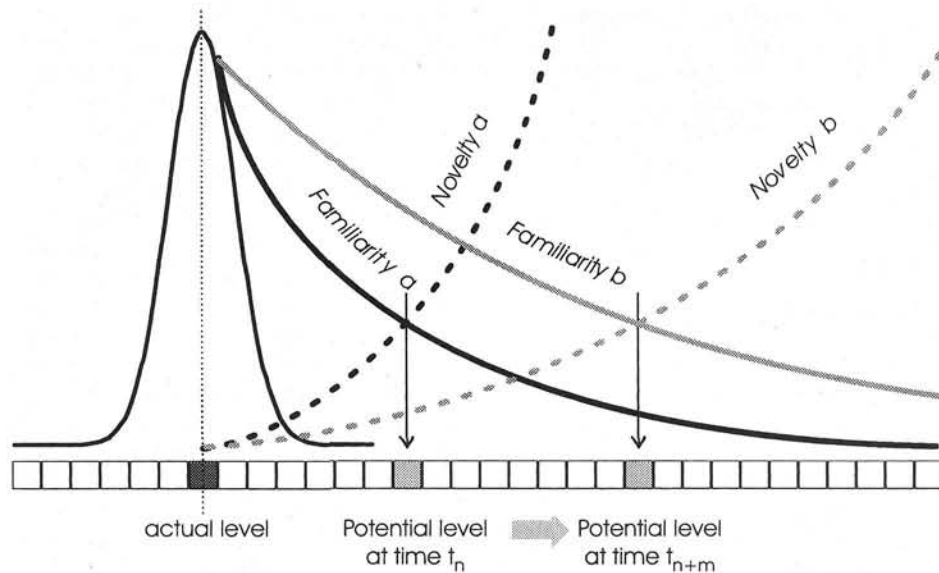


Figure 2. Cells at a distance from the actual level (dark gray cell) have an update probability that depends on familiarity (resemblance with the actual level) and on novelty; the cell where familiarity and novelty cross has the maximal update value, and thus corresponds with the most likely future actual level (potential level, State a); changes in the familiarity and novelty curve explain the shift of the optimal or potential level cell (State b).

function has a maximum value at the content that corresponds with the system's actual developmental level because that is, by definition, currently the most familiar content. Likewise, the novelty function has a minimum value at the system's current developmental level,

$$f_{\text{familiarity},M} = a \cdot b^{c \cdot \Delta M}$$

$$f_{\text{novelty},M} = d \cdot e^{f \cdot \Delta M}, \quad (4)$$

for $(a, b, c, d, e) > 0$, $e < 1$, and $f < 0$, and ΔM the distance between a content M in the array and the current developmental level. It is natural to assume that if both functions specify the system's preferences, the most preferred content is that for which the value of both functions is maximal. By "most preferred" I mean the content that, if experienced, causes the maximal learning or developmental effect (however small that effect may eventually be). This content (more precisely, its corresponding developing level) lies at the point at which the functions cross. Thus, the most preferred content, or the content for which the system is most sensitive, is a content with the following distance specification (or developmental level specification, for that matter):

$$\Delta M_m = M - \frac{\log\left(\frac{a}{d}\right)}{c \cdot \log(b) - f \cdot \log(e)}. \quad (5)$$

Assume that, as a consequence of teaching in the ZPD, a child becomes more familiar with contents (skills, rules, etc.) that are ahead of the child's current actual developmental level. Or, to give a different example, assume that a child has increas-

ing experience with a particular type of problem domain, which, in Piagetian terms, implies that the range of its accommodations to that problem domain has increased. This means that the familiarity function becomes less steep, that is, slightly more developmentally advanced contents become more familiar. Note that this does not necessarily mean that those contents become less attractive from a novelty point of view. It follows directly from Equation 4 that if familiarity with more developmentally advanced contents increases (expressed, for instance, by a decrease in the parameter c), the value of ΔM_m increases proportionally (see Figure 2). That is, with increasing familiarity with contents of higher developmental levels, the contents to which the system is most sensitive become developmentally more advanced. The same effect occurs if, for some reason or another, the novelty of more advanced contents becomes smaller (or if any combination of changes in novelty and familiarity occurs as a consequence of prolonged exposure or experience).

We have seen that maximal sensitivity to particular contents or developmental levels means that experiences at those levels have a maximal effect. The dynamic system allows for three possible effects: (a) the change in the weight associated with the contents in the internal array, (b) the change in the level at which maximal sensitivity occurs, and (c) the expansion of the internal array of contents by incorporating more developmentally advanced levels. I now discuss the first type of developmental effect, namely the change in the weights associated with the contents in the developmental array.

The system's development-and-change function: The weight update function. Each time the system has gone through an action/experience sequence (e.g., when a child has solved a particular problem in a particular way), the system's weight array

is updated. It is natural to assume that the maximal gain in weight is made at two places. One is the content corresponding with the actual output level, that is, the activated content A_t . This assumption is in line with Piaget's contention that assimilation leads to consolidation of the schema on which that assimilation was based. The other location at which a maximal gain in weight occurs is the point of maximal sensitivity to experience (accommodation, help, information, etc.), which is defined by the optimality equation (Equation 5). It is obvious that a content for which the system is maximally sensitive shows the fastest growth in its weight function, and thus represents the potential future developmental state P_t . In the Piagetian model, A_t represents the level incorporated in an assimilatory action at time t , and P_t represents the accommodation range, which is present in every action (in addition to the assimilation range, which is the A_t level; note again that A_t and P_t are action- and context-specific expressions of the system, not just general underlying properties). In the Vygotskian model, A_t corresponds with the actual level, and P_t with the potential level at the ZPD. The experiences associated with a particular activity do not necessarily actualize the whole ZPD. For instance, when instruction makes only little demands on a student, his or her experiences will stay far below his or her ZPD. In that case, the maximal update value lies not on P_t but on some stochastic function of P_t , that is, a randomly chosen point between A_t and P_t :

$$P'_t = A_t + (P_t - A_t)\xi_t, \quad (6)$$

for ξ_t a random number between 0 and 1 drawn from a uniform distribution.

By the same token, experiences do not always activate the whole accommodation range, that is the maximum or optimum of accommodation a child can achieve while activating a particular scheme. More generally, the effect of an action is not always equal to the maximal amount of information the system can eventually process, and thus Equation 6 describes a general condition (note that with the majority of simulations, the simplified condition $\xi_t = 1$ has been used).

The weight update function is determined in the following way. As a general rule, assume that the effect of an action, or an experience for that matter, consists of an increase in the action or experience potential. For instance, performing an action has a practice effect, increases its application range, increases the association strength between its assembly components, contributes to the internal equilibrium, or whatever other consolidating or learning effect one's particular theory specifies in such cases (see also Piaget's assimilation effect equation; Piaget, 1970b). In the current abstract model, we are interested in only the effect that is represented in the form of an increase or a decrease in the content's weight vector. As a rule, we let the weight increase or decrease by means of a logistic growth function, which is just a convenient way of letting the weight change within certain lower and upper limits. This function has a maximal value for contents that correspond with A_t and P_t (or its transformation P'_t), respectively.

Because the contents are ordered in terms of their distance and thus in terms of developmental or structural similarity, we may assume that the effect of an action not only applies to A_t and P_t (or P'_t) but also to all contents that are developmentally

similar to A_t and P_t (or P'_t). It is natural to assume that the magnitude of the effect decreases with decreasing similarity, that is, with increasing developmental distance from the currently activated content. The decrease actually turns into a negative number, that is, remote contents are negatively affected. This is a generalization of the support-and-competition notion put forward in a variety of models (e.g., Siegler & Shipley's adaptive strategy choice model [ASCM; 1995]; Fischer's [K. W. Fischer & Bidell, 1997] and my model of connected growers, van Geert, 1993, 1994a, 1995b; or Grossberg's, 1991, ART models). It is important to note, however, that this function is affected by a considerable variety of context-determined, accidental factors. For instance, if a child solves a social problem by using some abstract principle, the effect may spread toward other social problems that require slightly lower or higher levels of social problem-solving development but not toward numerical problems, even if those problems can be solved by using the same abstract principle. This concept of a domain-specific spreading of effect is explicitly accounted for by Case's (1992a) notion of central conceptual structures, which represent a means of linking related developmental contents.

In summary, the weight update rate $r_{i(t)}$ for a level L_i is the sum of two stochastic functions that have their maximum values η and θ on the levels A_t and P_t (note that P_t can also be read as P'_t , depending on which type of function one has decided to use):

$$r_{i_t} = (\eta_s - e^{-\sigma_a \Delta A_{i_t}}) \cdot \xi_{A_{i_t}} + (\theta_s - e^{-\sigma_b \Delta P_{i_t}}) \cdot \xi_{P_{i_t}} \quad \text{for}$$

$$\Delta A_{i_t} = |L_{i_t} - A_{i_t}| \quad \text{and} \quad \Delta P_{i_t} = |L_{i_t} - P_{i_t}|, \quad (7a)$$

where growth rate r at time t equals the sum of the functions of η , θ , σ , and a stochastic variable ξ , which is a number drawn from a uniform distribution with minimum 0 and maximum 1; e is the base of the natural logarithm. Note that this equation specifies a roughly bimodal distribution of growth maximum values for two particular contents, namely the content corresponding with the actual developmental level and the content with maximal experiential sensitivity, that is, the potential (future) level.

An alternative equation with similar general properties but that also accounts for a stochastically distributed eventual negative effect of an action or experience on related contents has the following form:

$$r_{i_t} = \begin{cases} \xi_k \cdot \eta_s & \text{if } \xi_k < 1 - \sigma_a \Delta A_{i_t} \\ \xi_m \cdot \eta_c & \text{if } \xi_k \geq 1 - \sigma_a \Delta A_{i_t} \end{cases} + \begin{cases} \xi_n \cdot \theta_s & \text{if } \xi_n < 1 - \sigma_b \Delta P_{i_t} \\ \xi_m \cdot \theta_c & \text{if } \xi_n \geq 1 - \sigma_b \Delta P_{i_t} \end{cases}, \quad (7b)$$

which also specifies a stochastically deformed, bimodal distribution of growth rates across the internal content array.

Given a particular $r_{i(t)}$ for a content $c_{i(t)}$ in the internal array of contents, the change in the associated weight w_i is specified by a simple logistic growth function:

$$\Delta w_{it} = w_{it} \cdot r_{it} \cdot \left(1 - \frac{w_{it}}{\max}\right), \quad (8)$$

which has an asymptote at a predefined maximal level, $\max$, and which is similar to the standard sigmoid activation function used in the majority of neural network models (Plunket & Elman, 1997).

The system's development-and-change function: The array expansion function. Recall that the model specifies development in three different ways: (a) by updating the weight functions discussed above, (b) by increasing the level of the potential state (i.e., the optimal effect point or the system's developmental attractor state), and (c) by adding more developmentally advanced contents to the internal array.

Let me now discuss the second and third ways, which incorporate the truly progressive character of development. This progressive tendency aims at adapting the system's action possibilities to the properties or exigencies of the environment that the system lives in, which is what the notions of accommodation and the ZPD refer to. We have seen that the abstract model specifies all contents in terms of a distance number, which corresponds with any content from a set of possible actions, skills, experiences, and so forth that share the developmental level indicated by that distance number. It thus suffices that we specify a potential environmental input, which is transformed by the system into an experience E_t , which corresponds with an activated internal content at a developmental level c_t . We have seen that the system must be sensitive to experiences that lie at a specific distance ϕ beyond the current maximal content level. If an experience falls within ϕ , the array of contents or levels is updated in the following way: Let c_m be the maximum level in the internal array $\mathbf{I}$ at time t and E_t be an experience at a level E , such that $c_m < E_t < c_m + \phi$. The array is then expanded with a randomly chosen number of new array contents:

$$\Delta \mathbf{I} = (c_{m+k}, w_0; c_{m+1}, w_0; \dots c_{m+\phi E}, w_0) \quad \text{for} \\ f(\phi E) = \delta(E_t - c_m), \quad (9)$$

for w_0 the minimal weight level and $f(\phi E)$ a function of ϕ and E_t , and a constant $\delta \leq 1$.

The system's development-and-change function: The potential level update function. The next question is how the potential state P_t changes as a function of action and experience. Recall that the potential state P_t is nothing but the content for which the effect of experience (information, accommodation, help, etc.) is maximal. Equations 4 and 5 specify that the point of maximal sensitivity depends on the degree of familiarity with contents, or with their novelty, or with anything else that contributes to the attractiveness of the content for which the maximal sensitivity holds. Thus, any change in familiarity or novelty will cause a shift in the position of the maximal sensitivity point, and thus, trivially, in the position of the potential level on the internal array. Instead of modeling changes in familiarity or novelty, I have chosen a shortcut in the form of a direct update of the position of P_t as a consequence of experience (accommodation, help, etc.). As a general rule, P_t changes if the system has an experience that lies within a distance ϕ from the current value of P_t . It is important to note that, because the level of

that experience lies either below or above P_t , P_t may either regress or progress, depending on the nature of the experience. That is, experiences may also amount to adverse conditions and thus eventually lead to regressions in the actual level.

Two different learning principles have been distinguished in the model, that is, principles determining the P_t increase (or the increase in the maximal array level). The first is a linear and stochastic principle that states that, if for some reason an experience falls within an optimal distance from the current potential level P_t , the shift in P_t is just a randomized increase in the position of P_t ,

$$\Delta P_t = \delta \xi_t, \quad (10)$$

for δ a constant and ξ_t a random number from an even distribution larger than 0 and smaller than 1. This condition simulates progress based on some internal *constructive* process, the outcome of which is not determined by the content of the input (constructive development).

The second principle leads to less variable increase and is related to general learning in that it specifies a more gradual change in P_t . It specifies an increase as a consequence of an *instructive* process, the outcome of which is a function of the actual input. This principle assumes that the change ΔP_t is maximal with an experience at an optimal distance from P_t (or the maximal array value c_m) and decreases exponentially with experiences at larger distances from the current P_t ,

$$\Delta P_t = \gamma_1 \cdot \Delta E_t \cdot e^{-(\gamma_2 \Delta E_t)^2}, \quad (11)$$

for γ_1 and γ_2 constants and e , the base of the natural logarithm. Because the content of the experience can lie either below or above level P_t , Equation 11 also allows for regression in the level of the attractor state.

The system's input function. So far, I have discussed various functions that the system needs in order to change its developmental level as a way of adapting to the environment in response to its own actions and experiences. The next function to discuss relates to the way the environment specifies or selects the range of potential or likely experiences that the system will undergo. I shall define a *neutral environment* as one that provides all the possibilities required for the system's development but which does so in a way that does not particularly reckon with the system's actual needs. That is, the system has to retrieve and activate those developmental opportunities by its own actions, explorations, and so forth. This is more or less the traditional—more correctly, the “prototypical”—Piagetian position. A neutral environment can be specified in the form of a uniform distribution of random inputs that cover the whole range of potential inputs to the system, whatever its actual developmental level.

An educationally sensitive environment, however, adapts its learning opportunities to the developmental level of the system. The prototypical Vygotskian stance is that such an environment provides opportunities at the level of the ZPD. In system-specific terms, the ZPD defines the potential or attractor level of the system, which is the input level to which the system is most sensitive and that produces the maximal developmental or learning effect. This is the kind of input that we expect to find with explicit teaching. Given the uncertainty involved in

determining the potential level of a participant—in view of the fact that it is a covert property of the system that must be read out from its activities and the way it profits from experience—it is highly likely that an educationally sensitive environment will present learning experiences that spread around the system's level of maximally fruitful input. The educational quality of the environment can be specified in the form of the input distribution it may provide, given the system's actual and potential levels.

The educational quality of the environment, that is, its ability to present the system with maximally effective learning experiences, is defined by two major parameters. One is the probability that an activity of the system is directly guided by an educational input (ζ_p). In an environment where all activities are driven by teaching, that probability equals 1. The second parameter is the breadth of the range of inputs around the system's level of maximal learning effect (ζ_r). Thus, a "perfect" teaching environment is characterized by $\zeta_p \approx 1$ and $\zeta_r \approx 0$.²

The empirical operationalization of the parameters. An important issue that has not been addressed thus far concerns the question of how to empirically measure the parameters described in the current model. What is the empirical meaning, for instance, of the central parameters, namely the "conservative" parameter η , the "progressive" parameter θ , and the spreading parameter σ ? First, it should be noted that these parameters characterize certain general, abstract properties of developing systems and should not be identified with isolated psychological variables. Having said this, however, the question remains how to find empirical indicators, however indirect they may be, of those parameters. These indicators must be independent of the outcome function. That is, if a specific set of parameter values is supposed to be responsible for a particular developmental pattern, the presence of those parameters must be measurable by empirical variables other than the predicted patterns themselves. With regard to the conservative tendency, for instance, Elbers (1986a) justly remarked that cognitive obstinacy and tenacity is an unexplored topic, in spite of its theoretical importance. He suggested an indicator of such obstinacy, which consists of determining how often children spontaneously return to the same topic. This return behavior indicates how strongly they are affected by their own goals instead of those set by the educational environment. The issue of conservative versus progressive tendencies can also be studied in domains where behavior is explicitly rule or scheme governed. Examples are language and drawing where grammatical and visual schemes, respectively, play an important role. In language, strong conservatism leads to overgeneralization, the frequency of which differs from child to child (Marcus et al., 1992). In drawing, children rely on visual schemes that are sometimes hard to change, depending on the context and the children's age (Light & Simmons, 1983; Madden, 1986). In general, the "trainability" of children with regard to some particular cognitive content can be used as an indicator of how stable or unstable the content in question is (van der Maas, 1993; van der Maas & Molenaar, 1992). Finally, the issue of the relationship between the child's actual and potential levels has been addressed explicitly by Vygotsky, who suggested that the distance should be measured between tasks carried out with and without the help of more competent others (Vygotsky, 1933, as cited in Van der Veer & Valsiner, 1991;

Vygotsky, 1978) and by K. W. Fischer, Pipp, and Bullock (1984), who made a distinction between levels found in testing conditions with and without supportive help (see also Guthke & Stein, 1996).

Empirically, the effect-spreading parameter σ is closely related to the issue of transfer, that is, the ease with which knowledge or training from one context is generalized or transmitted to another. Transfer differs among cognitive domains (Brown, 1990) and is affected by factors like metacognition (Alexander, Carr, & Schwanenflugel, 1995). Transfer has been measured in a variety of tasks, for instance, memory strategies, where considerable individual and age differences have been found (Best, 1993; Block, 1993). It can be concluded that empirical indicators for the model's parameters can be found, although it is unlikely that the parameters can be reduced to single psychological variables.

Summary—The actual model: Parameters and process. The most important parameters are (a) those that affect the changes in the weights of the developmental levels, (b) those that affect the shift in the potential level and the corresponding expansion of the developmental levels array, and (c) those that determine the nature of the environmental input. Because we are using Equation 7b—which specifies an increase in the levels close to the "winning" level and decline in distant levels—as the default option, we need weight increase and decrease parameters based on the current actual level (η_s and η_c) and weight increase and decrease parameters based on the potential level (θ_s and θ_c). In addition, we need parameters that determine how far the effect of the weight increase of the winning actual and potential levels spreads out to their neighbors (σ_a and σ_e). To be able to simulate processes in which a considerable part of the experiences that the developing system undergoes are not directly caused by that system's self-induced activities, we add a parameter α that specifies the probability that an experience is self-induced.

The second group of parameters governs the updating of the potential level and the expansion of the levels array. A crucial parameter is ϕ , the optimal distance of experiential contents from the current potential and maximal levels. If an experience falls within that optimal distance (for instance, a problem that is slightly ahead of the person's current level of understanding), the potential or maximal levels, or both, are updated. Two ways of updating the potential level have been distinguished. One is the linear model, which requires only one update parameter δ and which simulates a basically constructive process. The other is the Gaussian learning curve model, which requires a height parameter γ_1 , a width parameter γ_2 , and a parameter that affects the influence of negative values of the learning rate γ_3 and which simulates a basically instructional process.

The third group of parameters governs the distribution of

² In the actual simulation model, there is an additional parameter for simulating the effect of a decision error in establishing the potential level, that is, the level of maximal experiential sensitivity or learning effect. It turns out that with the degree of error variation applied in the present simulations, this parameter adds nothing to the general range parameter. Furthermore, the simulation model uses an additional parameter that turns the "zoped" condition on or off. The off-condition corresponds to the zoped-probability variable set to zero.

experiences (in terms of the developmental levels they address or affect). The model distinguishes three input conditions or cases. The simplest case is that of the uniform distribution of experiences. The second is the Piagetian case (which has been used for the simulation of the stage models), which has an inverse U-shaped distribution of the optimal distances across the time span covered by a developmental process. An on-off parameter distinguishes between the two cases. The third is the Vygotskian case, where the environment adapts its inputs to the needs of the developing system, that is, it presents the system with inputs that are in the vicinity of the developmental level to which it is currently most sensitive. These so-called ZPD parameters are the aforementioned range parameter ζ_r and the ZPD-probability parameter ζ_p (see Footnote 2).

In addition to these major parameters there are a few others, for instance, a parameter that allows us to manipulate the width of the filter function specified in Equation 2a, ω , and a parameter that allows us to manipulate the position of the maximal learning effect point relative to the potential level P_i (Equation 6). By default, the parameter sets the value of ξ_i to 1.

It is important to note that the model has no parameter that explicitly specifies time or the length of temporal intervals in terms of a numerical parameter value. It works with steps that simulate discrete activities or experiences. Assuming that activities are, on average, evenly distributed across the time span, temporal duration is thus approximated by the number of steps a particular model entails.³ The model has also no absolute numerical standard for developmental distance, or to put it more precisely, the representation of developmental distance is done by a dimensionless number. Thus, any distance range covered by a particular model is numerically specified by a range between 0 and 1. Specific parameter settings, however, may correspond on average with a longer or shorter developmental time span or, for that matter, with broader or smaller content ranges. For instance, we may assume that, on average, the rate and magnitude of the array and potential level changes are inversely related to the length of the modeled developmental process. That is, the magnitude of the change (relative to the size of the entire developmental array) caused by a single action or experience will on average be smaller, in long-term than in short-term processes. We may also assume that the width of the effect spreading is, on average, inversely related to the magnitude of the potential distances between contents that a particular model intends to cover. Thus, a high spreading factor corresponds, in general, with relatively small distances between the levels in the developmental array.⁴

The general question that I now pursue further is Does the dynamic model built along the principles explained in this section indeed produce sequences of outputs (i.e., developmental trajectories) that have the general qualitative and quantitative properties described by the theories that are in some way or another inspired by Piaget with an eventual expansion in the direction of Vygotsky's model?

I begin with a discussion of a general and important question, namely whether development is continuous or discontinuous. I first present examples of empirical evidence about this issue, conclude that both forms of development occur, and then show how the dynamic model explains both continuity and discontinuity. The next section is devoted to "strong" discontinuities.

One form is found in the neo-Piagetian models of long-term development, the other in models of domain-specific two- or three-stage development and learning. I then discuss discrete multimodal developmental patterns found in cognitive strategy models and in studies of microdevelopmental change.

Development: Continuous, Discontinuous, or Both?

Qualitatively, a process is continuous between time t and $t + n$ if its properties and parameters⁵ at time t are similar to those at time $t + n$. A logistic growth process, for instance, is continuous because its parameters and properties are similar across all points of the growth curve, notwithstanding the fact that some such processes (for instance, one with a cubic exponent) produce very steep and sudden accelerations (van Geert, 1994a, 1995a). Qualitatively, a process is discontinuous between t and $t + n$ if its properties and parameters are different. For instance, if at time $t + m$ a growth process is influenced by a newly emerging supportive factor, thus causing a change in the growth direction, rate, or asymptote, then that process is qualitatively discontinuous even if its growth is perfectly smooth and gradual.

Quantitatively, a process is discontinuous if there exists a region—that is not trivially small—of possible levels that are never (or at most, only accidentally) occupied. If the process has a rigorous mathematical specification, it can eventually be shown that the inaccessible region consists of points that are mathematically unstable (see, e.g., catastrophe models, Hopkins & van der Maas, 1998).

Multidimensionality and (Dis)continuity in the Object Concept: A Case Study

Multidimensionality. In the section *Developmental space and the developmental distance dimension*, I explained the concept of a developmental hyperspace, in which any developmental process can be specified in terms of a single dimension, namely the distance it travels in that space. The choice of such a one-dimensional representation of development, however, could easily, but mistakenly, be seen as a plea for a one-dimensional and eventually also continuous conceptualization of development. I have already explained that this single dimension is in fact only a distance specified across an abstract descriptive dimension of arbitrary complexity. In order to provide this abstract distance concept with empirical flesh and bones, I refer to the research on the development of the object concept, which is one of the most frequently studied aspects of cognitive development.

Any particular object concept task administered to a particu-

³ It is also probable, however, that the level of aggregation at which activities must be described in order to explain change at the level of a long-term developmental process is higher than that of a short-term process, which implies that the relationship between duration and number of steps is not just trivially linear.

⁴ The actual computational model is written in the form of a Visual Basic module that runs under the widely used spreadsheet program Excel and is available from Paul van Geert. The sets of parameter values on which the model simulations are based can be found in the Appendix.

⁵ It is explicitly understood that the parameters must remain equal, not necessarily the parameter values.

lar infant is in fact a point in a multidimensional space that consists of dimensions that distinguish task properties on the one hand and dimensions that distinguish subject properties on the other hand. For instance, as far as the difficulty of the task is concerned, it makes a difference whether the occluding object is opaque or transparent (Bremner & Knowles, 1984; Yates & Bremner, 1988), whether the object is of a 'classic' or a sortal object kind (Xu & Carey, 1996), whether the object's motion is continuous and smooth or not (Spelke, Kestenbaum, Simons, & Wein, 1995), whether the object is a physical object or a person, and if it is a person, whether it is a familiar person or a stranger (Bigelow, MacDonald, & MacDonald, 1995; Legerstee, 1994) or whether objects or persons are mobile or not (Cossette-Ricard, 1986), or, finally, whether the task involves a real object or a shadow (Cameron & Gallup, 1988). If we accept that, on average, solving more difficult tasks requires a higher developmental level of problem-solving skills, the object-related aspect of the tasks alone is specified by (at least) six physically more or less independent dimensions (corresponding with the properties manipulated in the experiments). In this six-dimensional hyperspace, the developing child travels a specified distance and does so in a particular quantitative way.

More dimensions can be added (recall that they need not be independent from a psychological point of view, i.e., it is likely that some of the dimensions act as prerequisites to others; this fact does not affect the principle of a distance specified in a multidimensional space). With the so-called A-not-B problem, in which 8- to 12-month-old infants persist in searching for an object in its previously familiar hiding place, success in solving the task depends on properties such as the number of hiding places or containers involved (Sophian, 1985; Wellman, Cross, & Bartsch, 1987), the spatial arrangement of the objects (Sophian, 1986), the physical differences between the containers (Wellman et al., 1987), and the delay between the visible replacement of the preferred object and the subsequent searching activity of the infant (Diamond, 1985; Wellman, et al., 1987).

In addition to these dimensions specifying object properties, there are a number of components in the task that require specific understanding or belief systems (or any capacity that makes a person act as if there was a particular understanding or belief). Examples of such components are the infant's understanding (or not) of the deletion component, that is, that the hidden object is actually taken away (Sophian & Yengo, 1985a, 1985b), the understanding of visible versus invisible movement (Sophian, 1985), and the understanding of object relations (Wishart & Bower, 1984). In a series of experiments with older children and adults, Subbotskii (1990, 1991a, 1991b; Subbotskii & Trommsdorff, 1992) has shown that the later expression of the belief in object permanence involves not only complex cultural belief systems but also social aspects such as persuasion and the demonstration of (tricked) exceptions.

In addition to these rather specific properties of the task, objects, and conceptual components involved, there are more general perceptual aspects that explain why certain tasks are solved earlier on the developmental time scale than others. Visual display properties play a role (Johnson & Aslin, 1995; Johnson & Nanez, 1995; Kellman & Shipley, 1991) and so does the presence of sounds (Bigelow, 1986; Legerstee, 1994). Visually impaired children, that is, children who are missing

the visual information channel, develop the object concept more or less along the same lines as nonvisually handicapped children but do so at considerably older ages (Bigelow, 1986, 1990; Rogers & Puchalski, 1988). The difficulty of a particular task and, consequently, the age at which such a task can be solved also depends on a variety of internal or subjective factors, which basically amount to central resource dimensions of the information-processing system. For instance, the capacity to direct one's attention and to overcome distracting aspects in the object concept tasks relates to the child's attention economy (Hood & Willatts, 1986; Horobin & Acredolo, 1986; Sophian & Yengo, 1985a, 1985b; Triana & Pasnak, 1986). Another important parameter is the child's memory capacity (Bjork & Cummings, 1984; Diamond, 1985; Schacter, 1986). Other subjective factors involve perceptual preference (Breuer, 1985), strategies for selecting information (Sophian & Sage, 1985) and coordinating different aspects (Haake & Somerville, 1985), persistence and perseverance (Sophian, 1986), and motivation (Lingle & Lingle, 1981). In general, the less the object concept task requires from the infant's mental resources, the earlier the infant will express what at least looks like object understanding. This principle is illustrated by the experiments done by Baillargeon and collaborators (Baillargeon, 1986, 1987a, 1987b, 1991; Baillargeon & DeVos, 1991; Baillargeon & Graber, 1987; Baillargeon, Graber, DeVos, & Black, 1990; Baillargeon, Spelke, & Wasserman, 1985), who have shown that if a task requires only selective attention or perceptual preference, certain components of object understanding occur as young as 2 months of age. A closer look at the effect sizes of the habituation experiments, and of most, if not all, of the object concept experiments, however, shows that the capacities reported are far from the stable, consolidated properties of the infants involved. There exist considerable intra-subject and intersubject differences that speak of the effect of a large number of unknown or uncontrolled variables.

The conclusion is that each of these task and subject variables defines a dimension in a descriptive space specifying possible variation in the expression of the child's understanding and representation of object properties.⁶ The number of dimensions (aspects, variables, etc.) reviewed here is about 20. A concrete task with a set of particular task properties, solved by a child with particular subject properties (in terms of attention and effort invested, specific knowledge, etc.) is thus represented by a single point in a 20-dimensional space of properties relating to the development of the object concept. With repeated test trials, that point may move within a certain region, depending on objective and subjective conditions.

(Dis)continuity. Thelen and Smith (1994) stated that, across all the task contexts, there is "small and progressive change" (p. 287), suggesting that, given the multidimensional nature of the object-concept tasks, development is continuous. By means of data taken from the aforementioned meta-analytic study on the development of the object concept, more particularly the A-not-B error (Wellman et al., 1987), it can be shown, however, that this conclusion does not necessarily follow from the multidimensionality and

⁶ It is not my intention to claim, though, that these variables represent anything like "features" of the child's object concept, which would amount to an oversimplified interpretation of the results of the empirical research and of the nature of the object concept; what matters is that these variables affect the expression of object understanding in the child.

context-dependency premise. The data were adapted from a considerable number of studies and consist of average percentages correct scores for three age groups (8, 9, and 10 months), which performed the A-not-B task in four delay conditions (0, 1, 3, and 5 s) and with either two or three locations. Although the details go well beyond the scope of the present article (see van Geert, 1998b), a growth analysis was performed that proceeded in the following way. First, the observed and predicted data from the Wellman et al. study were combined and extrapolated where necessary into series of "corrected" scores for each of the conditions. Next, linear, logistic, restricted, and exponential growth models were fitted on the data. For each of the growth curves, age-of-onset and age-of-100%-correct scores could easily be calculated. It was found that the best fitting curves yielded ages of (approximately) 100% correct that were far beyond the age at which such scores may be empirically expected. Only the exponential curve produced a realistic 100%-correct age. The exponential curve, however, yielded a particularly bad fit that ran completely against the course of the empirical curves. It was concluded that, in order to reach the 100%-correct level (which is a trivial accomplishment with older infants), a qualitative discontinuity must be introduced. That is, a developmental factor or variable not yet present in the object concept performance of 10-month-old infants must be introduced somewhere along the track in order to account for the empirically established age at which children can solve the A-not-B problem without difficulties. Whether this qualitative discontinuity also involves a quantitative discontinuity unfortunately cannot be answered on the basis of the available data.

Further Evidence for Continuity and Discontinuity in Infancy

In their analysis of the transition to reaching behavior, Thelen et al. (1993) showed that this transition contains discontinuous phase shift aspects (the transition to reaching itself), as well as continuous aspects, for instance, continuous changes in visual location and body control. Wimmers (1996; Wimmers, Beek, Savelsbergh, & Hopkins, 1996; Wimmers, Savelsbergh, van der Kamp, & Hartelman, 1998) concentrated on the actual shift to reaching and found substantial evidence that the transition is indeed a matter of discontinuous change. In their classic study on transitions in early mental development, McCall, Eichorn, and Hogarty (1977) found evidence for five major stages. The stages are characterized by what Uzgiris (1977) called a dominant characteristic instead of a structural transformation, which thus suggests that the stages are discontinuous in the qualitative sense discussed earlier. McCall et al. found that an important indicator of a stage transition was a change in the correlational patterns. Lewis and Ash (1992) found evidence for a stage shift in sensorimotor performance around the age of 4 months. Support for discontinuities, eventually in the form of rapid changes, as related to early brain development can be found in the study by Bell and Fox (1994).

Language development is another domain in which we find evidence both for discontinuities and continuity. The so-called "vocabulary spurt," for instance, is a well-known example of a discontinuity (Dromi, 1987; Goldfield & Reznick, 1990; Reznick & Goldfield, 1992). However, the spurt can be modeled in the form of an almost ideal S-shaped curve, which is a continuous curve (Cooney & Constanas, 1993; van Geert, 1991). A growth analysis

of such continuous curves, however, shows evidence for underlying qualitative discontinuities, such as the introduction of new structural possibilities, for instance, syntax (van Geert, 1991, 1993, 1994a, 1995b). In an analysis of the emergence of so-called "function words" (which is a closed word class containing determiners, prepositions, and so forth), we found considerable individual differences. Some children showed a sudden jump from almost no function words to a frequency that was close to that of adult language, whereas others progressed in a much more gradual fashion, eventually punctuated by strong oscillations (Ruhland, 1998; Ruhland & van Geert, 1998).

In summary, the evidence suggests that continuities and discontinuities exist alongside one another, although it is not clear from the data whether the eventual discontinuities are real and not just significant accelerations in underlying continuous processes.

Evidence for Continuity and Discontinuity in Childhood

The child's understanding of conservation of number or magnitude is another example of a frequently studied phenomenon comparable to object permanence. The research literature clearly shows that, similar to object permanence, conservation is not a single discovery, but a multidimensional process. That is, many variables affect the child's solving of conservation problems. For instance, there is ample evidence on the positive effect of supportive understanding or skills, such as the child's understanding of number (Hodges & French, 1988; McEvoy & O'Moore, 1991; Sophian, 1995), the nature of matter (Au, Siddle, & Rollins, 1993; Macnamara & Austin, 1993; Rosen & Rozin, 1993; Stavy & Stachel, 1985), and several other skills or types of knowledge that are necessary to answer the conservation problem (Bijstra, van Geert, & Jackson, 1989; Lautrey, Mullet, & Pacques, 1989; Lister, Leach, & Walch, 1989; Miller, 1989; Wheldall & Benner, 1993). Finally, there exists evidence that solving the conservation problem requires a correct understanding of the social, communicative, and pragmatic aspects of the conservation task itself (Eames, Shorrocks, & Thomlinson, 1990; Elbers, 1986a; Elbers, Wiegiersma, Brand, & Vroon, 1991; Galpert & Dockrell, 1995; Hanrahan, Yelin, & Rapagna, 1987; Light, Gorsuch, & Newman, 1987; Moore & Frye, 1986; Pratt, 1988). Older children and adults who have acquired a stable conservation concept are still sensitive to misleading clues, which lead them to giving nonconservation answers (Chandler & Lalonde, 1994; Winer, Craig, & Weinbaum, 1992; Winer, Hemphill, & Craig, 1988; Winer & McGlone, 1993).

It is likely that within this developmental space, there exist subspaces where the development of conservation indeed shows the form of a phase shift, that is, a discontinuity. Convincing evidence for such a discontinuity in the understanding of conservation in a (semi)classical task format has been found by van der Maas (1993; see also van der Maas & Molenaar, 1992, 1996). The van der Maas time-serial data are corroborated by the bimodally distributed scores from Bentler's classic cross-sectional study (Bentler, 1970).

Bimodal or, in general, mixed distributions provide good evidence of discontinuous change on the level of cross-sectional data. In an analysis of score distributions on Piaget's water-level task, Thomas and Turner (1991) and Thomas and Lohaus (1993) found a convincing bimodal distribution of accurate

versus inaccurate performers (I discuss this research later in the section *The water-level task*). Similar results have been obtained with class inclusion and reversal shifts (Tabor & Kendler, 1981). Recently, Hosenfeld, van der Maas, and van den Boom (1997a, 1997b) found a number of particularly clear indicators of discontinuous change in the development of analogical reasoning in 6- to 8-year-old children.

Discontinuities are not necessarily represented in the form of bimodal or mixed distributions. They can also take the form of growth spurts in task performance at certain ages, which have been found, among others, through reflective judgment tasks (Kitchener, Lynch, Fischer, & Wood, 1993) and the understanding of arithmetic concepts (K. W. Fischer, Hand, & Russell, 1984). Growth spurts can be modeled as continuous processes in which the spurt is either a property of the growth dynamic itself, of some external supporting factor, or both (K. W. Fischer & Rose, 1994; van Geert, 1991, 1994a). The combination of continuity and discontinuity and of growth and the creation of novel forms (skills) in cyclical patterns of change is characteristic of the neo-Piagetian models (Case & Okamoto, 1996; K. W. Fischer & Bidell, 1997; Thatcher, 1994).

Patterns of continuity and discontinuity have also been found in brain development, for instance, in the growth of electroencephalogram (EEG) power (Hudspeth & Pribram, 1992; Matousek & Petersen, 1973; see K. W. Fischer & Rose, 1994; and van der Molen & Molenaar, 1994, for discussion). Thatcher (1991, 1992, 1994) found three major cycles of neocortical reorganization between the ages of 1.5 and 14 years and growth spurts in mean EEG coherence between birth and 16 years. There exists an interesting relationship between these spurts and cycles and discontinuities of spurts in cognitive and behavioral development, which, however, does not imply that cognitive stages or levels are unidirectionally caused by changes in brain growth (K. W. Fischer & Rose, 1994; see also Dawson & Fischer, 1994, for a series of studies focusing on brain-behavior relationships during development).

Continuity and Discontinuity in the Formal Model

The dynamic model produces a maximally fast continuous linear increase in its output level if the following condition holds:

$$E_{t+1} = E_t + \phi, \quad (12)$$

that is, if the level incorporated in the next input E_{t+1} equals the level of the preceding output E_t plus the optimal distance ϕ , that is, the distance between the actual and the potential level that causes the highest progress. It is extremely unlikely, however, that an environment could monitor a developmental system with the accuracy required by this linear change condition. If the guidance and support conditions are more realistic, in the sense of less precise and all-encompassing, the model produces a pattern of continuous change that takes the form of a relatively broad band more or less equally distributed across an S-shaped regression line (Figure 3). The curve can be fitted by a logistic growth model (on the basis of 20 cases, the average percentage explained variance expressed by r^2 is .93 and for the model vs. the moving average of the data curve—with a window of $t = 20$ —the average $r^2 = .98$).

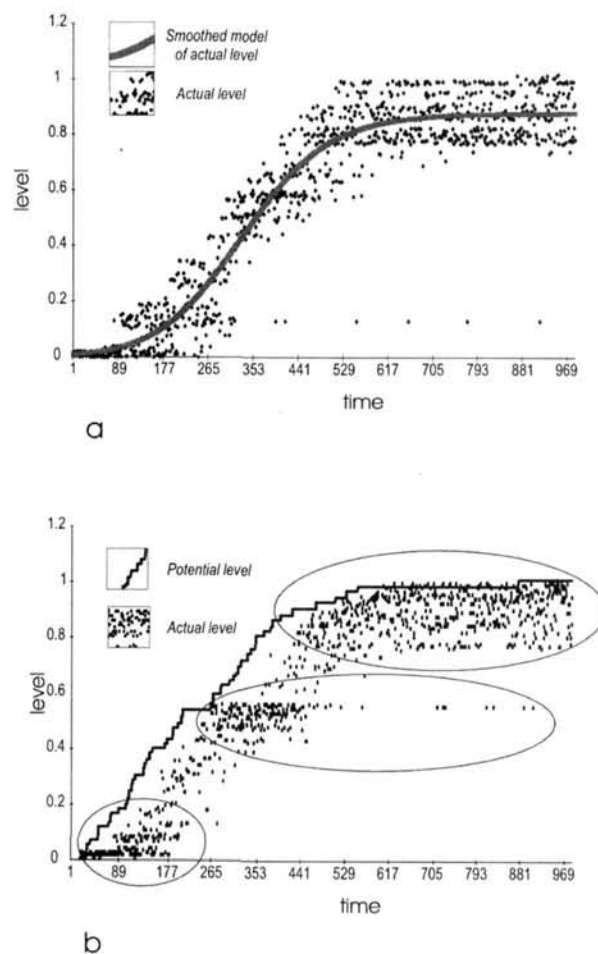


Figure 3. With a normal input function, experience-driven learning leads to a classic S-shaped growth pattern; the actual developmental levels expressed in the system's actions occur in a band around the idealized S-shaped growth curve (a). The consecutive outputs often cluster in groups (b).

This set of parameters represents a primarily environment-driven process of learning within a relatively small content and time domain. It can be obtained with a form of support that is (a) relatively frequent but not too frequent (in the case of the model that produced Figure 3, the frequency of help is 50% of the system's activities or experiences),⁷ (b) of relatively high quality (the educational environment makes no serious errors in its estimation of the kind of support that the developing

⁷ The description of the conditions may seem unnecessarily vague. The reason for doing so is that the model is driven by stochastic factors, which explains why particular results of the simulations—like the current S-shape—are randomly distributed across a range of parameter values. The minimal and maximal points of these ranges are model-specific values (see the enclosed simulation model). The vague linguistic categories by which those ranges are indicated could eventually be replaced by membership functions in a fuzzy logic rule base (Ross, 1995).

system needs). As a third condition, it is assumed that (c) the system is relatively progressive, which means that the effect of the attractor or potential state update parameters θ_s and θ_c is relatively large in comparison to the consolidating parameters that update the weight of the actual level (η_s and η_c). That is, developmental progress is mainly caused by the help and information given. The fourth condition is that (d) the developmental process takes place on a relatively small-to-medium scale in terms of the developmental distance bridged (which means that the effect spreading specified by the parameters σ_a and σ_e is relatively large). In summary, these conditions are more indicative of an instruction-driven, relatively short-term process, rather than a long-term process of autoregulated, constructive developmental processes.

Finally, it should be noted that the distribution of actual-level points is not entirely uniform across the continuous S-shaped curve fitted onto the data. The actual levels often occur in the form of clusters, which prefigure the explicit stagelike progressions that I discuss in the next section.

Long-Term Structural Change and Discontinuities: Piagetian and Neo-Piagetian Stage Models

Stagnation and Imitation

Piaget (1970b) claimed that "if assimilation alone were involved in development, there would be no variations in the child's structures. Therefore he would not acquire new content and would not develop further" (p. 707). And "conversely, when accommodation prevails over assimilation . . . representation . . . evolves in the direction of imitation" (p. 709). The implicit assumption made by Piaget is that if assimilation and accommodation are in balance, the child's structures will show a combination of invariance and variation, that is to say, a trajectory consisting of stages (or phases or levels) punctuated by sudden jumps. I consider Piaget's model as the archformat of a broader class of models, notably the neo-Piagetian models, that view development as a succession of levels, tiers, or phases that are qualitatively distinct and discontinuous, without claiming that such levels necessarily apply to all aspects of development at the same time. Recall that the dynamic model applies to only those developmental contents that are connected by supportive or competitive relationships and that it makes no empirical claim as to which (or how many) contents they are. Thus when I describe the Piagetian model I refer to only its major qualitative properties and not to its alleged claim that stages should be all-encompassing structures (or not; see Chapman, 1988).

Model Parameters

Given that the first simulation described in the previous section was based on characteristics of a relatively short-term instruction-driven process, the parameters for the Piagetian model should refer to long-term interactions that are more explicitly driven by the system's internal properties and states.

First, I assume that there exists a balance between assimilatory (conservative) and accommodatory (progressive) factors (i.e., rate parameters governing the growth of the weight factors

of the actual level, η_s and η_c , and the attractor or potential level, θ_s and θ_c).

Second, the long-term character of the process implies that the average progress caused by a single activity or experience is small in comparison to the entire length of the developmental distance that must be traveled. There exist several ways in which this long-term character can be represented in model-specific terms. To begin with, the accommodation rate, that is, the amount of new knowledge that can in principle be extracted from a particular action or experience, is set to a low value. The eventual gain or progress in the attractor state caused by a single action or experience is small in comparison to the overall developmental trajectory, thus suggesting a long-term process (which is expressed by a low value of δ). In addition, the spread of the positive effect of an action or experience (σ_a and σ_e) on related contents (skills, knowledge, schemes, etc.) is small, thus suggesting a large distance between the opposite sides of the developmental dimension (which, in the Piagetian case, varies between simple sensorimotor action and formal operational thinking or, in the case of a neo-Piagetian model like K. W. Fischer's, between single reflexes and general principles for integrating systems of abstractions).

Third, the developmental process is considered child-determined, which implies that there is no particular environmental or educational adaptation to the child's specific developmental needs (the ZPD or ζ parameters are set to low adaptive values). That is, the developing system selects its own resources and conditions for progress from a nonspecific supply of possibilities. Note that this assumption does not fit in with the neo-Piagetian postulate that education and teaching are important developmental factors (Bidell & Fischer, 1994; Case, 1992b). I show later that stage shifts also occur if educational guidance plays a significant role in the process.

Fourth, the model governing the eventual expansion of the knowledge system must allow for a large variation in the magnitude of learning or experience effects. In addition, these effects are primarily child-driven, that is, not necessarily proportional to properties of the input. These conditions can be realized by using the stochastic linear equation (see Equation 10), which simulates in a highly simplified way the fact that the outcome of a constructive process, although definitely based on experiences, does not necessarily proportionately reflect the content of those experiences.

Fifth, the update probability is an inverted U-shaped function of the attractor state level, implying that it is relatively low with low attractor values, about maximal in the middle, and low again toward the maximal values of the attractor state. The reasons behind the choice for a U-shaped form are as follow. In the Piagetian and neo-Piagetian models, low levels correspond with reflexes and reflexlike sensorimotor adaptations, which are strongly affected by biological maturation and thus less sensitive to experience than later, information-based rule-like adaptations (K. W. Fischer, 1980b). The highest levels (the stage of formal operational thinking, or Fischer's tier of principles) are less sensitive to experiences because the experiences that can alter the attractor states are strongly dependent on explicit, formal instruction and do not, or scarcely, occur in the form of daily events and experiences (note that Piaget accepted that the occurrence of formal operational

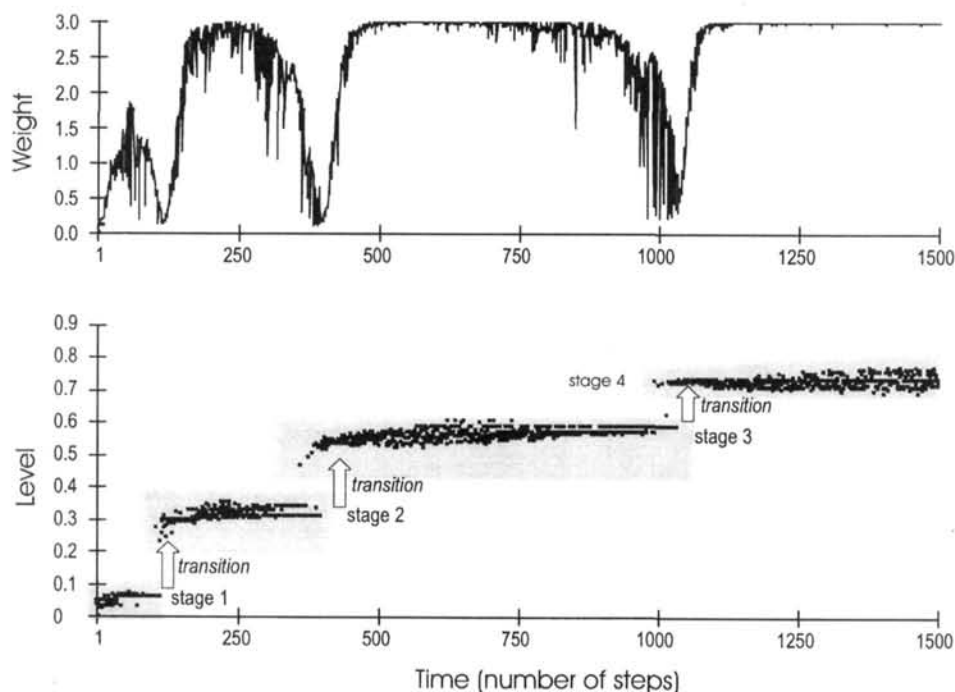


Figure 4. A characteristic four-stage developmental pattern resulting from the Piagetian parameter set; during the transitions, the weights decline to almost zero.

thinking depended on culture and schooling; Dasen, 1974; Dasen & Heron, 1981; Piaget, 1966/1974).⁸

Diachronous Multimodality: Sudden Jumps and Discontinuities

In the starting phase, the model's output levels remain at approximately the same initial level or they fluctuate within a relatively narrow band. They then suddenly jump to a significantly higher level, which remains stable until they jump again toward a still higher level. This process repeats itself a few times, dependent on the exact parameter values, although the four-stage pattern occurs most frequently. Figure 4 shows a characteristic four-stage pattern, based on balanced assimilatory and accommodatory parameters (note that, because of the way the model operates, the best balance is achieved when the accommodation parameter is about twice as large as the assimilation parameter). In light of these results, we must test whether Piaget's claim that if assimilation is lacking, that is, if only accommodation occurs, the input will be imitated, is also supported by the present model. We find that if the assimilation-parameter values are set to 0 (with all other values similar to those from Figure 4), the stagewise pattern disappears. Whether this boils down to a literal imitation of the input is unclear. The curve does however follow the form of a so-called restricted growth function, which is based on an input format where the probability that a new input occurs diminishes asymptotically (De Sapio, 1978; van Geert, 1991). If the assimilation parameters remain unaltered and the accommodation-parameter values are set to 0, the resulting trajectory consists of a continuous stable state, that is, no change occurs at all.

Catastrophe theory provides a set of criteria that enables us to distinguish discontinuities (jumps, for instance, those produced by the current Piagetian parameter values) from accelerations (Gillmore, 1981; Hosenfeld et al., 1997b; van der Maas & Molenaar, 1992). Three criteria refer to the jump-wise pattern of change (additional criteria are discussed later). One criterion is *sudden jump*, which is intuitively clear from the time plots (Figure 4). A second criterion is *bimodality*. If we plot the successive output levels not by time but by magnitude we find a clear bimodal distribution of levels around the suspected jump. A particularly strong criterion is the existence of a so-called *inaccessible region*, that is, values between the lower and the higher level do not occur (in a stochastic catastrophe model, their occurrence should be rare; see Figure 4).

The weight functions associated with the array of developmental contents correspond with the inverse of the so-called potential function describing the equilibrium levels in the cusp catastrophe (Jackson, 1991; van der Maas & Molenaar, 1992). In the vicinity of a jump, the potential function has two local minima that correspond with the two possible equilibrium levels at that point. At the jump, the developmental weight function

⁸ A simple way for stochastically determining the update probability p_u of the attractor state P_t in accordance with this inverted-U-shape principle is the equation

$$p_u = p(|\xi_t - P_t| > \phi), \quad (13)$$

which specifies that the update probability is the probability that the absolute difference between a random number ξ_t and the current potential level P_t is larger than the optimal distance ϕ .

has two local maxima, which are functionally similar with the minima of the potential function (see Figure 5). Because the output is in fact a stochastic function of the weights, we should expect a short period of bimodal behavior in the vicinity of a jump, that is, a random succession of behaviors at both equilibrium levels. I analyze this phenomenon later because it applies not only to Piagetian phenomena such as horizontal décalage, but also to transitional phenomena in general.

If we observe the individual trajectories produced by the current set of parameters, the number of levels is typically four (in slightly more than 80% of the simulated cases; about 20% have three levels, given the chosen parameter values). The present view on stages, however, is that if they occur, they are not like the all-encompassing changes attributed to Piaget. Rather, there exist significant interindividual and intraindividual differences (K. W. Fischer & Bidell, 1997; Flavell, 1982). The intraindividual differences refer, among others, to the fact that the stages are domain specific (Levin, 1986). In spite of the variability,

however, the (neo-Piagetian) stages are still found to occur (Bidell & Fischer, 1992; Case, 1992a; Turiel & Davidson, 1986). Does the model conserve the four-stage structure if we simulate a database consisting of different individuals or, for that matter, different independent developmental domains?

A simulation of 80 separate cases, based on the balanced parameter values, shows that summed data representing different individuals or developmental domains indeed conserve the stage-like structure of the individual patterns. The frequency plot (based on all the individual scores in each of the 80 cases) is shown in Figure 6. Stages 1 and 2 from the individual cases are clearly recognizable in the form of separate bell-shaped distributions, whereas Stages 3 and 4 overlap but still retain a peak at the beginning and one at the end of the overlapping frequency region. Overlapping stage levels belonging to separate developmental domains could eventually also explain the phenomenon of horizontal décalage because they allow for the simultaneous occurrence of performance characteristics at distinct levels.

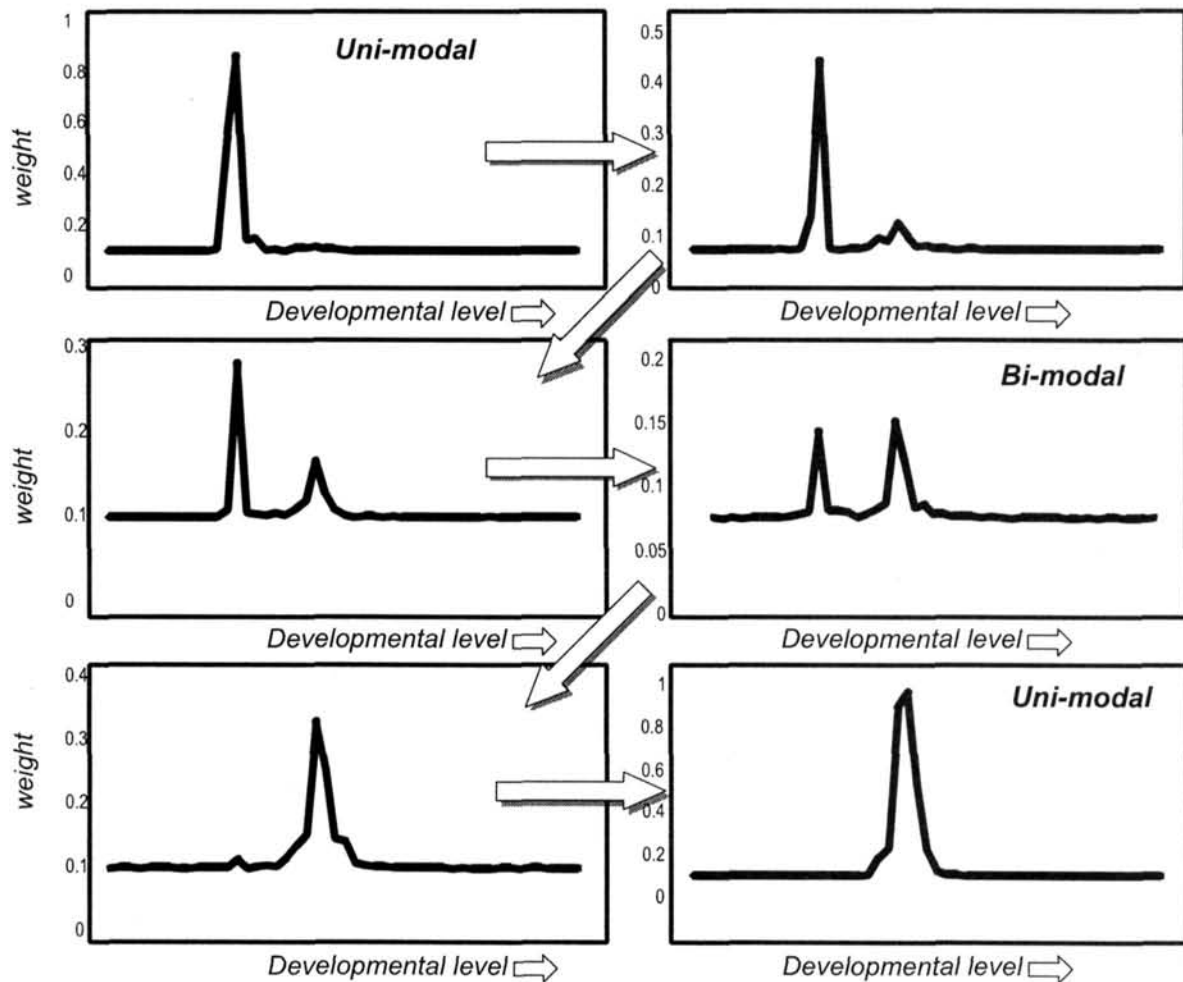


Figure 5. Before the transition, there exists a unimodal weight peak across the array of contents. The transition is announced by the emergence of a second peak, which does not correspond with a bimodal action pattern. During the transition, both weight peaks are approximately similar and produce a bimodal action pattern (both levels occur). After the transition, the old peak disappears, restoring the original unimodal pattern with a new, higher level.

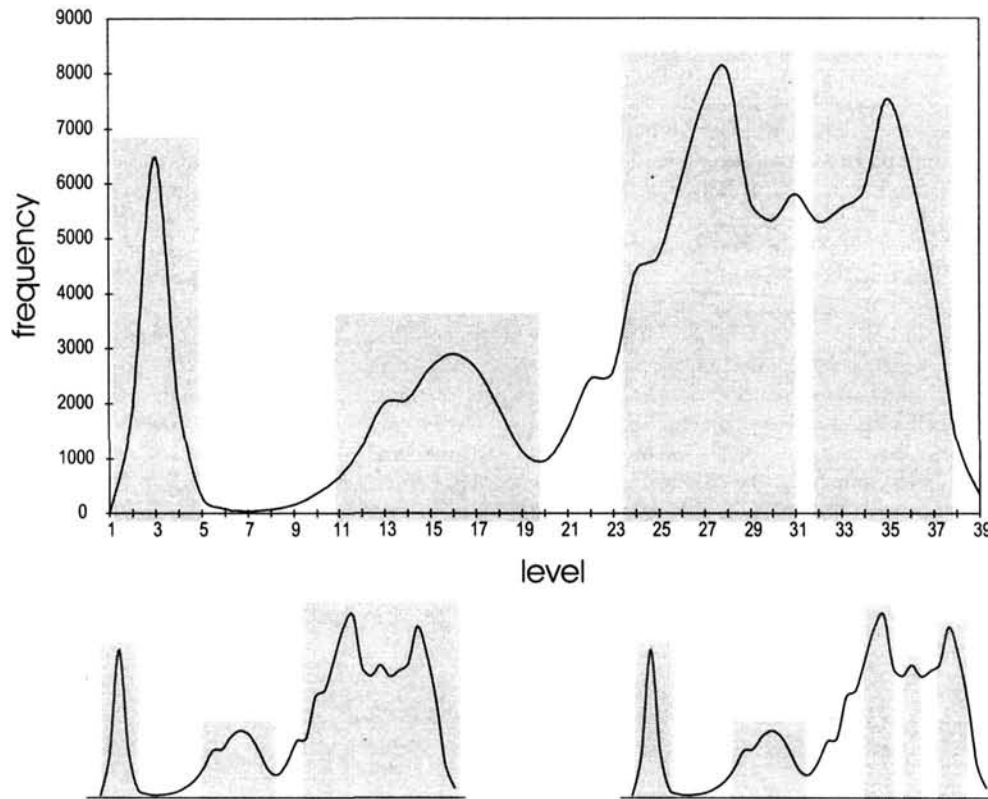


Figure 6. A frequency count of the actual level outputs occurring in 80 simulations based on the Piagetian parameters; the histogram shows four marked peaks, corresponding with the four stages in the individual simulations. It is also possible, however, to view the histogram as the representation of either a three-stage process or a five-stage process, depending on the criterion chosen.

Are (Neo-)Piagetian Stage Models Fractal?

Why are there only a small number of stages? Within the (critical) Piagetian tradition, theorists agree on a number of three to five major stages. Piaget himself easily switched between a number of three to six stages (see section *Basic Principles Adapted From Piagetian Theory*). The number 4, plus or minus 1, seems a canonical number as far as the distinction of major stages is concerned. Freud, for instance, described five stages, Kohlberg distinguished six levels of moral development, and Bruner found three levels in the development of symbolic thinking. The neo-Piagetian models of Case (1990, 1992a, 1992b) and Fischer (K. W. Fischer & Rose, 1994) distinguish four or five major stages or tiers, respectively, each with three substages (or four, depending on how they are ordered). Other examples are Karmiloff-Smith's (1986) three phases in linguistic and cognitive development, the five stages of development in infancy described in McCall et al. (1977), Selman's five levels of perspective taking (Selman, 1980), Fischer's four organizational stages in animal learning (K. W. Fischer, 1980a), Gal'perin's (1969) four levels in the formation of mental acts, and Klausmeier's (1976) four stages in concept formation. Of course, there are also notorious exceptions, such as Erikson's (1950) eight stages.

If human development is just a continuous stream of events, the separation into stages could just be a matter of bookkeeping that is probably determined by cultural conventions (Brainerd, 1978). In antiquity, for instance, life was divided into 4 stages, in accordance with Pythagorean principles. Medieval thought held that the number of stages in life ranged from 3 to 12, with a certain preference for 7. The number of stages was seen as analogous to that of other important phenomena, such as the number of seasons or months (Boom, 1994). The question now is whether the number of stages is indeed just a pragmatic, culture-specific way of cutting an otherwise more or less continuous stream of change into manageable chunks, or whether the number of 4 (plus or minus 2) is "real," that is, based on the intrinsic dynamics of processes characteristic of development. The latter is likely, because the number of stages resulting from simulations based on the Piagetian parameter set is about 3 to 5 (recall that it is called the Piagetian set because it produces the characteristic Piagetian 4-stage pattern. It goes without saying that this coincidence between the outcomes of the dynamic model and the classic stage theories says nothing about the empirical truth of the number of stages).

A deeper reason for this small number of stages could lie in the fact that processes of the kind described by a dynamic that operates on a tension between conservative and progressive

forces produces shifts or changes of a *fractal* nature. Fractal processes are characterized by so-called scale invariance, that is, properties at smaller timescales resemble those from higher timescales. An interesting property of fractal change processes is that the frequency of shifts or changes is a power function of their magnitude (Bak & Chen, 1991; Schroeder, 1991; van Geert, 1994a, 1994c). In the dynamic model, the magnitude of change is simply defined as the absolute difference between two consecutive output levels (i.e., two consecutive expressions of the system's actual level). A frequency count of the magnitudes of change in the (neo-)Piagetian model shows that magnitude and frequency are indeed connected by a simple power function of the form $N_M = c.1/M^p$ (i.e., the number N of changes of magnitude M is an inverse function of that magnitude to an exponent p ; Bak & Chen, 1991; Schroeder, 1991; van Geert, 1994a, 1994c; r^2 between data and model is .96). Given this power function, the number of shifts of considerable magnitude is small (say, in the order of three to five). This finding does not support claims about the exact number of major stages: It just tells us that, given the dynamics of this kind of process, we should expect two to six major shifts and many small, (quasi-)continuous changes.

The logarithmic distribution of stage shift intervals. The power distribution of frequencies of different magnitudes is related to another aspect of development that is also indicative of a power-law-governed pattern of change, namely the increasing length of the stages (phases, levels, tiers, etc.). The logarithmic plot of the average length of the stages or levels described in a variety of stage or phase models approximates a straight line, which shows that the increase of the stage or phase duration is governed by a power function. Examples are Selman's (1980) model of perspective taking, Gesell's (1954) phases in the development of grasping, Erikson's (1950) stages of identity formation, Freud's psychosexual stages, Piaget's stages of cognitive development, Fischer's K. W. Fischer & Bidell, (1997) tiers and levels and Case's (1992a) general developmental stages and stages in the development of memory (Case, 1985). This pattern of change is characteristic of growth phenomena in general (McMahon & Bonner, 1983) and relates to a similar, potentially fractal pattern of increase.

It can be shown that the dynamic model of the neo-Piagetian stages produces a similar log-linear plot of stage intervals (see van Geert, 1998a, for details). The correspondence between the relative ages of stage shifts of the Piagetian model (and of the neo-Piagetian models, for that matter) with the simulated data (for the present set of model parameters) suggests that the exponential increase of stage duration is an emergent phenomenon of the dynamics. It is likely that the real transitions are codetermined by control variables such as the growth of working memory or brain growth, which are not incorporated in the dynamic model.

Transitory and Substage Phenomena

Pretransitory regressions. In the overview of Piaget's theory, it was shown that a stage is characterized by a distinction between a substage of preparation followed by a substage of completion or consolidation (Piaget, 1947, 1955, 1970b). These substages are closely linked to the process of equilibration: The

new stage begins with a low equilibration level and gradually moves toward complete equilibrium (Piaget, 1975). The most characteristic property of equilibrium is stability, that is, the capacity to compensate for perturbations (Piaget, 1964, 1970b; Flavell, 1963). In the dynamic model, there is no separate variable representing the level of equilibrium. However, the weight function as it is defined in the model comes close to it. Recall that the weight function specifies the level of consolidation of an associated cognitive or developmental content. That is, contents with a maximal weight are maximally consolidated, which probably implies that they correspond with fast and secure problem-solving processes.

We have seen that, according to Piaget (1970b), repeated exercise of a particular type of activity leads, by way of assimilation, to "an increase of material or of efficiency in operation" (p. 707). Put more generally, this increase could imply several things: (a) automatization of that activity (or parts thereof) and thus higher processing and performance speed, (b) an increase in mastery and thus a decrease in errors, and (c) an increase in generalization and thus a broadening of the range of problems and occasions to which that particular type of activity is applicable. Because the weight associated with a particular content (skill, rule, activity pattern, etc.) represents the content's consolidation level and determines its activation probability, we may also take the weight as an indication of how fast, secure, or stable the corresponding content (skill, rule, etc.) can be performed. Thus, if a particular weight increases, we may assume that the activities associated with that particular weight vector will show an increase in speed, generality, and mastery, or whatever else a particular theory considers to be a sign of consolidation of a specific developmental level. Conversely, if the weight decreases, those activities may suffer from slowing down and show an increase of errors or effort needed to accomplish a task, increased reaction time, and loss of contexts and ranges of applications. If we plot the weights associated with the levels of the output activities and compare them with the output activity levels themselves, we observe that during the discontinuity the associated weight levels drop to about their minimal value. This means that around the jump the system shows a considerable increase in regression-related behavioral qualities such as increased error, loss of generality, and slowing down (provided, of course, that is what low weights are associated with). According to van der Maas and Molenaar (1992), such behavioral properties are likely candidates for formal catastrophe flags such as critical slowing down and divergence.

The current Piagetian model parameters typically produce weight functions that oscillate with the emergence and disappearance of the stages (see Figure 4). These marked regressions do not occur or are considerably less extreme in non-Piagetian models where, for instance, the ZPD parameters are active or in which the parameters governing the rate of acquiring new contents are different. That is to say, strong regressions are typical of the Piagetian model simulation, not of discontinuities per se.

The cyclical pattern of change is also explicitly considered in Fischer's neo-Piagetian model of development. K. W. Fischer and Bidell (1997) provided evidence for a pattern of spurts eventually followed by regressions in the form of temporary drops in performance (see also K. W. Fischer & Kennedy, 1996; K. W. Fischer & Pipp, 1984). The spurts and regressions can

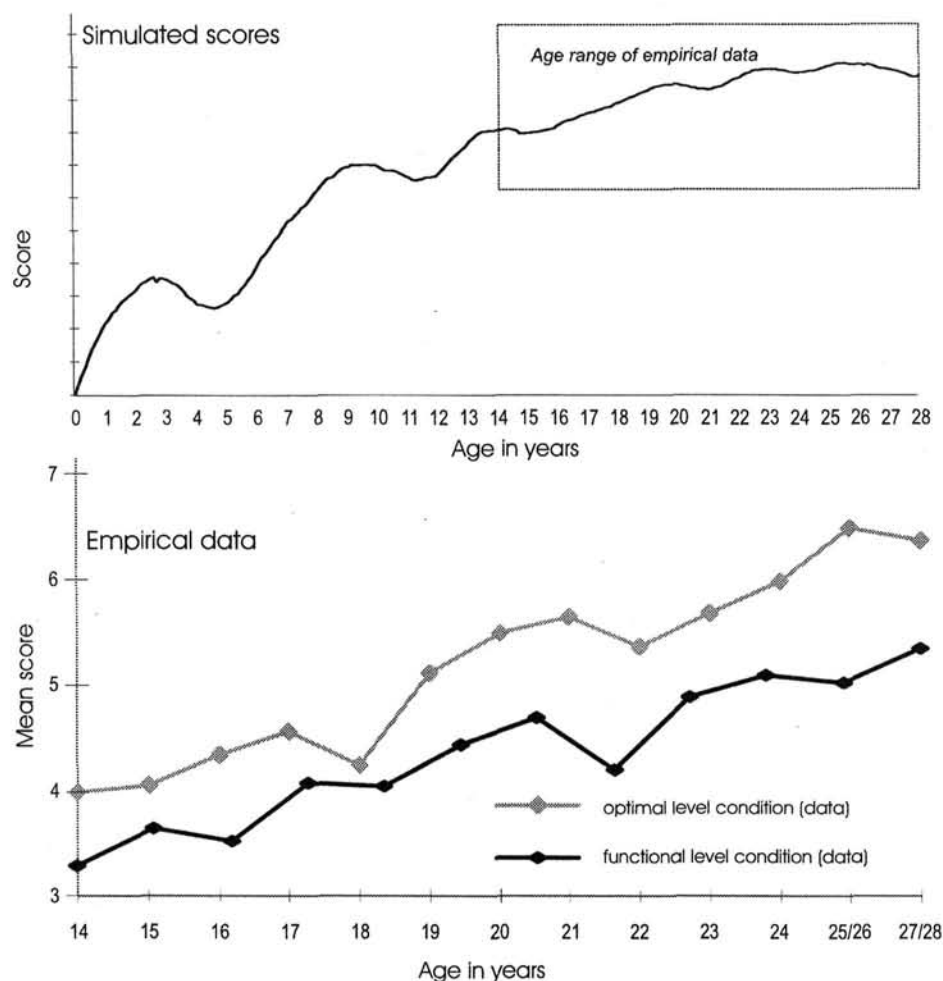


Figure 7. A simulation of test scores based on the product of output levels and their associated weights; the scores show a temporary dip during the transitional phase, comparable to those found by Kitchener et al. (1993) and Fischer (K. W. Fischer & Kennedy, 1996; K. W. Fischer & Pipp, 1984).

be modeled in the form of growth models of interacting competing and supporting growers (van Geert, 1994a; K. W. Fischer & Bidell, 1997). The present model, which combines shifts in developmental level with a cyclical weight pattern, provides an alternative or complementary explanation. If the weights indeed correspond with the speed and stability of a performance, we may assume that test scores vary with the weights associated with the cognitive contents tested (or more precisely, they vary with whatever is the psychological analog of the weight function). In a study by Kitchener et al. (1993) of Fischer's model of the development of reflective judgment, test scores were averaged for tasks characteristic of a specific developmental level. We may assume that the scores depend not only on the existence of skills representative of that particular level in the cognitive system of a subject but also on the stability, degree of consolidation, et cetera, of those skills. Assume for the sake of illustration that the weights explain between one fourth and one third of the variance in the test scores (and skill level itself explains the remaining 66% to 75%; it goes without saying that these are

just reasonable assumptions—in fact, there is a broad range of such proportions that results in qualitatively similar outcomes). The average scores of 12 simulated developmental trajectories are represented in Figure 7 and are compared with those of the original empirical study (note that the original study made a cross-sectional comparison between age groups of 12 participants, whereas the simulation is based on averages of 12 simulated longitudinal studies).

Horizontal décalage. We have seen that Piaget used the concept of horizontal décalage to refer to the fact that a stage shift eventually takes place within a limited range and is completed only later. For instance, children apply concrete operational thinking to conservation of substance years before they are able to apply it to conservation of volume (Piaget & Inhelder, 1941). Put differently, there exist two levels of functioning at the same time whose expressions depend on the kind of problems the child is confronted with (e.g., either mass or volume problems). The horizontal décalage phenomenon can be explained in two different ways. First, it could be an illustration

of the fact that development is domain specific and that the model of the development in a single individual is in fact a concatenation of several independent developmental trajectories, as explained in the section on *Diachronous Multimodality: Sudden Jumps and Discontinuities*. Second, horizontal décalage could imply that, within a domain, some contents are assimilated earlier to a new stage than others, thus implying the simultaneous existence of different levels, or multimodality. Catastrophe theory expects the occurrence of multimodality or, more particularly, bimodality in the vicinity of a sudden jump (Gillmore, 1981). It is interesting to see that the present model produces such temporary bimodality as a natural outcome of the stage shifts. That is, a sudden jump toward a new equilibrium level is often preceded by an intermediary stage in which the system's output randomly oscillates between the new and the old level. In a series of 20 simulations based on the Piagetian parameter set, 16 showed at least one clear instance of bimodality (later I explain cases in which the overlap stage lasts much longer than in the present case). This bimodality is based on the fact that the jump toward a new level is characterized by the existence of two simultaneous weight peaks instead of one. The occurrence of oscillations directly preceding a transition has also been noticed by Flavell and Wohlwill (1969). It also explains why individual differences between children lose their stability in the vicinity of transition points, a finding reported by McCall et al. (1977; note, however, that if the drop in stability merely refers to the pattern of correlations in a group of subjects, it may also be explained by the fact that individual children reach higher levels at different ages; the point is that the drop in stability is not necessarily explained by individual differences but also follows from the dynamic phenomena themselves).

It should be noted that the length of the bimodal periods is not very great and that it is more or less arbitrary whether or when they occur, unless we apply Equation 6, which is the stochastic version of the equation that determines the accommodation range. Other sets of model parameters (discussed in the section *Strategies and Multimodality*) produce much clearer cases of prolonged bi- and even multimodality.

Substages. The occurrence of bi- and multimodality is a general phenomenon that is part of the dynamics based on the general developmental mechanisms discussed earlier. It may express itself in the form of transitional bimodality (or horizontal décalage), in the form of coexisting cognitive strategies (discussed later), and in the form of overlapping substages. Piaget described substage phenomena in various domains. Examples are the substages of circular reactions during the first, that is, the sensorimotor period (Piaget, 1952); the substages in the emergence of conservation (Piaget, 1957); and the six stages in the construction of the object and of space, which are themselves substages of the sensorimotor period (Piaget, 1954). Substages play an even more important role in the neo-Piagetian models (Case, 1992a; K. W. Fischer & Rose, 1994). If the stages simulated in the model are stages in the literal, unimodal sense of the word, we might expect that, also taken into consideration the stochastic nature of the model, all outputs (actions, etc.) in a particular stage are approximately normally distributed around an average that is characteristic of the stage at issue. However, what we find looks more like an illustration of what I called the fractal nature of the developmental process: In models with

sharp discontinuities, we find stages that consist of a small number of overlapping sublevels or substages, that is, concentrations of values in narrow bands (see also Figure 8). Again, this finding does not imply that the model simulates how the real substages, if any occur empirically, come about. What the finding implies is that the organization of stages into substages is a natural property of the dynamics producing patterns of sharply distinguished stages and discontinuities.

From General Stage Models to Domain-Specific (Two-)State Models

Given the fact that the dynamic model is thus neutral with regard to which range of cognitive contents it encompasses or which stretch of time it simulates, it should be possible to simulate different types of changes, for instance, discontinuous changes occurring in smaller content ranges and time periods, provided the parameters are adapted to the time and content aspects of the new model application. I first review some of the evidence on such small-scale discontinuities.

Evidence for Domain-Specific (Two-)State Models

Brainerd (1993) claimed that there is convincing evidence for the fact that memory and learning are a matter of discontinuous change across two or three states (I take the initial level of performance also as a state; see also Brainerd, 1985; Brainerd, Howe, & DesRochers, 1982). Data on the acquisition of Piagetian concepts involving length, number, quantity, class inclusion, and proportionality also support the two- and three-state pattern (Brainerd, 1993).

Thomas and Lohaus's (1993) study on Piaget and Inhelder's (1956) water-level problem point in the same direction. In the

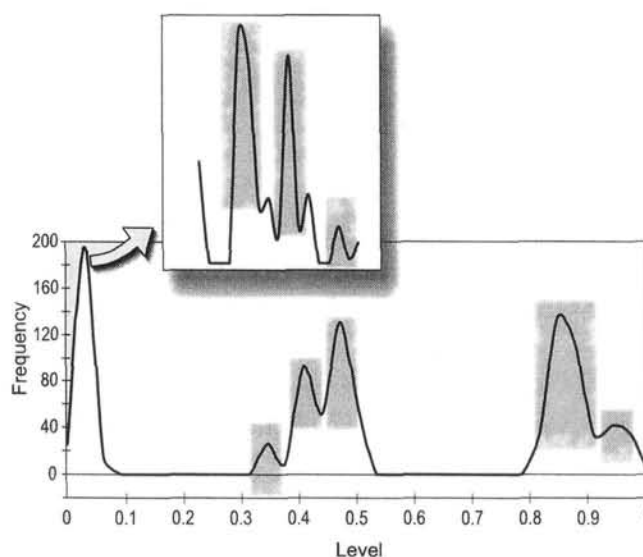


Figure 8. Substages, separate levels within major stages, result from the "fractal" character of transition processes, based on the negative exponential relationship between frequency of occurrence and the magnitude of the shifts; the three major stages resulting from the present simulation each consist of three to four substages or sublevels.

water-level problem, participants have to draw the water level of a tilted container. They have to understand that the level is oblique to the line of gravitation, irrespective of the orientation of the container. The cross-sectional study involved 579 youngsters in an age range from 6 to 17 years. Ninety-two percent of the participants were between 8 and 16 years; the average age of the children was 11 years, 10 months. The response patterns of the task (and also that of a related task involving a van on a slope) were clearly bimodally distributed. This fact demonstrates that development in this field is a discrete stage process (Thomas & Lohaus, 1993; Thomas & Turner, 1991). The study also showed that the response pattern of the poor performers indicates that they follow a constrained random rule. This rule involves a trial-and-error construction of horizontal and vertical axes and corresponds with an intermediate stage already described by Piaget and Inhelder (1956).

van der Maas and Molenaar (1996; see also van der Maas, 1993) tested conservation understanding in 101 children in a series of 11 repeated sessions during 7 months. The test was implemented on a computer in the children's classrooms. It consisted of four different types of liquid conservation questions, had a maximal score of 8, and a minimal score of 0. At the start of the experiment the ages of the children varied between 6.2 and 10.6 years. Twenty-four transitional children were found, that is, children who showed the development from lack of conservation to conservation understanding during the 7 months the experiment lasted. The results provide support for a discontinuity interpretation of conservation development. The data demonstrate a sharp increase in the score, which is most clearly shown if the data are corrected for the lag in transition moment. A plot of the frequencies of the scores of the 24 participants shows a bimodal distribution, very similar to the bimodal distribution found in the water-level experiment by Thomas and Lohaus (1993; see Figure 9). The occurrence of transitional strategies is interpreted as evidence of so-called anomalous variance, which the authors relate to oscillations in the scores and the existence of intermediary forms of reasoning in the vicinity of the transition. Comparable evidence for a bimodal pattern of change in analogical reasoning has recently been discussed by Hosenfeld et al. (1997a, 1997b).

A Model of Two-State Change: Parameter Decisions

In order to determine a set of parameter values for the simulation of the Thomas and Lohaus (1993) and the van der Maas and Molenaar (1996) studies, we have to consider the differences in timescale and content range between those studies and, for instance, the simulation of the Piaget model. First, both studies deal with contents and problems that are within the cognitive attractor range of the majority of the participants or, to put it differently, within their ZPD. For instance, the water-level task involves the ability to detach oneself from the suggestive orientation of the container and to take a second, absolute reference frame into account. Piaget and Inhelder (1956) saw the water-level task as an instance of concrete operational thinking, and the upper level of the task should in principle be reached at the age of approximately 9 years. As far as conservation is concerned, the transitional participants are, by definition, within reach of the level of understanding necessary to solve the com-

puterized conservation task and apply the required cognitive strategies. For instance, they can be expected to understand the distinction between appearance and reality (Flavell, 1986).

We may assume, first, that the attractor state is in principle defined by 100% correct responses (i.e., there are no real cognitive difficulties in the conservation task that would prevent participants from reaching a 100% correct score) and, second, that the attractor state is either already present in the system (qua attractor state or potential level, i.e., not in the form of the actual state) or, more likely, that it will very soon become established in the system in the form of a potential state, as a consequence of action and experiences with conservation-related problems. These assumptions imply that we must determine the parameters governing the shift in attractor state in such a way that the gap between the initial and final attractor state level is bridged in a short time span. This is done by setting the δ or γ parameters to values that guarantee a fast change in the attractor state level, given appropriate experiences.

Second, we may assume that the range of contents (skills, knowledge, rules, etc.) relating to the present tasks is considerably smaller than the overall range that we used with the Piagetian model and that, in principle, spanned the whole range of cognitive contents. As far as this point is concerned, however, there is a difference between the water-level and the conservation task. The water-level task is administered only once, whereas the conservation task is measured repeatedly. Assuming that the children are not familiar with the water-level task (meaning that it is not a task they perform regularly), our model must consider the change in some underlying broad range of skills, rules, or understanding that covers the kind of thinking required in this particular task. In the cross-sectional study, the score on the water-level task is a single estimation of the developmental level of this range of skills or knowledge. The van der Maas and Molenaar (1996) conservation experiment, however, involves a repeated confrontation with a relatively simple problem. The participants answered eight conservation problems 11 times, which means that we can model the process as a series of 88 steps (each step is an answer to a conservation problem). Because it is not certain that the children also experienced each question as a separate activity, we must assume that the number of steps lies between a minimum of 44 (four kinds of problems, administered 11 times) and 88 (four kinds of problems, two problems each, 11 tests). This means that the width of the range of cognitive contents involved in the water level on the one hand and the conservation task on the other is probably considerably different. The wide range of contents (rules, skills, etc.) expressed in a single administration of the water-level experiment implies that the spread of the effect of a single action involving those contents (rules, skills, etc.) is significantly smaller than the spread of the effect in a much more confined range of contents (a small range means, by definition, more interaction or stronger connection between the components of that range). This difference can be expressed by means of the effect-spreading parameters σ_a and σ_c : the wider the spreading effect, the smaller (on average) the range of contents covered by the model.

Third, the tasks differ in the age ranges involved. The time span of the conservation task is represented by (a maximum of) 88 steps, lasting 7 months, whereas the number of steps of the water-level model should be considerably bigger, given the wider

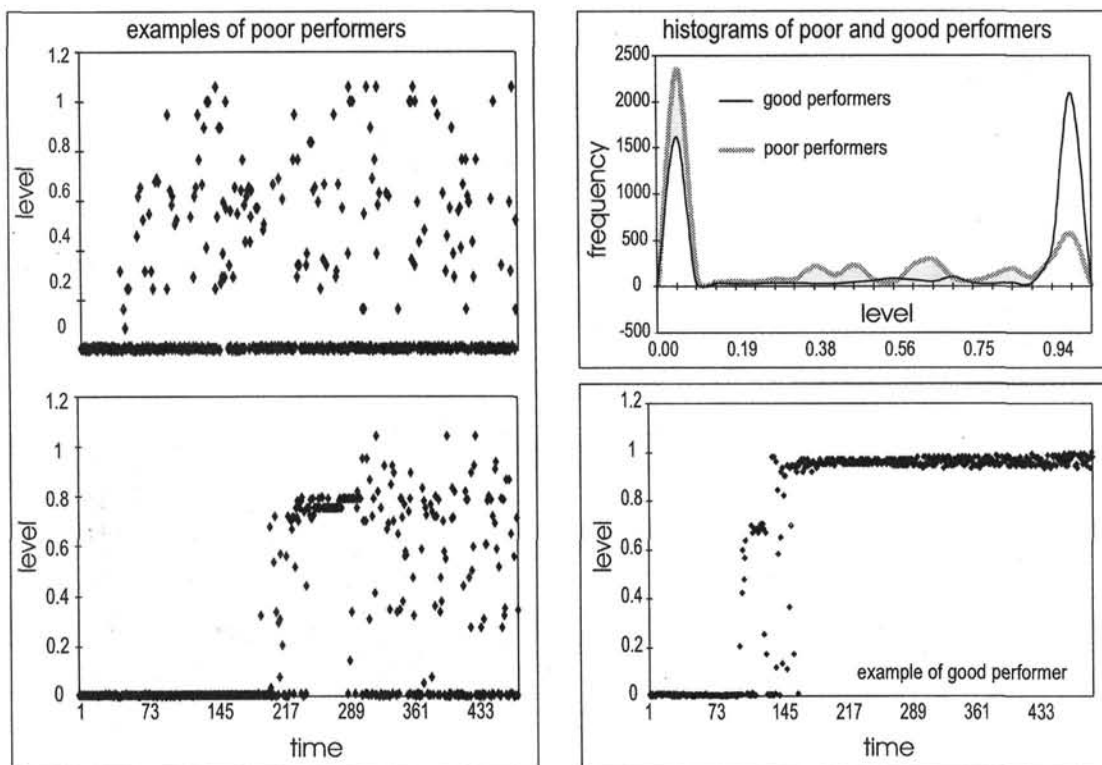


Figure 9. A simulation of the bimodal score distribution of Thomas and Lohaus's (1993) study on the water-level task; at the left are examples of poor performers, characterized by a high variation in response patterns; bottom-right is an example of a good performer, whose scores show a transition between wrong and correct problem solving, with high response variance only during the transition; the histogram of 10 simulated good and 10 poor performers shows the bimodal score distribution of good performers and the unimodal distribution of poor performers; the latter distribution has a "tail" to the right, representing the variety of problem-solving strategies and levels.

age range, from 8 to 16 years (which resulted in a somewhat arbitrarily chosen number of 500 steps for the water-level task).

Finally, tasks of this kind are probably very conservative because they involve a kind of self-fulfilling prophecy. This is especially clear with the conservation task (see also van Geert, 1994a). If one believes that the amount of liquid changes if it is poured into a container with a different shape, then each time one sees a liquid change its shape the belief state is automatically confirmed (simply because the indicators of the belief in changing amounts, namely the different shapes of the liquid column after pouring, are by definition present). Moreover, conservation is an interesting problem because, under ordinary circumstances, children are not explicitly trained in it. A similar reasoning holds for the water-level task, provided it involves the ability to change reference frames. If one focuses on a particular reference frame, then an eventual alternative frame is not taken into consideration. This conservative tendency probably explains why these tasks are so difficult, notwithstanding the fact that the cognitive principles on which they rest are within the reach of the participants' understanding. This conservative tendency can easily be represented by making the conservative parameter significantly larger than the progressive one.

A Model of Two-State Change: Simulation Results

The water-level task. The simulation of the water-level task data shows a subgroup of good performers and a subgroup of poor performers. The good performers make the predicted jump-like shift from incorrect to correct performance, with a relatively short interval of randomized responses (see Figure 9), which have already been noticed by Piaget and Inhelder (1956). The bad performers do not reach the correct level within the time range of the simulation (and prolongation of the time range shows that this is not just an accidental cutoff effect). They show a completely randomized response pattern, that is, their response levels occur on any intermediary level between the initial and the maximal level. This finding is in accordance with what Thomas and Lohaus (1993) found with the bad performers and that they termed the "constrained random rule" performance pattern. Figure 9 shows simulated data from a typical good performer in comparison with simulated data from a bad performer. The frequency count of the response levels of good and bad performers is also characteristically different and is in accordance with the Thomas and Lohaus data (the frequencies are based on 10 simulated participants in each performance

class). Simulated good performers show a characteristic bimodal score distribution, bad performers have a peak on the initial level and an approximately even distribution for the higher levels.

What explains the distinction between good and bad performers in the model? The current simulation suggests that the distinction between a good and a bad performer can be explained by the parameter that determines the speed with which the attractor state escapes from the actual development state. In the present set of models, that speed is high anyway, but once it crosses a certain threshold, the outcomes of the simulations become characteristic of poor performers. A simulation of 50 cases based on a stepwise increase in one of the parameters that governs the rate of change of the attractor state (the γ_1 parameter) shows that the parameter explains 28% of the variance in the average scores (all levels from a simulation divided by the number of steps; $F = 18.68$, $p < 1E - 4$).

The psychological meaning of this escape rate is that participants who are confronted with the kind of reference frame problem from the water-level task know that it can be solved by using more advanced knowledge or skills, for instance, logical reasoning or operational thinking, which they probably know from school or formal training. In order to be able to profit from this attractor state, the distance between the actual and the attractor level should not exceed certain (broad) limits, enabling the child to discover potential similarities and relationships between that more advanced level on the one hand and the current task on the other. If we assume that the attractor level, indicated by logical or operational thinking, is primarily determined by schooling and formal education, the rate with which that level changes is about similar for all (usually intelligent) children participating in the same educational system. However, for a particular group of children, namely those who are less capable of individually inferring similarities or relationships between logical or operational principles on the one hand and the present task on the other, the rate of change of the logical or formal attractor level, which they see exemplified in scholarly tasks, for instance, is too big (in comparison with their rate of self-induced application of that attractor level to new problems). That is, they are not able to make the link between the (unfamiliar) water-level task on the one hand and what they know about geometry, logical reasoning, and so forth on the other hand. They get lost and continue to perform poorly on this relatively simple task. This explanation of the effect of the escape rate of the attractor state follows directly from the model simulations, but empirically it is entirely speculative, although not without intuitive plausibility. Further empirical tests are needed in order to decide whether it is correct.

The longitudinal conservation task. The simulation of the longitudinal conservation data shows that the chosen parameter values produce the empirically observed pattern in the van der Maas and Molenaar (1996) data. Recall that the major difference between this and the previous simulation lies in the number of steps (simulated time span) and the parameter that determines the size of the cognitive content range to which the simulated process applies. In the case of repeated conservation tests, this range is considerably smaller than with the water-level task, which probably referred to the development of a broad range of cognitive contents related to reference frames, such as geome-

try and physical knowledge. The range is indirectly determined by the width of the spreading of experience or learning effects, which is considerably larger in the conservation than in the water-level model (on average, the larger the spreading, the more restricted the range of cognitive contents to which it applies). The frequency count of the data for 20 simulated subjects shows the characteristic bimodal pattern (see Figure 10) found with

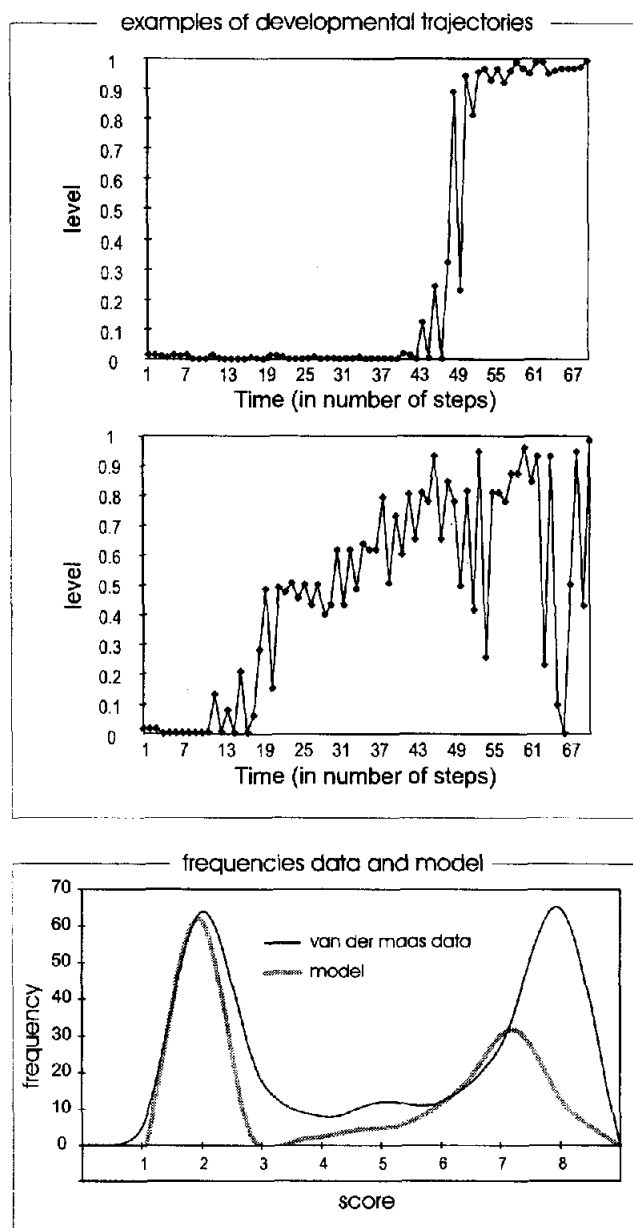


Figure 10. A simulation of the bimodal distribution of the van der Maas and Molenaar (1996) conservation data; examples of trajectories are on the left side. Both sudden jumps and strong oscillations toward correct understanding occur; the relative overrepresentation of the non-conservation condition in the histogram of the simulated data is an artifact (the empirical data series was not long enough to cover a significant amount of the nonconservation period).

the 24 transitional participants in the empirical study. Note that the peaks in the simulated data have a different height than those of the empirical data: There is a relative overrepresentation of the initial level. This is most likely an artifact of the transitional participant selection procedure or of the fact that the number of available test sessions differed per participant. Plots of simulated individual sequences show the transitional state of oscillatory behavior and intermediate response levels that is also found in the original data (Figure 10).

Note that both the water-level and the conservation simulation are based on highly conservative systems (i.e., systems where the parameter, η , determining the growth of the weight of the actual level content has a high value). That is, they involve what could be called self-fulfilling prophecies. I have argued that a belief in nonconservation, for instance, determines the nature of the experiences with conservation problems and thus amplifies that belief. This property explains why these conservative systems follow the basic two-stage pattern found in the data and in the simulations. If we increase the effect of alternative experiences, that is, of information that does not comply with the (wrong) beliefs, we should find three or four instead of the two states of the conservative system development. This increase can be obtained by increasing the value of the progressive parameter θ , which determines the effect of attractor-state-defined experiences. This increase results in patterns of three to four distinct states, which have been found by Brainerd (1993) and K. W. Fischer (1980a).

Strategies and Multimodality

Cognitive Strategies and Multimodal Activity Patterns

According to Siegler (1994), development should be viewed as a process of change in the distribution of different ways of thinking, in the emergence of new ways that find a place next to the already existing ones, or in the eventual loss of old forms of acting. In his analysis of microgenetic change, Siegler (1995) sketched a model of overlapping waves of strategies. Each strategy represents a particular form of (in this case numerical) problem solving. Strategies can be compared with stages or levels in that a stage or level represents a particular, generalized way of acting, separated from another stage or level by a discontinuity. I have explained that, in the current model, strategies, or stages and levels for that matter, are not represented in terms of their qualitative properties and distinctions but only in terms of their developmental distance. If strategies differ only in terms of content and not in terms of developmental distance from one another, it appears that they cannot be represented by the present dynamic model, because the latter confines itself to developmental level differences. However, the association between the ordering in the model's internal array on the one hand and the concept of developmental distance on the other is by no means obligatory: The array can in fact represent any distance specified in any space of descriptive dimensions, not only those that code for developmental differences. However, if we use the dimension to represent a distinction in terms of contents, it does not similarly evoke a developmental distance interpretation, and vice versa. For the sake of simplicity, I continue to use the distance dimension in terms of developmental distances.

Before proceeding to the question of whether the occurrence of simultaneously occurring discrete levels can be simulated by the current model, I first discuss some important features of a model of strategy development introduced by Siegler and Shrager (1984) and Siegler and Shipley (1995). This model, the ASCM, ascribes a specific strength to each strategy. The strength represents the strategy's accuracy and speed when applied to specific problems. The probability that a strategy will be selected depends on the level of the strategy's strength. In addition, Siegler and Shipley (1995) assumed competitive relationships between different strategies and, conversely, a supportive relationship between the strategy and itself in that successful application of the strategy increases its strength (see also Siegler, 1989). In previous articles, I have shown that a growth model of mutual support and competition can indeed explain the increase and eventual decline of strategies (van Geert, 1991, 1994a). A disadvantage of the growth model, however, is that it does not specify how new strategies come about.

Siegler and Shipley's (1995) model has two important features in common with the general dynamic model discussed in this article: It specifies the probability that a strategy will be selected in terms of its strength, which corresponds with the weight function in the present model, and it assumes supportive and competitive relationships as a function of the distance (i.e., dissimilarity) between the strategies involved. This latter principle is mirrored by the dynamic model's negative rate (competition) factors η_c and θ_c , the actual values of which are a function of developmental distance. In the preceding sections, however, we have seen that our model produces successive stages or levels, or, eventually, successive strategies for that matter, but not sets of coexisting levels or strategies that emerge and eventually wane. The point is that these models used competition parameters that were in fact too strong: As soon as a new level emerged, it outperformed the older level, that is, it exerted a competitive force strong enough to make the old level disappear more or less suddenly. We have also seen that in a number of cases the old level lasted for a little while, thus producing a pattern of bimodal behaviors that marked the transition to a higher level.

In order to obtain a pattern of distinct, coexisting levels simulating distinct but coexisting strategies, the following parameter values can be used. First, the competitive strength parameters η_c and θ_c must be considerably lower than those used with sequential patterns. This means that competitors (e.g., alternative ways for solving comparable problems) are less negatively affected by the success of a specific strategy or skill level than in the cases discussed previously. Second, the model implicitly assumes that the cognitive contents related in a system of competitive and supportive relationships are relatively close or similar to one another. This assumption follows directly from the fact that the model deals with strategies, that is, alternative ways for solving problems within a specific domain, such as numbers. This implies a wide range of mutual interaction between contents at different levels. In other words, the effect of an action or an experience on the underlying knowledge or rules (or whatever mechanism is held responsible for a particular activity) must be relatively wide, which is obtained by setting the spreading parameters σ_a and σ_r to a low value (which produces a wide spreading effect). In addition, strategies or rules are highly dependent on success information. They are frequently used

(i.e., each time a problem occurs that requires that particular strategy) and altered in function of the results or effects. Put differently, there exists a relatively high update rate (i.e., a high rate of learning new problem types and problem-solving possibilities), which, however, must be compensated by a relatively low progress rate. Thus the update parameter ϕ is set to a relatively high value, whereas the progress rate (how much can a strategy change after each experience) is set to a moderate value (which is either δ in the case of a linear learning model or γ_1 and γ_2 in the case of the Gaussian model).

Figure 11 shows an example of a process, producing several coexisting discrete levels of action patterns (problem-solving skills, rules, strategies, etc.). The number of separate bands is typically in the order of two to six.

With reduced competitive parameter values it is also easy to simulate the emergence of pairs of strategies, that is, two separate bands of action types that lie at the opposite sides of the developmental distance dimension (or content dimension, if one decides to view the distance dimension as a reference to contents rather than developmental distance). This case simulates the emergence of a new strategy out of an old one within a single conceptual dimension.

The simulations clearly demonstrate the fact that the new strategy is like an offshoot of the old one: The old strategy first

increases its level of complexity (i.e., its developmental level) in a continuous fashion. When it has reached a local maximum, the new strategy branches off in the form of a discontinuity. With the new strategy growing, the old one falls back on a less advanced level (see Figure 11, bands 1 and 2 in particular). The simulated data also support Siegler and Crowley's (1991) observation that the generalization of the new strategy takes place after its actual discovery: In Figure 12, for instance, a new strategy band emerges at some intermediary level and then grows in a continuous fashion toward its maximum. In some cases, the old strategy grows toward the new one in an almost continuous fashion, that is, without a discontinuous jump, after which they split apart (see Figure 12a).

With the emergence of a new strategy, the old one falls back on a less advanced level and eventually disappears almost completely. This is what we expect with a sequence of strategies that increase in correctness or developmental level. For instance, the use of a word inversion rule in forming *wh*-questions or the use of two-word sentences in early language is an example of this type of strategy or rule emergence (van Geert, 1991): The old rule disappears as the new one takes over. A similar pattern is found with the strategies or rules used in solving balance scale and conservation tasks (Siegler, 1981). However, in this particular case, there exist intermediary levels between

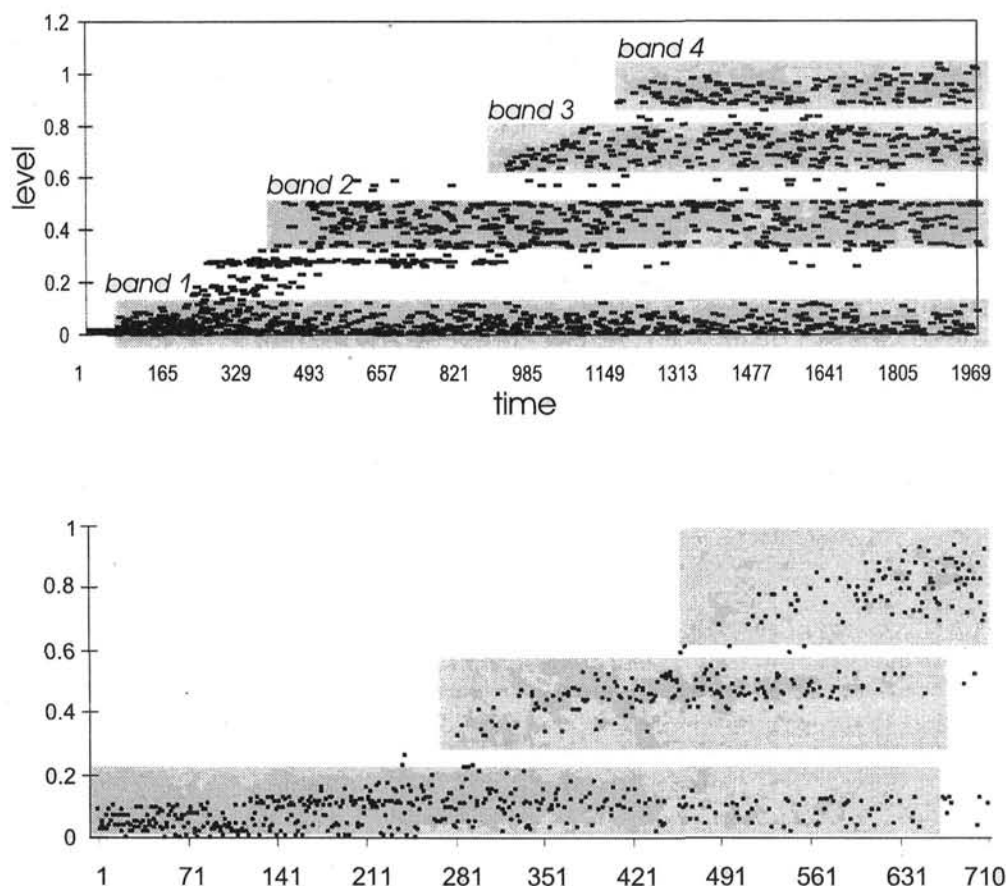


Figure 11. Under conditions of "relaxed" competition, development takes the form of bands of co-occurring levels. Graphs result from different random variables operating during each simulation step.

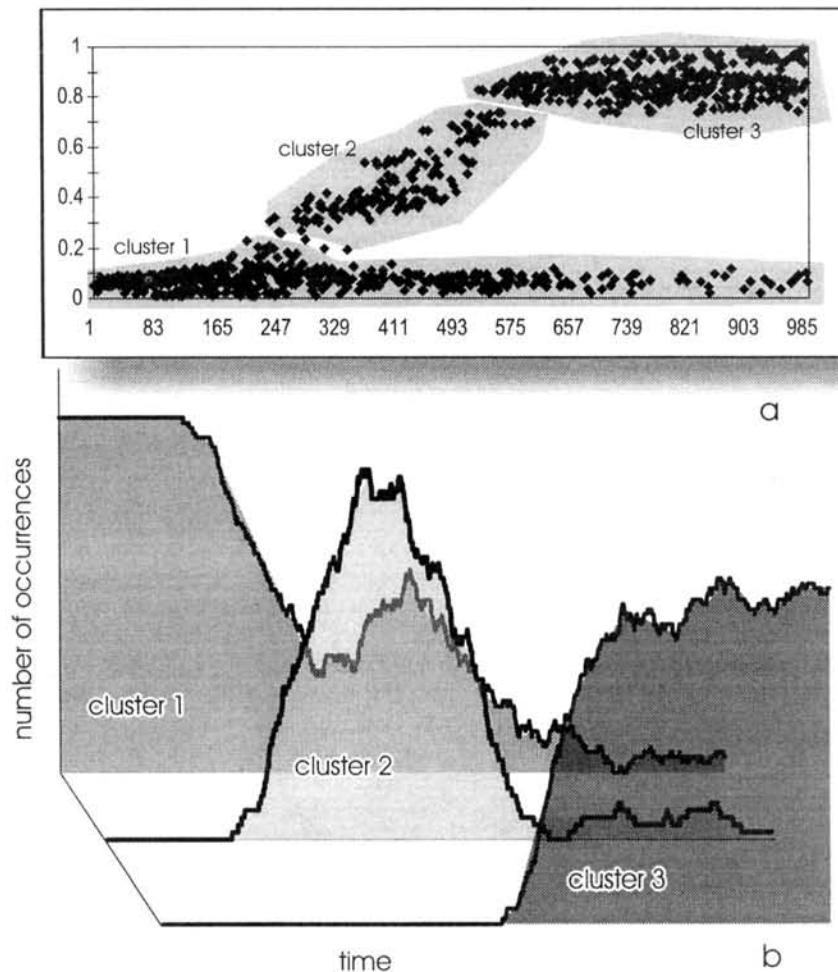


Figure 12. A frequency count of the occurrence of outputs from three clusters (1, 2, & 3); the frequency of occurrence corresponds with the "wave" pattern described by Siegler (1995).

the simplest and the most advanced rules. This is a pattern that can be most clearly found if we use the stochastic potential level update function (Equation 6; see Figure 12). A limitation of the present simulations is that they often produce only one major discontinuity, which is eventually subdivided into smaller discontinuities, that is, nested strategies. On the other hand, an interesting feature of the simulated patterns is that they demonstrate the predicted wavelike appearance and disappearance of strategies or rules (Siegler, 1995, 1996; see Figure 12b).

Given the results obtained with these particular parameter values, I should caution against a possible misunderstanding of what this particular model actually implies with regard to the emergence of strategies. It is obvious that a considerable majority of the strategies that a person uses are highly specific and result from examples given, teaching, and so forth. The present model does not specify particular contents or skills that are transmitted to developing participants, unlike, for instance, neural network models that present the learning system with specific examples or exemplars. It follows then that the present model cannot explain the transmission and appropriation of specific strategies or skills, for example. However, what the model does

show—and that is the main reason why it is presented in the context of a discussion of strategy development—is that the formation of clusters is an intrinsic property of the kind of dynamics discussed in this section, that is, it is an emergent property of this particular process of change and exchange with the environment. Once such clusters of patterns emerge—or at least have the tendency to emerge—they are likely to be affected by selective processes that filter out the less adequate solutions and shape the clusters in accordance with external models. Such selective and shaping processes could well be established in the form of setting examples, teaching, and so forth.

This particular pattern of strategies in the form of bands of simultaneously occurring clusters can also be viewed as a representation of a long-term developmental process consisting of discrete but simultaneous levels. It is likely that for the highest levels to occur, the task or contextual conditions must be optimally supportive. Likewise, more adverse task conditions will confine the participant's performance to lower levels of the developmental spectrum. If participants are presented with testing conditions that help them achieve at their maximal level, such as in Fischer's (K. W. Fischer & Kennedy, 1984; K. W. Fischer & Pipp, 1984;

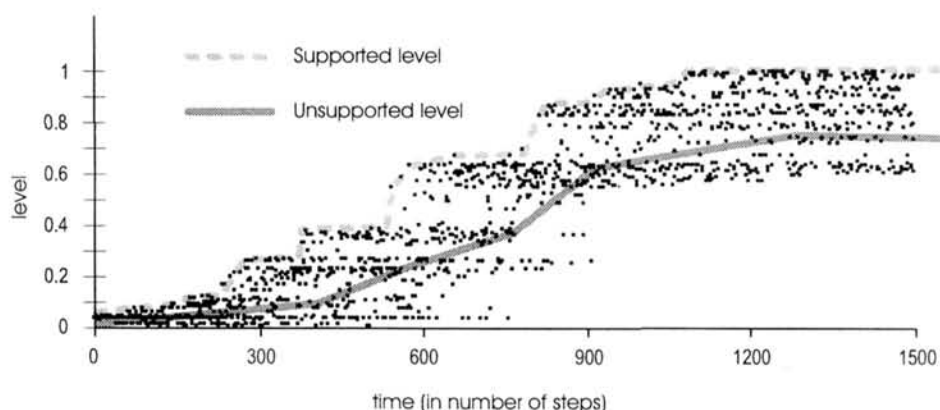


Figure 13. Under supportive testing conditions, the scores are likely to follow the maximal levels, whereas unsupported testing conditions reveal scores at considerably lower levels.

Kitchener et al., 1993) help-and-support testing conditions, the test scores will follow the maximal developmental level, which is often characterized by a stepwise increase. Under unsupported test conditions, the participant's scores are more likely to be confined by one of the lower level bands (see Figure 13).

Multimodality and Transitions

Strategies are structurally or qualitatively distinct formats of thinking or action that coexist for a considerable amount of time. At the other end of the scale, there are developmentally distinct and discontinuous formats of action (stages, phases, tiers, etc.) that literally jump from one stage to another, without ever showing any overlap. Many cases of successive level transition, however, will probably show some form of temporary coexistence of the old and the new levels. I have already mentioned Piaget's concept of horizontal décalage, which implies that different levels of cognitive functioning can coexist with both levels operating on different content or problem domains.

Goldin-Meadow et al. (1993) and Alibali and Goldin-Meadow (1993) have readdressed the issue of transient multimodality in developmental transitions (by multimodality I understand the simultaneous existence of developmentally distinct patterns of actions, skills, knowledge, etc.). Multimodality as an indicator of transition means that the simultaneous existence of these distinct levels indicates a period of transition between the unimodal occurrence of the more primitive level on the one hand and the unimodal occurrence of the more advanced level on the other. Goldin-Meadow et al. (1993) characterized transitional stages as stages in which multiple hypotheses, that is, hypotheses at distinct levels, are considered simultaneously. During the transition period, these simultaneous hypotheses are explicitly expressed in the form of distinct representations or communications. For instance, children solving problems of mathematical equivalence express one, more primitive strategy, in the form of their verbal justification and a more advanced strategy in the form of an accompanying gesture (Alibali & Goldin-Meadow, 1993; see also Perry, Church, & Goldin-Meadow, 1988). This so-called gesture-speech mismatch is an observable indicator of a transitional stage and of the ZPD,

because the gesture refers to a potential level, that is, a future unimodal developmental state. The observation of gesture-speech mismatch thus allows educators to calibrate their input to the child's level of understanding (the ZPD) and thus to promote faster or better cognitive growth (Goldin-Meadow et al., 1993).

In the dynamic model presented in this article, temporary multimodality is a naturally occurring phenomenon. Assuming a competitive parameter value θ_1 produces strict succession between two different levels and a value θ_2 produces a pattern of long-term coexistence as discussed in the section *Cognitive Strategies and Multimodal Activity Patterns*, values between θ_1 and θ_2 produce intervals of temporary overlap of varying length between distinct levels. The overlap can be observed in the form of a random succession of actions on the more primitive and on the more advanced level (see Figure 14).⁹

It turns out that multimodality is not in all cases indicative of a jump toward a higher level. Goldin-Meadow et al. (1993) discussed evidence showing that, in conservation development, some children evolve from a transitory stage in which more than one hypothesis was present to one with only one hypothesis that was nevertheless incorrect. This occurs if children do not receive the right kind of help and information during the transitory, multimodal state. Because the dynamic model operates with a purely stochastic distribution of optimally adapted input (based on the ZPD parameters), there must be accidental cases in which the input is not sufficiently adapted to the system's developmental attractor. In such cases, the system will first move into a bimodal transitory phase and then fall back onto its original less-developed unimodal action pattern. A series of 50 simu-

⁹ Unfortunately, the dynamic model allows only one expression of a skill, concept, understanding, and so forth at a time (for no other than mere technical reasons). Because the successive transitional multimodality of the present model is based on the fact that the weight functions of the distinct levels are about equal, it makes sense that, if the model would indeed allow more than one representation of a level at a time, those representations would be dissimilar half the time and thus show a representational mismatch of the kind observed by Goldin-Meadow and her collaborators.

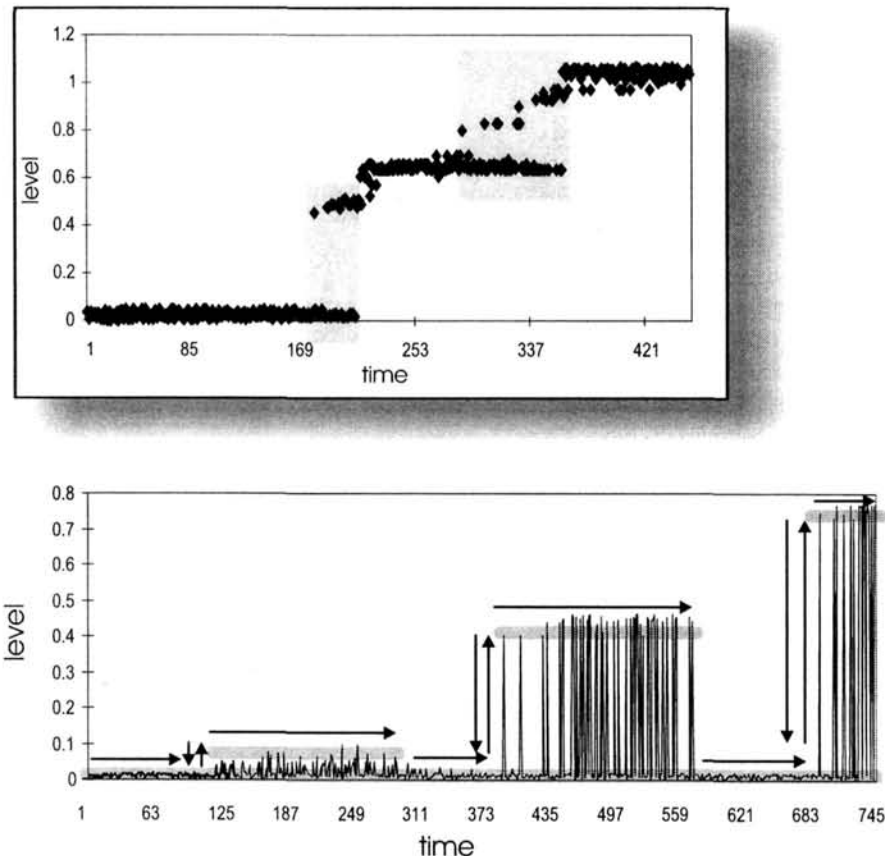


Figure 14. Bimodal response patterns occur in the vicinity of a transition, where both response modalities are about equally probable; if responses can be bimodal themselves (e.g., when they contain a verbal and a gestural component), a mismatch may occur between both modalities; in the present model, the mismatch takes the form of a succession of different response modalities. In some cases, the bimodality may temporarily disappear, announcing the emergence of a higher level (lower panel).

lations has shown that such regressions to the older level indeed occur in one third of the cases (see Figure 14).¹⁰

Variability and Transition in Microdevelopment

The notion that people function at different levels of development, for instance, in the form of distinct rules, hypotheses, or strategies, and that this variability is an essential factor in promoting development, features prominently in the work of Siegler (1994) and Goldin-Meadow et al. (1993). It is also central in the work of Granott (1993, 1996), K. W. Fischer and Granott (1995), K. W. Fischer, Bullock, Rotenberg, & Raya (1993); K. W. Fischer, Knight, & Van Parys (1993), K. W. Fischer and Bidell (1997), and K. W. Fischer and Kennedy (1996).

In their microdevelopmental studies, Fischer and Granott have tried to capture the nature and sequence of developmental stage progression. The results of their studies, in which adult dyads have to find an explanation for the behavior of small robots, show that the microdevelopmental trajectory is all but a neat progression toward the highest attainable level. Instead, progress occurs in the form of highly variable action levels, often showing

temporary regressions to a lower level. Granott (1996) stated that development is basically a matter of fluctuating responses within a band or range that, as a whole, progresses toward higher level functioning. This range, within which the participants' actions vary, constitutes the observable expression of what Granott (1996) called the "zone of current development," which is the microdevelopmental analog of Vygotsky's ZPD. Temporary regressions, which occur almost by necessity, given that development is a process of fluctuation within a range, not only occur in micro- but also in macrodevelopment (K. W. Fischer et al., 1993; K. W. Fischer & Kennedy, 1996; K. W. Fischer, Pipp, & Bullock, 1984; K. W. Fischer & Rose, 1994). Although the basic principles of variation, of multimodality, of the developmental significance of multimodality, and a ZPD-like range concept of the driving force of development are very

¹⁰ This result is based on a definition of regression as a series of at least a few higher levels followed by a series of at least a few lower levels, provided any higher level is absent at the moment of the regression. The exact number of regressions found depends, of course, on the meaning of "few" in this respect. I used two or more as criterion.

similar to those advanced by Goldin-Meadow et al. (1993), K. W. Fischer and Granott (1995) are less interested in the question of transition per se and more in the overall resulting pattern of development.

If the underlying principles are indeed similar, we expect that the dynamic model producing the local transition phenomena described by Goldin-Meadow et al. (1993) also produces the overall developmental pattern described by K. W. Fischer and Granott (1995). Given the figures of the developmental trajectories associated with the Goldin-Meadow et al. (1993) model, it is probably no big surprise that the overall process has indeed several of the qualitative properties described by Fischer and Granott, such as a global stagewise development, the existence of relatively broad ranges instead of narrow levels, the occurrence of multimodality (i.e., the simultaneous existence of actions at distinct developmental levels), and the occurrence of temporary regressions (see Figure 15).

A major difference between simulations based on the Goldin-Meadow et al. (1993) and the Granott (1996) models is that the former concentrated on the occurrence of local bimodality, which occurs in various places during a continued developmental process, whereas the latter focused on the entire microdevelopmental trajectory, which, in the experiments carried out by Granott, consisted of about 60 to 70 registered activities of dyads. The latter means that the simulation model must capture a qualitatively similar process with the same number of steps, each step thus representing a simulated action. The figures (e.g., Figure 15) are based on simulations with 100 steps, covering the entire range of possible levels. Note that there is considerably more variation in the results obtained with the simulation model than in the microdevelopmental data in that the simulated data range from two-step to multistep cases. In addition, only about half of the simulated patterns resemble the results of K. W. Fischer and Granott's (1995) empirical microdevelopmental patterns (11 out of 20 simulated cases; of the 11 cases, 5 showed the required broad range of fluctuation only partially). That is, the microgenetic pattern found in the empirical studies is probably only a subset of the patterns generated by this particular set of parameter values, given the stochastic nature of the process.

Conclusion and Discussion

The primary aim of the present study was to investigate whether a dynamic computational model implementing classic developmental principles could simulate processes of self-organization in which a number of characteristic quantitative developmental phenomena naturally emerge. These principles concern interactions between conservative and progressive forces, the first consolidating and magnifying the status quo, the others shaping the system in accordance with environmental inputs. The resulting phenomena concern the occurrence of both continuity and discontinuity in development, the succession of discontinuous shifts and temporary equilibria, and the occurrence of prolonged multimodality and variability. It was shown that the model in general produced those phenomena as emergent properties under parameter conditions that were supposed to match the empirical conditions under which they naturally occur. This simulation success notwithstanding, we should be aware of the fact that a large number of questions still remain unanswered.

Drawbacks and Potential Problems

The problem of the psychological reality of the parameters. In an earlier section, I introduced the problem of which psychological phenomena correspond with the abstract parameters distinguished in the model. I made several suggestions as to which empirical indicators could approximate those parameters. A major problem, which far exceeds the scope of this article, concerns the question by what mechanisms and processes the parameter values characteristic of real developmental processes are set and selected. Eventually, the problem is not one of the present model only but rather applies to any broad and abstract model of development that introduces general mechanisms that can eventually be specified for particular contexts, but maybe not in the form of context-free underlying psychological and environmental variables.

The problem of model falsification. Verification, validation, and confirmation of numerical simulation models pose complicated questions (Oreskes, Schrader-Frechette, & Belitz, 1994). One of the problems with dynamic simulation models containing a considerable number of parameters is that they are eventually overdetermined or overpowerful. That is, it is possible—though not necessarily likely—that they can simulate not only the patterns that have been found empirically but also patterns that do not occur in real situations. With models that have such a strong stochastic component as the one presented here, it is difficult to formally demonstrate that there are certain classes of outcomes they cannot produce. At any rate, such demonstration falls outside the scope of this article. On the other hand, if the model is capable of simulating any pattern that could eventually be empirically found, it runs the danger that it cannot in principle be falsified.

An answer to this criticism is that the model does indeed make specific predictions, namely specific developmental outcomes under specific parameter conditions, which correspond to empirically observable independent variables. It is therefore falsifiable under the predictions that it makes. As an example, let me take the simulation of the van der Maas (1993) conservation data. Because the simulation is post hoc, the model cannot be falsified by those data. However, the model claims that the results should be obtained under a specific set of parameter values and thus makes a number of specific predictions about independent variables that can in principle be tested in a replication of the van der Maas study. For instance, the model assumes that the computerized conservation tasks are new and unfamiliar to the children and require considerable processing time and effort at the start, which follow from the low initial weights with which the model starts. The model predicts that processing time and effort will decrease with repeated exercise and will increase again at the moment the child increases his or her understanding of the conservation principle and thus begins to increase his or her test score. Time and effort will again decrease as the child has more and more items correct at repeated test sessions. Next, the model predicts that there exists a moderately strong conservative tendency in those children who show the sudden jump during the series of repeated tests. All these predictions follow directly from the simulated levels and associated weights. Assuming that we have agreed on empirical indicators of conservative tendency with regard to the problem contexts

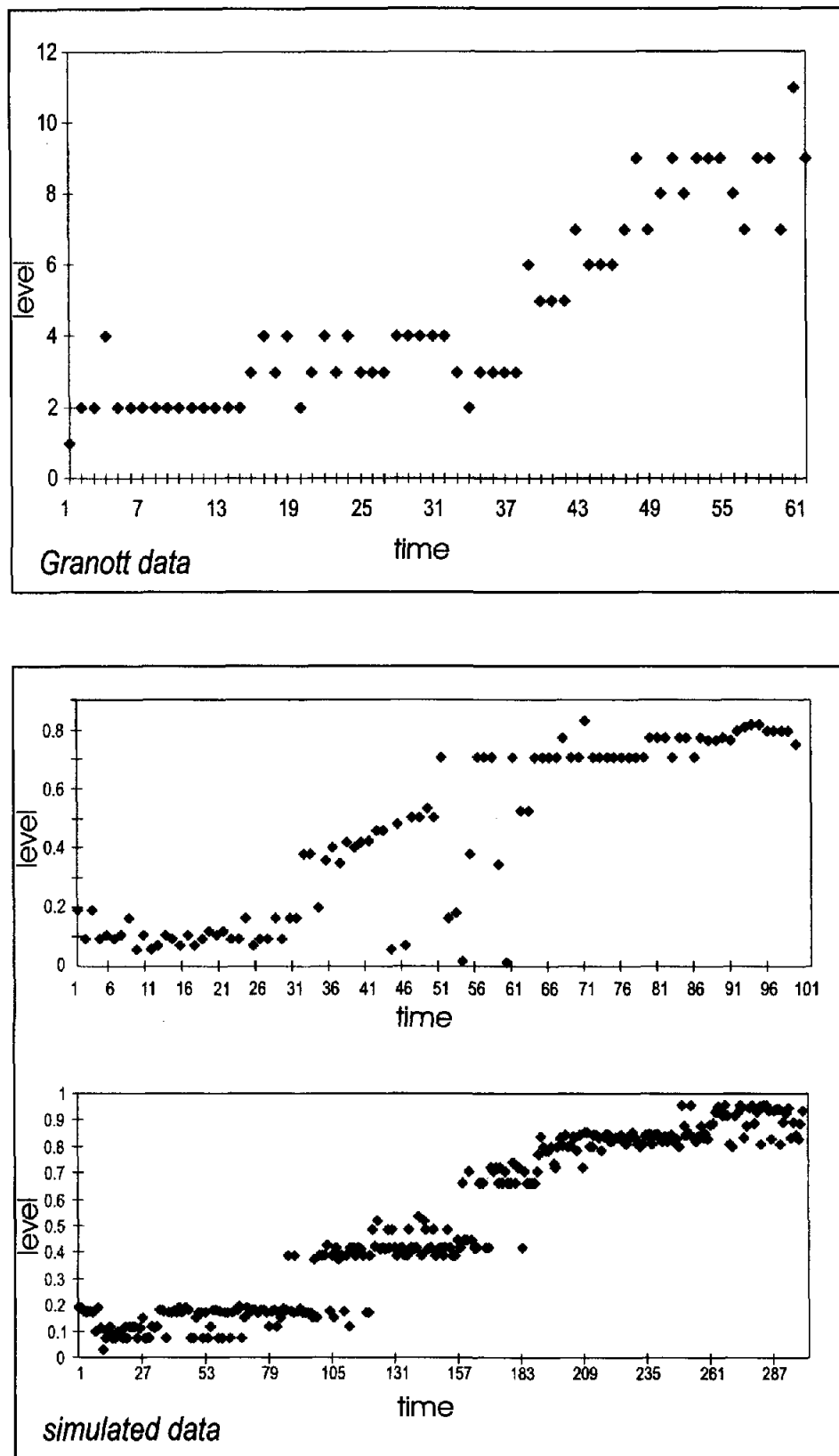


Figure 15. In K. W. Fischer and Granott's (1995) data from microdevelopmental problem-solving studies, the microdevelopmental pattern takes the form of a band of alternating but generally related levels (top). Simulations show that these bands occur more or less naturally, given the right parameter conditions; eventually, the developmental pattern may show considerable regressions (middle), which have also been found in the empirical studies.

presented in our test—like cognitive tenacity, coming back to earlier problems, solutions, and so forth—and assuming that the population under study shows a sufficiently broad distribution of those variables, we can make the following predictions. First, children who have high scores on the “conservation conservatism” variable will, on average, make significantly fewer shifts to conservation understanding during the test series than those with lower scores. If they do make the shift, it should be in the form of an evident discontinuous change. Second, we predict that children whose conservation conservatism levels are low will show patterns of increase that are significantly more gradual than those of children with higher conservatism levels. That is, we predict significantly more intermediate conservation test scores with the low conservative than with the moderate-to-high conservative children. If those predictions are not supported, the model can be considered falsified. It could nevertheless be “repaired” by making additional assumptions or by introducing additional parameters and principles. Although such ad hoc repairs could in principle be adequate and correct, they also contribute to making the model weaker and less persuasive (Oreskes et al., 1994). Finally, it should be noted that a dynamic model is not the endpoint of a research trajectory and that its contribution must be judged relative to other models and empirical data.

The problem of parameter estimation. If one specifies a mathematical model of a psychological phenomenon, it should be possible, first, to estimate the values of the parameters involved, given a particular empirical data set, and, second, to determine whether the parameters make a significant contribution to the explanation of the phenomenon. A considerable problem that often arises with complicated dynamic models is that it is far from trivial to show that they are instantiations of simpler models whose parameters can, in principle, be estimated. The way to proceed in these cases is to make plausible that the complicated model shows sufficient qualitative similarities with a specific canonical model to warrant a description of the complicated model by parameters borrowed from the simpler one (e.g., Thom, 1977). This means that parameter estimation, if possible, must be carried out in an indirect way, which is unfortunately far less rigorous than with regression models, for example.

This indirect approach to parameter estimation—which also relates to the preceding issue, namely that of falsification of the model—proceeds as follows. I again assume that we agree on a set of reasonably valid empirical indicators of classes of parameter conditions, for instance, that indicators for conservative versus progressive aspects of learning or developmental progress. Given certain such indicators in a particular learning or developmental process, we can set up a series of simulations based on corresponding sets of parameter values. That is, we use the model to predict a distribution of developmental trajectories by running repeated stochastic model simulations and by altering the parameter values within the confinements given by the empirical indicators (i.e., the independent variables). Next, it must be shown that this distribution of simulated trajectories has certain unique properties that do not occur in simulations based on parameter conditions that differ significantly from those found in our empirical sample. Examples of such properties are the occurrence of 2-, 3-, or n -modal distributions of levels, continuous S-shaped change, discontinuities character-

ized by inaccessible regions, and so forth. Standard fitting techniques can be used in order to test whether the data from the empirically found trajectories correspond to those predicted from the model simulations based on the specified parameters. For instance, we can use mixture distributions techniques to test for the occurrence of n -modal distributions (Everitt & Hand, 1981; see for examples Ruhland, 1998; van der Maas, 1993; Wimmers, 1996). In order to distinguish continuous from bimodal discontinuous models, stochastic catastrophe fitting techniques can be used (Cobb, 1978, 1981; Guastello, 1992, 1995, 1997; Hartelman, 1997; Hartelman, van der Maas, & Molenaar, 1998). Another technique for distinguishing continuity from discontinuity in the data is offered by the Saltus model (G. H. Fischer, 1992; Wilson, 1989). It goes without saying that this qualitative approximation of model fitting is still very different from a reliable, direct quantitative estimation of parameter values based on a mathematical and statistical theory of the data, but at the moment such approximation is probably as much as we may hope for.

In summary, if we have an empirical data set in which we can identify a number of independent variables that indirectly correspond to a specific range of parameter values, we can run simulations of the model to investigate whether these parameter values produce the empirically observed developmental pattern. If this is so and if we can show that within certain confinements there exists no alternative parameter set that produces a similar outcome, we can be confident that the current parameter set provides a reasonable characterization of the empirical data at issue. Note, however, that the latter requirement, which shows that there are no alternative parameter sets or models that do the same job, is notoriously difficult to achieve (Oreskes et al., 1994, related this problem to the issue of nonuniqueness or underdetermination of models).

The Potential Contribution of Dynamic Modeling to Developmental Theory Building

With regard to theory building, one implication of the current attempt at building a dynamic model is that it makes sense to look both at the old and the new theories with the aim of integrating their most basic and generalized principles. The present attempt at doing so showed that such integration is possible and that the results are in accordance with a great variety of empirical phenomena found by taking different theoretical models as a starting point.

The second implication is that the present attempt at building an integrative model has proven the possibility of transforming a basic conceptual model into a computational model. The computational model shows that, under stochastic conditions, the major empirical predictions and findings of the original theories and models result as a consequence of the self-organizational principles governing the dynamic model.

Third, the present dynamic model specifies various empirical predictions, for instance, predictions regarding the outcomes of processes with properties such as different strengths of conservative versus progressive (environment-oriented) tendencies in learning, different ways of adapting the environmental input to the observable developmental level of children, and so forth. Given the complexity of the interactions involved in develop-

mental processes, the dynamic model functions as a deductive procedure, a modeling tool without which the inference of specific predictions out of constellations of possible empirical conditions would be hardly possible. Because the dynamic model pretends to be more than just a neutral modeling tool, but also contains the fundamentals of an integrative developmental theory, it can be falsified by falsifying its empirical predictions.

Finally, it goes without saying that the present model is only an initial attempt at bringing together general developmental principles in a dynamic framework in order to see whether they do indeed lead to a first approximation of a number of developmental phenomena that, however limited, deserve to be further explored.

References

- Alexander, J. M., Carr, M., & Schwanenflugel, P. J. (1995). Development of metacognition in gifted children: Directions for future research. *Developmental Review, 15*, 1–37.
- Alibali, M. W., & Goldin-Meadow, S. (1993). Gesture-speech mismatch and mechanisms of learning: What the hands reveal about a child's state of mind. *Cognitive Psychology, 25*, 468–523.
- Aljaafreh, A., & Lantolf, J. P. (1994). Negative feedback as regulation and second language learning in the zone of proximal development: Sociocultural theory and second language learning [Special issue]. *Modern Language Journal, 78*, 465–483.
- Au, T. K., Sidle, A. L., & Rollins, K. B. (1993). Developing an intuitive understanding of conservation and contamination: Invisible particles as a plausible mechanism. *Developmental Psychology, 29*, 286–299.
- Baillargeon, R. (1986). Representing the existence and the location of hidden objects: Object permanence in 6- and 8-month-old infants. *Cognition, 23*, 21–41.
- Baillargeon, R. (1987a). Object permanence in 3 1/2- and 4 1/2-month-old infants. *Developmental Psychology, 23*, 655–664.
- Baillargeon, R. (1987b). Young infants' reasoning about the physical and spatial properties of a hidden object. *Cognitive Development, 2*, 179–200.
- Baillargeon, R. (1991). Reasoning about the height and location of a hidden object in 4.5- and 6.5-month-old infants. *Cognition, 38*, 13–42.
- Baillargeon, R., & DeVos, J. (1991). Object permanence in young infants: Further evidence. *Child Development, 62*, 1227–1246.
- Baillargeon, R., & Graber, M. (1987). Where's the rabbit? 5.5-month-old infants' representation of the height of a hidden object. *Cognitive Development, 2*, 375–392.
- Baillargeon, R., Graber, M., DeVos, J., & Black, J. (1990). Why do young infants fail to search for hidden objects? *Cognition, 36*, 255–284.
- Baillargeon, R., Spelke, E. S., & Wasserman, S. (1985). Object permanence in five-month-old infants. *Cognition, 20*, 191–208.
- Bak, P., & Chen, K. (1991, January). Self-organizing criticality. *Scientific American, 265*(1), 46–53.
- Bak, P., & Sneppen, K. (1993). Punctuated equilibrium and criticality in a simple model of evolution. *Physical Review Letters, 71*, 4083–4086.
- Bechtel, W., & Abrahamsen, A. (1991). *Connectionism and the mind*. Oxford, England: Basil Blackwell.
- Beilin, H. (1994). Mechanisms in the explanation of developmental change. In H. W. Reese (Ed.), *Advances in child development and behavior* (Vol. 25, pp. 327–352). San Diego, CA: Academic Press.
- Bell, M. A., & Fox, N. A. (1994). Brain development over the first year of life: Relations between electroencephalographic frequency and coherence and cognitive and affective behaviors. In G. Dawson & K. Fischer (Eds.), *Human behavior and the developing brain* (pp. 314–345). New York: Guilford Press.
- Bentler, P. M. (1970). Evidence regarding stages in the development of conservation. *Perceptual and Motor Skills, 31*, 855–859.
- Bernicot, J. (1994). Speech acts in young children: Vygotsky's contribution. Learning and development: Contributions from Vygotsky [Special issue]. *European Journal of Psychology of Education, 9*, 311–319.
- Best, D. L. (1993). Inducing children to generate mnemonic organizational strategies: An examination of long-term retention and materials. *Developmental Psychology, 29*, 324–336.
- Bidell, T. R. (1988). Vygotsky, Piaget, and the dialectic of development. *Human Development, 31*, 329–345.
- Bidell, T. R. (1990). *Mechanisms of cognitive development in problem solving: A structural integration approach*. Unpublished doctoral dissertation, Harvard University.
- Bidell, T. R., & Fischer, K. W. (1992). Beyond the stage debate: Action, structure and variability in Piagetian theory and research. In R. Sternberg & C. Berg (Eds.), *Intellectual development* (pp. 100–140). New York: Cambridge University Press.
- Bidell, T. R., & Fischer, K. W. (1994). Developmental transitions in children's early on-line planning. In M. M. Haith, J. B. Benson, R. J. Roberts, & B. F. Pennington (Eds.), *The development of future-oriented processes* (pp. 141–176). Chicago: University of Chicago Press.
- Bigelow, A. E. (1986). The development of reaching in blind children. *British Journal of Developmental Psychology, 4*, 355–366.
- Bigelow, A. (1990). Relationship between the development of language and thought in young blind children. *Journal of Visual Impairment and Blindness, 84*, 414–419.
- Bigelow, A. E., MacDonald, D., & MacDonald, L. (1995). The development of infants' search for their mothers, unfamiliar people, and objects. *Merrill Palmer Quarterly, 41*, 191–208.
- Bijstra, J., Van Geert, P., & Jackson, S. (1989). Conservation and the appearance–reality distinction: What do children really know and what do they answer? *British Journal of Developmental Psychology, 7*, 43–53.
- Bjork, E. L., & Cummings, E. M. (1984). Infant search errors: Stage of concept development or stage of memory development. *Memory and Cognition, 12*, 1–19.
- Block, C. C. (1993). Strategy instruction in a literature-based reading program: Strategies instruction [Special issue]. *Elementary School Journal, 94*, 139–151.
- Boerlijst, M., & Hogeweg, P. (1992). Self-structuring and selection: Spiral waves as a substrate for prebiotic evolution. In C. G. Langton, C. Taylor, J. D. Farmer, & S. Rasmussen (Eds.), *Artificial life II* (pp. 255–276). Redwood City, CA: Addison-Wesley.
- Bogartz, R. S. (1994). The future of dynamic systems models in developmental psychology in the light of the past. *Journal of Experimental Child Psychology, 58*(2) 289–319.
- Boom, J. (1991). Collective development and the learning paradox. *Human Development, 34*, 273–287.
- Boom, J. (1994). Developmental patterns and the individual. *Tijdschrift voor Ontwikkelingspsychologie, 21*, 225–238.
- Brainerd, C. J. (1978). The stage question in cognitive developmental theory. *Behavioral and Brain Sciences, 1*, 173–182.
- Brainerd, C. J. (1985). Three-state models of memory development: A review of recent advances in statistical methodology. *Journal of Experimental Child Psychology, 40*, 375–394.
- Brainerd, C. J. (1993). Cognitive development is abrupt (but not stage-like). *Monographs of the Society for Research in Child Development, 58*(9, Serial No. 237), 179–190.
- Brainerd, C. J., Howe, M. L., & Desrochers, A. (1982). The general

- theory of two-stage learning: A mathematical review with illustrations from memory development. *Psychological Bulletin*, 91, 634–665.
- Bremner, J. G., & Knowles, L. S. (1984). Piagetian Stage IV search errors with an object that is directly accessible both visually and manually. *Perception*, 13, 307–314.
- Breuer, K. H. (1985). Intentionality of perception in early infancy. *Human Development*, 28, 71–83.
- Brissaud, R. (1994). Teaching and development: Solving “missing addend” problems using subtraction. Learning and development: Contributions from Vygotsky [Special issue]. *European Journal of Psychology of Education*, 9, 343–365.
- Brown, A. L. (1990). Domain-specific principles affect learning and transfer in children. *Cognitive Science*, 14, 107–133.
- Cameron, P. A., & Gallup, G. G. (1988). Shadow recognition in human infants. *Infant Behavior and Development*, 11, 465–471.
- Campione, J. C., Brown, A. L., Ferrara, R. A., & Bryant, N. R. (1984). The Zone of Proximal Development: Implications for individual differences and learning. In B. Rogoff & J. V. Wertsch (Eds.), *Children's learning in the "Zone of Proximal Development"* (pp. 45–64). San Francisco: Jossey-Bass.
- Camras, L. A. (1992). Expressive development and basic emotions. *Cognition and Emotion*, 6(3–4), 269–283.
- Carpenter, G. A. (1991). Neural network models for pattern recognition and associative memory. In G. A. Carpenter & S. Grossberg (Eds.), *Pattern recognition by self-organizing neural networks* (pp. 1–34). Cambridge, MA: Bradford Books with MIT Press.
- Case, R. (1985). *Intellectual development from birth to adulthood*. New York: Academic Press.
- Case, R. (1990). Neo-Piagetian theories of child development. In R. Sternberg & C. Berg (Eds.), *Intellectual development* (pp. 161–196). New York: Cambridge University Press.
- Case, R. (1992a). *The mind's staircase: Exploring the conceptual underpinnings of children's thought and knowledge*. Hillsdale, NJ: Erlbaum.
- Case, R. (1992b). Neo-Piagetian theories of child development. In Sternberg, R. J. & C. A. Berg (Eds.), *Intellectual development* (pp. 161–196). New York: Cambridge University Press.
- Case, R., & Okamoto, Y. (1996). The role of central conceptual structures in the development of children's thought. *Monographs of the Society for Research in Child Development*, 61(1–2, Serial No. 246).
- Chandler, M. J., & Lalonde, C. E. (1994). Surprising, magical and miraculous turns of events: Children's reactions to violations of their early theories of mind and matter: Magic [Special issue]. *British Journal of Developmental Psychology*, 12, 83–95.
- Chapman, M. (1988). *Constructive evolution*. New York: Cambridge University Press.
- Chapman, M. (1992). Equilibration and the dialectics of organization. In H. Beilin & P. Pufall (Eds.), *Piaget's theory: Prospects and possibilities* (pp. 34–59). Hillsdale, NJ: Erlbaum.
- Chomsky, N. (1980). *Rules and representations*. Oxford, England: Basil Blackwell.
- Cobb, L. (1978). Stochastic catastrophe models and multimodal distributions. *Behavioral Science*, 23, 360–373.
- Cobb, L. (1981). Parameter estimation for the cusp catastrophe model. *Behavioral Science*, 26, 75–78.
- Cooney, J. B., & Constan, M. A. (1993). On determining the dimension of discontinuities in human learning and development. In M. L. Howe & R. Pasnak (Eds.), *Emerging themes in cognitive development. Vol. I: Foundations* (pp. 154–191). New York: Springer.
- Cooney, J. B., & Troyer, R. (1994). A dynamic model of reaction time in a short-term memory task. *Journal of Experimental Child Psychology*, 58, 200–226.
- Cossette-Ricard, M. (1986). *Le décalage dans le développement de la notion d'objet: Mobilité et immobilité de l'objet*. [Décalage in the development of the object concept: Mobility and immobility of the object]. *Canadian Journal of Psychology*, 40, 272–281.
- Dasen, P. R. (1974). Cross-cultural Piagetian psychology: A summary. In J. W. Berry & P. R. Dasen (Eds.), *Culture and cognition: Readings in cross-cultural psychology* (Vol. 1, pp. 409–423). London: Academic Press.
- Dasen, P. R., & Heron, A. (1981). Cross-cultural tests of Piaget's theory. In H. C. Triandis & W. W. Lambert (Eds.), *Handbook of cross-cultural psychology* (Vol. 4, pp. 295–341). Boston: Allyn & Bacon.
- Dawson, G., & Fischer, K. W. (1994). *Human behavior and the developing brain* (pp. 314–345). New York: Guilford Press.
- De Guerrero, M. C. M., & Villamil, O. S. (1994). Social-cognitive dimensions of interaction in L2 peer revision. *Sociocultural theory and second language learning* [Special issue]. *Modern Language Journal*, 78(4), 484–496.
- De Sapio, R. (1978). *Calculus for the life sciences*. San Francisco: Freeman.
- Diamond, A. (1985). Development of the ability to use recall to guide action, as indicated by infants' performance on AB. *Child Development*, 56, 868–883.
- Diaz, R. M., Neal, C. J., & Vachio, A. (1991). Maternal teaching in the zone of proximal development: A comparison of low- and high-risk dyads. *Merrill Palmer Quarterly*, 37, 83–107.
- Dromi, E. (1987). *Early lexical development*. New York: Cambridge University Press.
- Eames, D., Shorrocks, D., & Tomlinson, P. (1990). Naughty animals or naughty experimenters? Conservation accidents revisited with video-stimulated commentary. *British Journal of Developmental Psychology*, 8, 25–37.
- Eckerman, C. O., & Stein, M. R. (1990). How imitation begets imitation and toddlers' generation of games. *Developmental Psychology*, 26, 370–378.
- Elbers, E. (1986a). Children's theories and developmental theory. In P. van Geert (Ed.), *Theory building in developmental psychology* (pp. 365–403). Amsterdam: North Holland.
- Elbers, E. (1986b). Interaction and instruction in the conservation experiment. *European Journal of Psychology of Education*, 1, 77–89.
- Elbers, E., Wiegersma, S., Brand, N., & Vroon, P. A. (1991). Response alternation as an artifact in conservation research. *Journal of Genetic Psychology*, 152, 47–55.
- Elman, J. L. (1993). Learning and development in neural networks: The importance of starting small. *Cognition*, 48, 71–99.
- Elman, J. L. (1995). Language as a dynamical system. In R. F. Port & T. Van Gelder (Eds.), *Mind as motion: Explorations in the dynamics of cognition* (pp. 195–226). Cambridge, MA: Bradford Book with MIT Press.
- Elman, J. L., Bates, E. A., Johnson, M. H., Karmiloff-Smith, A., Parisi, D., & Plunkett, K. (1996). *Rethinking innateness: A connectionist perspective on development*. Cambridge, MA: Bradford Books with MIT Press.
- Erikson, E. (1950). *Childhood and society*. New York: Norton.
- Everitt, B. S., & Hand, D. J. (1981). *Finite mixture distributions*. London: Chapman & Hall.
- Fischer, G. H. (1992). The “Saltus model” revisited. *Methodika*, 6, 87–98.
- Fischer, G. H., & Molenaar, I. W. (1995). *Rasch models: Foundations, recent developments and applications*. New York: Springer.
- Fischer, K. W. (1980a). Learning and problem solving as the development of organized behavior. *Journal of Structural Learning*, 6, 253–267.
- Fischer, K. W. (1980b). A theory of cognitive development: The control and construction of hierarchies of skills. *Psychological Review*, 87, 477–531.
- Fischer, K. W., & Bidell, T. R. (1997). Dynamic development of psycho-

- logical structures in action and thought. In R. M. Lerner (Ed.) & W. Damon (Series Ed.), *Handbook of child psychology: Vol 1: Theoretical models of human development* (5th ed., pp. 467–561). New York: Wiley.
- Fischer, K. W., Bullock, D., Rotenberg, E. J., & Raya, P. (1993). The dynamics of competence: How context contributes directly to skill. In R. H. Wozniak & K. W. Fischer (Eds.), *Development in context: Acting and thinking in specific environments* (pp. 93–117). Hillsdale, NJ: Erlbaum.
- Fischer, K. W., & Granott, N. (1995). Beyond one-dimensional change: Multiple, concurrent, socially distributed processes in learning and development. *Human Development*, 38(6), 302–314.
- Fischer, K. W., Hand, H. H., & Russell, S. L. (1984). The development of abstractions in adolescence and adulthood. In M. Commons, F. A. Richards, & C. Armon (Eds.), *Beyond formal operations* (pp. 43–73). New York: Praeger.
- Fischer, K. W., & Kennedy, B. P. (1996). Tools for analyzing the many shapes of development. The case of self-in-representations in Korea. In A. Renninger & E. Amsel (Eds.), *Processes of development* (pp. 117–152). Hillsdale, NJ: Erlbaum.
- Fischer, K. W., Knight, C. C., & Van-Parys, M. (1993). Analyzing diversity in developmental pathways: Methods and concepts. In R. Case & W. Edelstein (Eds.), *The new structuralism in cognitive development: Theory and research on individual pathways. Contributions to human development* (Vol. 23, pp. 33–56). Basel, Switzerland: S. Krager.
- Fischer, K. W., & Pipp, S. L. (1984). Processes of cognitive development: Optimal level and skill acquisition. In R. J. Sternberg (Ed.), *Mechanisms of cognitive development* (pp. 45–80). New York: Freeman.
- Fischer, K. W., Pipp, S. L., & Bullock, D. (1984). Detecting discontinuities in development: Methods and measurement. In R. N. Emde & R. Harmon (Eds.), *Continuities and discontinuities in development* (pp. 95–121). Norwood, NJ: Ablex.
- Fischer, K. W., & Rose, S. P. (1994). Development of coordination of components in brain and behavior: A framework for theory and research. In G. Dawson & K. W. Fischer (Eds.), *Human behavior and the developing brain* (pp. 3–66). New York: Guilford.
- Flavell, J. H. (1963). *The developmental psychology of Jean Piaget*. New York: Van Nostrand.
- Flavell, J. H. (1982). On cognitive development. *Child Development*, 53, 1–10.
- Flavell, J. H. (1986). The development of children's knowledge about the appearance–reality distinction. *American Psychologist*, 41, 418–425.
- Flavell, J. H., & Wohlwill, J. F. (1969). Formal and functional aspects of cognitive development. In D. Elkind & J. H. Flavell (Eds.), *Studies in cognitive development: Studies in honour of J. Piaget* (pp. 67–120). New York: Oxford University Press.
- Fodor, J. A. (1975). *Language of thought*. New York: Crowell.
- Fogel, A. (1993). *Developing through relationships: Origins of communication, self and culture*. New York: Harvester Wheatsheaf.
- Fogel, A., Nwokah, E., Dedo, J. Y., & Messinger, D. (1992). Social process theory of emotion: A dynamic systems approach. *Social Development*, 1, 122–142.
- Fogel, A., & Reimers, M. (1989). On the psychobiology of emotions and their development. *Monographs of the Society for Research in Child Development*, 54(1–2, Serial No. 219) 105–113.
- Fogel, A., & Thelen, E. (1987). Development of early expressive and communicative action: Reinterpreting the evidence from a dynamic systems perspective. *Developmental Psychology*, 23, 747–761.
- Gal'perin, P. (1969). Stages in the development of mental acts. In M. Cole & J. Maltzman (Eds.), *A handbook of contemporary Soviet psychology* (pp. 249–273). New York: Basic Books.
- Galpert, L., & Dockrell, J. (1995). Is understanding the experimenter's intention the clue to conservation ability? *International Journal of Behavioral Development*, 18, 505–517.
- Garcia, R. (1992). The structure of knowledge and the knowledge of structure. In H. Beilin & P. Pufall (Eds.), *Piaget's theory: Prospects and possibilities* (pp. 21–38). Hillsdale, NJ: Erlbaum.
- Gesell, A. (1954). The ontogenesis of infant behavior. In L. Carmichael (Ed.), *Manual of child psychology* (pp. 335–373). New York: Wiley.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston: Houghton Mifflin.
- Gillmore, R. (1981). *Catastrophe theory for scientists and engineers*. New York: Wiley.
- Glassman, M. (1994). All things being equal: The two roads of Piaget and Vygotsky. *Developmental Review*, 14, 186–214.
- Goldfield, B. A., & Reznick, J. S. (1990). Early lexical acquisition: Rate, content and the vocabulary spurt. *Journal of Child Language*, 17, 171–183.
- Goldin-Meadow, S., Alibali, M. W., & Breckinridge Church, R. (1993). Transitions in concept acquisition: Using the hand to read the mind. *Psychological Review*, 100, 279–297.
- Granott, N. (1993). Patterns of interaction in the co-construction of knowledge: Separate minds, joint efforts and weird creatures. In R. H. Wozniak & K. Fischer (Eds.), *Development in context: Acting and thinking in specific environments* (pp. 183–207). Hillsdale, NJ: Erlbaum.
- Granott, N. (1996). *The dynamics of problem solving: Rediscovery, variability and the complexity of making sense*. Manuscript submitted for publication.
- Griffin, P., & Cole, M. (1984). Current activity for the future: The Zoped. In B. Rogoff, & J. V. Wertsch (Eds.), *Children's learning in the "Zone of Proximal Development"* (pp. 45–64). San Francisco: Jossey-Bass.
- Grossberg, S. (1991). Nonlinear neural networks: Principles, mechanisms and architectures. *Neural Networks*, 1, 17–61.
- Gruber, H. E., & Vonèche, J. J. (1977). *The essential Piaget*. London: Routledge & Kegan Paul.
- Guastello, S. J. (1992). Clash of the paradigms: A critique of an examination of the polynomial regression technique for evaluating catastrophe theory hypotheses. *Psychological Bulletin*, 111, 375–379.
- Guastello, S. J. (1995). *Chaos, catastrophe, and human affairs: Applications of nonlinear dynamics to work, organizations, and social evolution*. Mahwah, NJ: Erlbaum.
- Guastello, S. J. (1997). Scientific insight need not conflict with scientific rigor. *Psychological Inquiry*, 8, 129–132.
- Guthke, J., & Stein, H. (1996). Are learning tests the better version of intelligence tests? *European Journal of Psychological Assessment*, 12, 1–13.
- Haake, R. J., & Somerville, S. C. (1985). Development of logical search skills in infancy. *Developmental Psychology*, 21, 176–186.
- Hanrahan, J. P., Yelin, M., & Rapagna, S. (1987). The effects of test materials and the order of presentation of the materials on young children's understanding of conservation of number. *British Journal of Educational Psychology*, 57, 407–411.
- Hare, M., & Elman, J. L. (1995). Learning and morphological change. *Cognition*, 56, 61–98.
- Hartelman, P. (1997). *Stochastic catastrophe theory*. Unpublished doctoral dissertation, University of Amsterdam.
- Hartelman, P., van der Maas, H., & Molenaar, P. (1998). Detecting and modelling developmental transitions. *British Journal of Developmental Psychology*, 16, 97–122.
- Hodges, R. M., & French, L. A. (1988). The effect of class and collection labels on cardinality, class-inclusion, and number conservation tasks. *Child Development*, 59(5), 1387–1396.
- Holland, J. H. (1975). *Adaptation in natural and artificial systems*. Ann Arbor, MI: University of Michigan Press.

- Holzman, L. (1995). Creating developmental learning environments: A Vygotskian practice. Lev S. Vygotsky and contemporary school psychology [Special issue]. *School Psychology International*, 16, 199–212.
- Hood, B., & Willatts, P. (1986). Reaching in the dark to an object's remembered position: Evidence for object permanence in 5-month-old infants. *British Journal of Developmental Psychology*, 4, 57–65.
- Hopkins, B., & van der Maas, H. (1998). Developmental transitions: So what's new? *British Journal of Developmental Psychology*, 16, 1–13.
- Horobin, K., & Acredolo, L. (1986). The role of attentiveness, mobility history, and separation of hiding sites on Stage IV search behavior. *Journal of Experimental Child Psychology*, 41, 114–127.
- Hosenfeld, B., van der Maas, H. L. J., & van den Boom, D. C. (1997a). Detecting bimodality in the analogical reasoning performance of elementary schoolchildren. *International Journal of Behavioral Development*, 20, 529–547.
- Hosenfeld, B., van der Maas, H. L. J., & van den Boom, D. C. (1997b). Indicators of discontinuous change in the development of analogical reasoning. *Journal of Experimental Child Psychology*, 64, 367–395.
- Howe, M. L., & Rabinowitz, F. M. (1994). Dynamic modeling, chaos, and cognitive development. *Journal of Experimental Child Psychology*, 58, 184–199.
- Hudspeth, W. J., & Pribram, K. H. (1992). Psychophysiological indices of cerebral maturation. *International Journal of Psychophysiology*, 12, 19–29.
- Jackson, E. A. (1991). *Perspectives of nonlinear dynamics* (Vols. 1–2). New York: Cambridge University Press.
- Johnson, S. P., & Aslin, R. N. (1995). Perception of object unity in 2-month-old infants. *Developmental Psychology*, 31, 739–745.
- Johnson, S. P., & Nanez, J. E. (1995). Young infants' perception of object unity in two-dimensional displays. *Infant Behavior and Development*, 18, 133–143.
- Juckes, T. J. (1991). Equilibration and the learning paradox. *Human Development*, 34, 261–272.
- Karmiloff-Smith, A. (1986). Stage/structure versus phase/process in modelling linguistic and cognitive development. In I. Levin (Ed.), *Stage and structure: Reopening the debate* (pp. 164–190). Norwood, NJ: Ablex.
- Karmiloff-Smith, A. (1996, September). *Contrasting approaches to infancy: Behaviorism, Piagetianism, nativism, connectionism*. Paper presented at the Centennial Conference of Jean Piaget's Birth, The Growing Mind, Geneva, Switzerland.
- Kauffman, S. A. (1992). *The origins of order: Self-organization and selection in evolution*. Oxford, England: Oxford University Press.
- Kellman, P. J., & Shipley, T. F. (1991). A theory of visual interpolation in object perception. *Cognitive Psychology*, 23, 141–221.
- Kitchener, K. S., Lynch, C. L., Fischer, K. W., & Wood, P. K. (1993). Developmental range of reflective judgment. The effect of contextual support and practice on developmental stage. *Developmental Psychology*, 29, 893–906.
- Klahr, D. (1995). Computational models of cognitive change: The state of the art. In T. S. Simon & G. S. Halford (Eds.), *Developing cognitive competence: New approaches to process modeling* (pp. 355–375). Hillsdale, NJ: Erlbaum.
- Klahr, D., & Wallace, J. G. (1976). *Cognitive development: An information processing approach*. Hillsdale, NJ: Erlbaum.
- Klausmeier, H. J. (1976). Conceptual development during the school years. In J. T. Levin & V. L. Allen (Eds.), *Cognitive learning in children: Theories and strategies* (pp. 5–29). New York: Academic Press.
- Kreindler, D. M., & Lumsden, C. J. (1994). Extracting a narrative's causal gist: A modeling study based on fuzzy trace theory. *Journal of Experimental Child Psychology*, 58, 227–251.
- Kruschke, J. K. (1996). Dimensional relevance shifts in category learning. *Connection Science Journal of Neural Computing, Artificial Intelligence and Cognitive Research*, 8(2), 225–247.
- Kuhn, D., Garcia-Mila, M., Zohar, A., & Andersen, C. (1995). Strategies of knowledge acquisition. *Monographs of the Society for Research in Child Development*, 60(4, Serial No. 245, 4–128).
- Langton, C. G. (1992). Life at the edge of chaos. In C. G. Langton, C. Taylor, J. D. Farmer, & S. Rasmussen (Eds.), *Artificial life II* (pp. 41–91). Redwood City, CA: Addison-Wesley.
- Lautrey, J., Mullet, E., & Paques, P. (1989). Judgments of quantity and conservation of quantity: The area of a rectangle. *Journal of Experimental Child Psychology*, 47, 193–209.
- Legerstee, M. (1994). The role of familiarity and sound in the development of person and object permanence. *British Journal of Developmental Psychology*, 12, 455–468.
- Levin, I. (1986). *Stage and structure: Reopening the debate*. Norwood, NJ: Ablex.
- Lewis, M. D. (1994). Cognition–emotion feedback and the self-organization of developmental paths. *Human Development*, 38, 71–102.
- Lewis, M. D. (1996). Self-organising cognitive appraisals. *Cognition and Emotion*, 10(1), 1–25.
- Lewis, M. D., & Ash, A. J. (1992). Evidence for a neo-Piagetian stage transition in early cognitive development. *International Journal of Behavioral Development*, 15(3), 337–358.
- Lewis, M. D., & Douglas, L. (1998). A dynamic systems approach to cognition–emotion interactions in development. In F. Mascolo & S. Griffin (Eds.), *What develops in emotional development* (pp. 159–188). New York: Plenum.
- Lidz, C. S. (1995). Dynamic assessment and the legacy of L. S. Vygotsky: Lev S. Vygotsky and contemporary school psychology [Special issue]. *School Psychology International*, 16, 143–153.
- Light, P. H., Gorsuch, C., & Newman, J. (1987). “Why do you ask?” Context and communication in the conservation task. *European Journal of Psychology of Education*, 2(1), 73–82.
- Light, P., & Simmons, B. (1983). The effects of a communication task upon the representation of depth relationships in young children's drawings. *Journal of Experimental Child Psychology*, 35(1), 81–92.
- Lillard, A. S. (1993). Pretend play skills and the child's theory of mind. *Child Development*, 64, 348–371.
- Lingle, K. M., & Lingle, J. H. (1981). Effects of selected object characteristics on object-permanence test performance. *Child Development*, 52, 367–369.
- Lister, C., Leach, C., & Walsh, M. (1989). The development of conservation concepts in children with visual impairment. *British Journal of Educational Psychology*, 59(2), 211–219.
- Lourenço, O., & Machado, A. (1996). In defense of Piaget's theory: A reply to 10 common criticisms. *Psychological Review*, 103, 143–164.
- Lumpe, A. T., & Staver, J. R. (1995). Peer collaboration and concept development: Learning about photosynthesis. *Journal of Research in Science Teaching*, 32, 71–98.
- Macnamara, J., & Austin, G. (1993). Physics and plasticine. *Canadian Psychology*, 34(3), 225–232.
- Madden, J. (1986). The effects of schemes on children's drawings of the results of transformations. *Child Development*, 57(4), 924–933.
- Marcus, G. F., Pinker, S., Ullman, M., Hollander, M., Rosen, J. T., & Xu, F. (1992). Overregularization in language acquisition. *Monographs of the Society for Research in Child Development*, 57(4, Serial No. 228).
- Marshall, D., Plunkett, K., & Harris, P. (1995). Developing object performance: A connectionist model. In J. D. Moore & J. F. Lehman (Eds.), *Proceedings of the seventeenth annual conference of the Cognitive Science Society*. Mahwah, NJ: Erlbaum.
- Matousek, M., & Petersen, I. (1973). Frequency analysis of the EEG in normal children and adolescents. In P. Kellaway & I. Petersen

- (Eds.), *Automation of clinical electroencephalography* (pp. 75–102). New York: Raven Press.
- McCall, R. B., Eichorn, D. H., & Hogarty, P. S. (1977). Transitions in early mental development. *Monographs of the Society for Research in Child Development*, 42(3, Serial No. 171).
- McCune, L. (1995). A normative study of representational play in the transition to language. *Developmental Psychology*, 31, 198–206.
- McEvoy, J., & O'Moore, A. M. (1991). Number conservation: A fair assessment of numerical understanding? *Irish Journal of Psychology*, 12, 325–337.
- McMahon, T. A., & Bonner, J. T. (1983). *On size and life*. New York: Freeman.
- Meins, E. (1997). Security of attachment and maternal tutoring strategies: Interaction within the zone of proximal development. *British Journal of Developmental Psychology*, 15(2), 129–144.
- Messick, D. M., & Liebrand, W. B. G. (1996). The new dynamical social psychology. *Psychological Inquiry*, 8(2), 135–137.
- Metcalfe, J. (1993). Novelty monitoring, metacognition, and control in a composite holographic associative recall model: Implications for Korsakoff amnesia. *Psychological Review*, 100, 3–22.
- Miller, K. F. (1989). Measurement as a tool for thought: The role of measuring procedures in children's understanding of quantitative invariance. *Developmental Psychology*, 25, 589–600.
- Moll, I. (1994). Reclaiming the natural line in Vygotsky's theory of cognitive development. *Human Development*, 37(6), 333–342.
- Moore, C., & Frye, D. (1986). The effect of experimenter's intention on the child's understanding of conservation. *Cognition*, 22(3), 283–298.
- Nowak, A., Szamrej, J., & Latane, B. (1990). From private attitude to public opinion: A dynamic theory of social impact. *Psychological Review*, 97, 362–376.
- Oreskes, N., Shrader-Frechette, K., & Belitz, K. (1994). Verification, validation and confirmation of numerical models in the earth sciences. *Science*, 263, 641–646.
- Pacifici, C., & Bearison, D. J. (1991). Development of children's self-regulations in idealized and mother-child interactions. *Cognitive Development*, 6(3), 261–277.
- Parisi, D. (1996, September). *Piaget and artificial life*. Paper presented at the Centennial Conference of Jean Piaget's Birth, The Growing Mind, Geneva, Switzerland.
- Pellegrini, A. D., Perlmuter, J. C., Galda, L., & Brody, G. H. (1990). Joint reading between Black Head Start children and their mothers: Minority children [Special issue]. *Child Development*, 61(2), 443–453.
- Perry, M., Church, R. B., & Goldin-Meadow, S. (1988). Transitional knowledge in the acquisition of concepts. *Cognitive Development*, 3(4), 359–400.
- Peters, J. M. (1996). Paired keyboards as a tool for Internet exploration of third grade students. *Journal of Educational Computing Research*, 14(3), 229–242.
- Piaget, J. (1947). *La psychologie de l'intelligence* [The psychology of intelligence]. Paris: Colin.
- Piaget, J. (1952). *The origins of intelligence in children* (M. Cook, Trans.). New York: International University Press. (Original work published 1935)
- Piaget, J. (1954). *The construction of reality in the child* (M. Cook, Trans.). New York: Basic Books. (Original work published 1937)
- Piaget, J. (1955). Les stades du développement intellectuel de l'enfant et de l'adolescent [The stages of intellectual development in the child and the adolescent]. In P. Osterrieth, J. Piaget, R. De Saussure, J. M. Tanner, H. Wallon, R. Zazzo, B. Inhelder, & A. Rey (Eds.), *Le Problème des stades en psychologie de l'enfant* (pp. 33–42). Paris: Presses Universitaires de France.
- Piaget, J. (1957). Logique et équilibre dans le comportement du sujet [Logic and equilibrium in the behavior of the subject]. In L. Apostel, B. Mandelbrot, & J. Piaget (Eds.), *Logique et Equilibre. Etudes d'Epistémologie génétique* (Vol. 2, pp. 27–117). Paris: Presses Universitaires de France.
- Piaget, J. (1964). *Six études de psychologie* [Six studies in psychology]. Genève, Switzerland: Gonthier.
- Piaget, J. (1970a). *Epistémologie génétique* [Genetic epistemology]. Paris: Presses Universitaires de France.
- Piaget, J. (1970b). Piaget's theory. In P. H. Mussen (Ed.), *Carmichael's manual of child psychology* (Vol. 1, 3rd ed., pp. 703–732). New York: Wiley.
- Piaget, J. (1973). *The child's conception of the world*. (J. Tomlinson & A. Tomlinson, Trans.). London: Paladin. (Original work published 1926)
- Piaget, J. (1974). Need and significance of cross-cultural studies in genetic psychology. In J. W. Berry & P. R. Dasen (Eds.), *Culture and cognition: Readings in cross cultural psychology* (pp. 299–310). London: Methuen. (Original work published 1966)
- Piaget, J. (1975). *L'Équilibration des structures cognitives. Problème central du développement* [The equilibration of cognitive structures. The central problem of development]. Paris: Presses Universitaires de France.
- Piaget, J., & Inhelder, B. (1941). *Le développement des quantités chez l'enfant* [The development of quantities in the child]. Neuchâtel, Switzerland: Delachaux et Niestlé.
- Piaget, J., & Inhelder, B. (1956). *The child's conception of space* (F. J. Langdon & J. L. Paul, Trans.). London: Routledge and Kegan Paul. (Original work published 1948)
- Plunkett, K., & Elman, J. L. (1997). *Exercises in rethinking innateness*. Cambridge, MA: Bradford Book with MIT Press.
- Pratt, C. (1988). The child's conception of the conservation task. *British Journal of Developmental Psychology*, 6(2), 157–167.
- Prusinkiewicz, P., & Lindenmeyer, A. (1990). *The algorithmic beauty of plants*. New York: Springer.
- Rabinowitz, F. M., Grant, M. J., Howe, M. L., & Walsh, C. (1994). Reasoning in middle childhood: A dynamic model of performance on transitivity tasks. *Journal of Experimental Child Psychology*, 58, 252–288.
- Raijmakers, M. E. J. (1996). *Epigenesis in neural network models of cognitive development*. Unpublished doctoral dissertation, University of Amsterdam.
- Raijmakers, M. E. J., & Molenaar, P. C. M. (1994). Simulating transitions in cognitive development by means of neural networks. *Tijdschrift voor Ontwikkelingspsychologie*, 21(3), 179–195.
- Raijmakers, M. E. J., van Koten, S., & Molenaar, P. C. M. (1996). On the validity of simulating stagewise development by means of PDP networks: Application of catastrophe analysis and experimental test of rule-like network performance. *Cognitive Science*, 20, 101–136.
- Reznick, J. S., & Goldfield, B. A. (1992). Rapid change in lexical development in comprehension and production. *Developmental Psychology*, 28, 406–413.
- Rogers, S. J., & Puchalski, C. B. (1988). Development of object permanence in visually impaired infants. *Journal of Visual Impairment and Blindness*, 82, 137–142.
- Rogoff, B., Malkin, C., & Gilbride, K. (1984). Interaction with babies as guidance in development. In B. Rogoff & J. V. Wertsch (Eds.), *Children's learning in the "Zone of Proximal Development"* (pp. 31–44). San Francisco: Jossey-Bass.
- Rosen, A. B., & Rozin, P. (1993). Now you see it, now you don't: The preschool child's conception of invisible particles in the context of dissolving. *Developmental Psychology*, 29, 300–311.
- Ross, T. J. (1995). *Fuzzy logic with engineering applications*. New York: McGraw-Hill.
- Rucker, R. (1995). *CAPOW: Evolving continuous-valued cellular au-*

- tomato for power simulation. San José, CA: Department of Mathematics and Computer Science, San José State University.
- Ruhland, R. (1998). *Going the distance: A non-linear approach to change in language development*. Unpublished doctoral dissertation, University of Groningen.
- Ruhland, R., & van Geert, P. (1998). Jumping into syntax: Transitions in the development of closed class words. *British Journal of Developmental Psychology*, 16, 65–96.
- Rumelhart, D. E., & McClelland, J. L. (1986). *Parallel distributed processing: Explorations in the microstructure of cognition*. Cambridge, MA: MIT Press.
- Sandefur, J. T. (1990). *Discrete dynamical systems: Theory and applications*. Oxford, England: Clarendon Press.
- Saxe, G. B., Gearhart, M., & Guberman, S. R. (1984). The social organization of early number development. In B. Rogoff, & J. V. Wertsch (Eds.), *Children's learning in the "Zone of Proximal Development"* (pp. 19–30). San Francisco: Jossey-Bass.
- Schacter, D. L. (1986). Mnemonic precedence in amnesic patients: An analogue of the AB error in infants? *Child Development*, 57, 816–823.
- Schneuwly, B. (1994). Contradiction and development: Vygotsky and paedology. Learning and development: Contributions from Vygotsky [Special issue]. *European Journal of Psychology of Education*, 9(4), 281–291.
- Schroeder, M. (1991). *Fractals, chaos and power laws*. New York: Freeman.
- Selman, R. L. (1980). *The growth of interpersonal understanding: Developmental and clinical analysis*. New York: Academic Press.
- Sera, M., & Smith, L. B. (1987). Big and little: Nominal and relative uses. *Cognitive Development*, 2, 89–112.
- Siegler, R. S. (1981). Developmental sequences within and between concepts. *Monographs of the Society for Research in Child Development*, 46, (Serial No. 189).
- Siegler, R. S. (1983). Information-processing approaches to development. In W. Kessen (Ed.), *Handbook of child psychology: Vol. 1. History, theory and methods* (pp. 129–211). New York: Wiley.
- Siegler, R. S. (1984). Mechanisms of cognitive growth: Variation and selection. In R. J. Sternberg (Ed.), *Mechanisms of cognitive development* (pp. 141–162). New York: Freeman.
- Siegler, R. S. (1989). Mechanisms of cognitive development. *Annual Review of Psychology*, 40, 353–379.
- Siegler, R. S. (1994). Cognitive variability: A key to understanding cognitive development. *Current Directions in Psychological Science*, 3, 1–5.
- Siegler, R. S. (1995). How does change occur: A microgenetic study of number conservation. *Cognitive Psychology*, 28, 225–273.
- Siegler, R. S. (1996). *Emerging minds: The process of change in children's thinking*. New York: Oxford University Press.
- Siegler, R. S., & Crowley, K. (1991). The micro-genetic method: A direct means for studying cognitive development. *American Psychologist*, 46, 606–620.
- Siegler, R. S., & Shipley, C. (1995). Variation, selection and cognitive change. In T. Simon & G. S. Halford (Eds.), *Developing cognitive competence: New approaches to process modeling* (pp. 31–76). Hillsdale, NJ: Erlbaum.
- Siegler, R. S., & Shrager, J. (1984). Strategy choices in addition and subtraction: How do children know what to do? In C. Sophian (Ed.), *Origins of cognitive skills* (pp. 229–293). Hillsdale, NJ: Erlbaum.
- Simon, T. S., & Klahr, D. (1995). A computational theory about children's learning about number conservation. In T. S. Simon & G. S. Halford (Eds.), *Developing cognitive competence: New approaches to process modeling* (pp. 315–353). Hillsdale, NJ: Erlbaum.
- Smith, L. B. (1997). Metaphors and methods: Variability and the study of word learning. In E. Amsel & K. A. Reninger (Eds.), *Change and development. Issues of theory, method and application* (pp. 153–172). Mahwah, NJ: Erlbaum.
- Smith, L. B., & Sera, M. (1992). A developmental analysis of the polar structure of dimensions. *Cognitive Psychology*, 24, 99–142.
- Sophian, C. (1985). Perseveration and infants' search: A comparison of two- and three-location tasks. *Developmental Psychology*, 21, 187–194.
- Sophian, C. (1986). Developments in infants' search for invisibly displaced objects. *Infant Behavior and Development*, 9, 15–25.
- Sophian, C. (1995). Representation and reasoning in early numerical development: Counting, conservation, and comparisons between sets. *Child Development*, 66(2), 559–577.
- Sophian, C., & Sage, S. (1985). Infants' search for hidden objects: Developing skills for using information selectively. *Infant Behavior and Development*, 8, 1–14.
- Sophian, C., & Yengo, L. (1985a). Infants' understanding of visible displacements. *Developmental Psychology*, 21, 932–941.
- Sophian, C., & Yengo, L. (1985b). Infants' search for visible objects: Implications for the interpretation of early search errors. *Journal of Experimental Child Psychology*, 40, 260–278.
- Spelke, E. S., Kestenbaum, R., Simons, D. J., & Wein, D. (1995). Spatiotemporal continuity, smoothness of motion and object identity in infancy. *British Journal of Developmental Psychology*, 13, 113–142.
- Spence, M. J. (1996). Young infants' long-term auditory memory: Evidence for changes in preference as a function of delay. *Developmental Psychobiology*, 29, 685–695.
- Stavy, R., & Stachel, D. (1985). Children's conception of changes in the state of matter: From solid to liquid. *Archives de Psychologie*, 53, 331–344.
- Subbotskii, E. V. (1990). The preschooler's conception of the permanence of an object (verbal and actual behavior). *Soviet Psychology*, 28, 42–67.
- Subbotskii, E. V. (1991a). Existence as a psychological problem: Object permanence in adults and preschool children. *International Journal of Behavioral Development*, 14, 67–82.
- Subbotskii, E. V. (1991b). A life span approach to object permanence. *Human Development*, 34, 125–137.
- Subbotski, E. V., & Trommsdorff, G. (1992). Object permanence in adults: A cross cultural perspective. *Psychologische Beitrage*, 34, 62–79.
- Tabor, L. E., & Kendler, T. S. (1981). Testing for developmental continuity or discontinuity: Class inclusion and reversal shifts. *Developmental Review*, 1, 330–343.
- Thatcher, R. (1994). Cyclic cortical reorganization: Origins of human cognitive development. In G. Dawson & K. Fischer (Eds.), *Human behaviour and the developing brain* (pp. 232–266). New York: Guilford.
- Thatcher, R. W. (1991). Maturation of the human frontal lobes: Physiological evidence for staging. Developmental consequences of early frontal lobe damage [Special issue]. *Developmental Neuropsychology*, 7, 397–419.
- Thatcher, R. W. (1992). Cyclic cortical reorganization during early childhood. The role of frontal lobe maturation in cognitive and social development [Special issue]. *Brain and Cognition*, 20(1), 24–50.
- Thelen, E. (1989). Self-organization in developmental processes: Can systems approaches work. In M. Gunnar & E. Thelen (Eds.), *Systems and development: The Minnesota Symposia on Child Psychology* (Vol. 22, pp. 17–117). Hillsdale, NJ: Erlbaum.
- Thelen, E. (1992). Development as a dynamic system. *Current Directions in Psychological Science*, 1(6), 189–193.
- Thelen, E. (1995a). Motor development: A new synthesis. *American Psychologist*, 50, 79–95.
- Thelen, E. (1995b). Time-scale dynamics and the development of an embodied cognition. In R. F. Port & T. Van Gelder (Eds.), *Mind as*

- motion: Explorations in the dynamics of cognition* (pp. 69–100). Cambridge, MA: Bradford Books with MIT Press.
- Thelen, E., Corbetta, D., Kamm, K., Spencer, J. P., Schneider, K., & Zernicke, R. F. (1993). The transition to reaching: Mapping intention and intrinsic dynamics. *Child Development*, 64, 1058–1098.
- Thelen, E., & Smith, L. B. (1994). *A dynamic systems approach to the development of cognition and action*. Cambridge, MA: Bradford Books with MIT Press.
- Thom, R. (1977). *Stabilité structurelle et morphogénèse: Essai d'une théorie générale des modèles* [Structural stability and morphogenesis: Toward a general theory of models]. Paris: InterEditions.
- Thomas, H. (1989). A binomial mixture model for classification performance: A commentary on Waxman, Chambers, Yntema and Gelman. *Journal of Experimental Child Psychology*, 48, 423–430.
- Thomas, H., & Lohaus, A. (1993). Modeling growth and individual differences in spatial tasks. *Monographs of the Society for Research in Child Development*, 58(9, Serial No. 237).
- Thomas, H., & Turner, G. F. W. (1991). Individual differences and development in water-level-task performance. *Journal of Experimental Child Psychology*, 51, 171–194.
- Triana, E., & Pasnak, R. (1986). A distraction task as an assessment of object permanence. *Infant Behavior and Development*, 9, 151–165.
- Tudge, J. R., & Winterhoff, P. A. (1993). Vygotsky, Piaget, and Bandura: Perspectives on the relations between the social world and cognitive development. *Human Development*, 36, 61–81.
- Tulving, E., Markowitsch, H. J., Craik, F. I. M., Habib, R., et al. (1996). Novelty and familiarity activations in PET studies of memory encoding and retrieval. *Cerebral Cortex*, 6, 71–79.
- Tulving, E., Markowitsch, H. J., Kapur, S., Habib, R., et al. (1994). Novelty encoding networks in the human brain: Positron emission tomography data. *Neuroreport: An International Journal for the Rapid Communication of Research in Neuroscience*, 5, 2525–2528.
- Turiel, E., & Davidson, P. (1986). Heterogeneity, inconsistency and asynchrony in the development of cognitive structures. In I. Levin (Ed.), *Stage and structure: Reopening the debate* (pp. 106–143). Norwood, NJ: Ablex.
- Turvey, M. T., & Fitzpatrick, P. (1993). Commentary: Development of perception–action systems and general principles of pattern formation. *Child Development*, 64, 1175–1190.
- Uzgiris, I. C. (1977). Commentary. *Monographs of the Society for Research in Child Development*, 171, 42 (3, Serial No. 171), 95–102.
- Vallacher, R. R., & Nowak, A. (1994). Dynamical systems: A tool for social psychology. In R. R. Vallacher & A. Nowak (Eds.), *Dynamical systems in social psychology* (pp. 17–54). New York: Academic Press.
- Valsiner, J. (1984). Construction of the Zone of Proximal Development in adult–child joint interaction: The socialization of meals. In B. Rogoff & J. V. Wertsch (Eds.), *Children's learning in the "Zone of Proximal Development"* (pp. 65–76). San Francisco: Jossey-Bass.
- Valsiner, J. (1988). *Developmental psychology in the Soviet Union*. Brighton, England: Harvester.
- Valsiner, J. (1994). Vygotskian dynamics of development: Comment. *Human Development*, 37, 366–369.
- van der Maas, H. (1993). *Catastrophe analysis of stage-wise cognitive development: Model, method and applications*. Doctoral dissertation, University of Amsterdam.
- van der Maas, H., & Molenaar, P. (1992). A catastrophe–theoretical approach to cognitive development. *Psychological Review*, 99, 395–417.
- van der Maas, H., & Molenaar, P. (1996). Catastrophe analysis of discontinuous development. In A. von Eye and C. C. Clogg (Eds.), *Categorical variables in developmental research: Methods of analysis* (pp. 77–105). New York: Academic Press.
- van der Molen, M. W., & Molenaar, P. C. M. (1994). Cognitive psychophysiology: A window to cognitive development and brain maturation. In G. Dawson & K. Fischer (Eds.), *Human behavior and the developing brain* (pp. 456–490). New York: Guilford Press.
- Van der Veer, R. (1986). Vygotsky's developmental psychology. *Psychological Reports*, 59(2), 527–536.
- Van der Veer, R. (1996). *L. Vygotskij: Cultuur en Ontwikkeling* [L. Vygotsky: Culture and development]. Amsterdam: Boom.
- Van der Veer, R., & Valsiner, J. (1991). Understanding Vygotsky: A quest for synthesis. Oxford, England: Basil Blackwell.
- van Geert, P. (1991). A dynamic systems model of cognitive and language growth. *Psychological Review*, 98, 3–53.
- van Geert, P. (1993). A dynamic systems model of cognitive growth: Competition and support under limited resource conditions. In L. B. Smith & E. Thelen (Eds.), *A dynamic systems approach to development: Applications* (pp. 265–331). Cambridge, MA: MIT Press.
- van Geert, P. (1994a). *Dynamic systems of development: Change between complexity and chaos*. New York: Prentice Hall with Harvester Wheatsheaf.
- van Geert, P. (1994b). Vygotskian dynamics of development. *Human Development*, 37, 346–365.
- van Geert, P. (1994c). Why are there four Piagetian stages? A speculative journey into developmental fractals and subcriticality. *Tijdschrift voor Ontwikkelingspsychologie*, 21, 197–214.
- van Geert, P. (1995a). Dimensions of change: A semantic and mathematical analysis of learning and development. *Human Development*, 38, 322–331.
- van Geert, P. (1995b). Growth dynamics in development. In R. F. Port & T. Van Gelder (Eds.), *Mind as motion: Explorations in the dynamics of cognition* (pp. 313–337). Cambridge, MA: Bradford Books with MIT Press.
- van Geert, P. (1997). Variability and fluctuation: A dynamic view. In E. Amsel & K. A. Renninger (Eds.), *Change and development: Issues of theory, method and application* (pp. 193–212). Mahwah, NJ: Erlbaum.
- van Geert, P. (1998a). *Are neo-Piagetian stage models fractal?* Groningen, the Netherlands: University of Groningen, Department of Psychology.
- van Geert, P. (1998b). *Evidence for qualitative discontinuity in the A-not-B object concept task: A growth analysis of the Wellman et al. (1987) study*. Groningen, the Netherlands: University of Groningen Department of Psychology.
- Vandewalle, N., & Ausloos, M. (1996). The screening of species in a Darwinistic tree-like model of evolution. *Physica D*, 90, 262–270.
- Verhofstadt, L., Vyt, A., & Van Geert, P. (1995). *Handboek ontwikkelingspsychologie* [Handbook of developmental psychology]. Houten, the Netherlands: Bohn, Stafleu en Van Loghum.
- Vonèche, J. J., & Aeschlimann, E. (1994). Theory integration versus theory selection in developmental psychology. In P. Van Geert, L. P. Mos, & W. J. Baker (Eds.), *Annals of theoretical psychology* (Vol. 10, pp. 151–188). New York: Plenum.
- Vygotsky, L. S. (1962). *Thought and language* (E. Hanfmann & G. Vakar, Trans.). Cambridge, MA: MIT Press.
- Vygotsky, L. S. (1966). Play and its role in the psychological development of the child. *Voprosy Psikhologii*, 12(6), 62–76.
- Vygotsky, L. S. (1978). *Mind in society*. London: Harvard University Press.
- Weil-Barais, A. (1994). Heuristic value of the notion of zone of proximal development in the study of child and adolescent construction of concepts in physics. Learning and development: Contributions from Vygotsky [Special issue]. *European Journal of Psychology of Education*, 9, 367–383.
- Wellman, H. M., Cross, D., & Bartsch, K. (1987). Infant search and object permanence: A meta-analysis of the A-not-B error. *Monographs of the Society for Research in Child Development*, 51(Serial No. 214), 1–51.

- Wertsch, J. (1985). *Vygotsky and the social formation of mind*. Cambridge, MA: Harvard University Press.
- Wertsch, J. V., & Tulviste, P. (1992). L. S. Vygotsky and contemporary developmental psychology. *Developmental Psychology*, 28, 548–557.
- Wheldall, K., & Benner, H. (1993). Conservation without conversation revisited: A replication and elaboration of the Wheldall–Poborca findings on the nonverbal assessment of conservation of liquid quantity. *Educational Psychology*, 13, 49–58.
- Whitehurst, G. J., Falco, F. L., Lonigan, C. J., Fischel, J. E., Debaryshe, B. D., Valdez-Menchaca, M. C., & Caulfield, M. (1988). Accelerating language development through picture book reading. *Developmental Psychology*, 24, 552–559.
- Wilson, M. (1989). Saltus: A psychometric model of discontinuity in cognitive development. *Psychological Bulletin*, 105, 276–289.
- Wimmers, R. H. (1996). *Grasping developmental change: Theory, methodology and data*. Doctoral dissertation, Free University of Amsterdam.
- Wimmers, R. H., Beek, P. J., Savelsbergh, G. J. P., & Hopkins, B. (1996). Developmental changes in action: Theoretical and methodological issues. *British Journal of Developmental Psychology*, 16, 45–63.
- Wimmers, R. H., Savelsbergh, G. J. P., van der Kamp, J., & Hartelman, P. (1998). A developmental transition in prehension modeled as a cusp catastrophe. *Developmental Psychobiology*, 32, 23–35.
- Winer, G. A., Craig, R. K., & Weinbaum, E. (1992). Adults' failure on misleading weight conservation tests: A developmental analysis. *Developmental Psychology*, 28, 109–120.
- Winer, G. A., Hemphill, J., & Craig, R. K. (1988). The effect of misleading questions in promoting nonconservation responses in children and adults. *Developmental Psychology*, 24, 197–202.
- Winer, G. A., & McGlone, C. (1993). On the uncertainty of conservation: Responses to misleading conservation questions. *Developmental Psychology*, 29, 760–769.
- Wishart, J. G., & Bower, T. G. (1984). Spatial relations and the object concept: A normative study. *Advances in Infancy Research*, 3, 57–123.
- Wishart, J. G., & Bower, T. G. (1985). A longitudinal study of the development of the object concept: Infancy [Special issue]. *British Journal of Developmental Psychology*, 3, 243–258.
- Xu, F., & Carey, S. (1996). Infants' metaphysics: The case of numerical identity. *Cognitive Psychology*, 30, 111–153.
- Yates, D. J., & Bremner, J. G. (1988). Conditions for Piagetian Stage IV search errors in a task using transparent occluders. *Infant Behavior and Development*, 11, 411–417.
- Zender, M. A., & Zender, B. F. (1974). Vygotsky's view about the age periodization of child development. *Human Development*, 17(1), 24–40.

Appendix

Parameter Sets of Simulations in Figures

Parameter sets	Figure 3	Figure 4	Figure 9	Figure 9(2)	Figure 10	Figure 11	Figure 12	Figure 13	Figure 14	Figure 15
Rate or R-group										
η Support	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
θ Support	0.3	0.2	0.1	0.1	0.1	0.3	0.3	0.2	0.3	0.3
η Competition	-0.001	-0.2	-0.25	-0.25	-0.25	-0.05	-0.05	-0.01	-0.1	-0.05
θ Competition	-0.3	-0.4	-0.25	-0.25	-0.25	-0.3	-0.3	-0.2	-0.3	-0.3
α Activity level	0.1	1	1	1	1	1	1	0.5	1	0.5
Spreading or S-group										
σa Spreading action	5	10	10	10	5	5	7	7	10	7
σe Spreading experience	5	10	10	10	5	5	7	7	10	7
Linear input	no	yes	no	no	no	no	yes	no	no	no
ϕ Optimal distance		0.1					0.2			
δ Expansion factor ^a		35					20			
Gaussian input	yes	no	yes	yes	yes	yes	no	yes	yes	yes
$\gamma 1$	0.5		2	4	1.54	1.5		0.6	0.5	0.5
$\gamma 3$	8.94427		8.94427	8.94427	8.94427	8.94427		8.94427	8.94427	8.94427
$\gamma 5$	0		0	0	0	0.5		1	0.5	0.5
Input format	Standard	Piaget	Standard	Standard	Standard	Standard	Standard	Standard	Standard	Standard
Zoped or Z-group										
ξo Zoped—on	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ξr Zoped—range	0.10		0.10	0.10	0.10	0.20	0.10	0.10	0.10	0.10
ξp Zoped—probability	0.50		1.00	1.00	1.00	0.10	0.20	0.20	0.40	0.80
ξm	1.00		1.00	1.00	1.00	1.00	0.20	1.00	1.00	1.00
Maximum zoped probability										
Number of steps	1,000	1,500	500	500	70	2,000	700	1,500	750	300

^a The given parameter value is the divisor of a fraction, which has 1 as the numerator and δ as its denominator.

Received February 8, 1996

Revision received March 31, 1998

Accepted March 31, 1998 ■

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